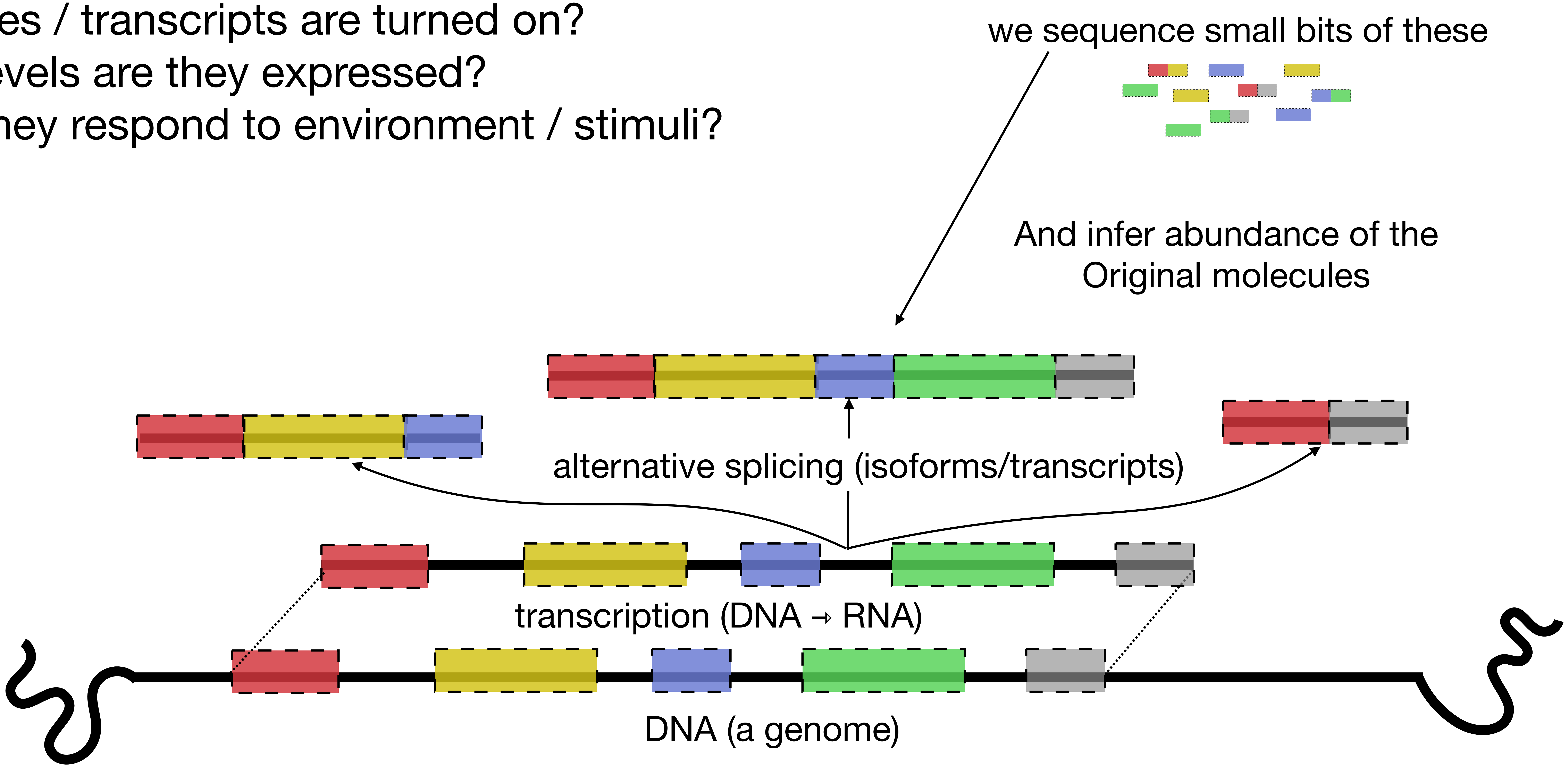
Computational and inferential challenges in the processing of single-cell RNA-seq data

UNIVERSITY OF MARYLAND

RNA-seq basics in one slide

Lets us (among other things) quantify expression in a tissue:

- What genes / transcripts are turned on?
- At what levels are they expressed?
- How do they respond to environment / stimuli?



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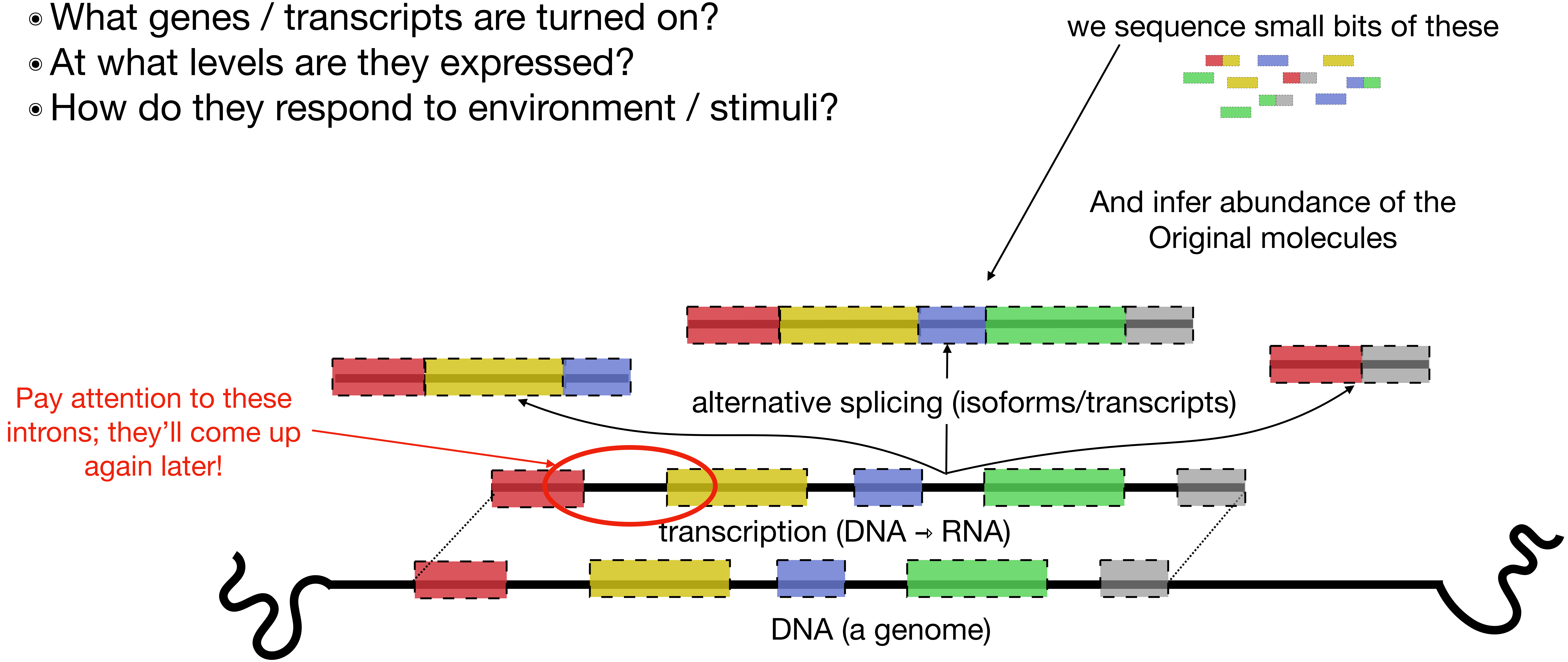
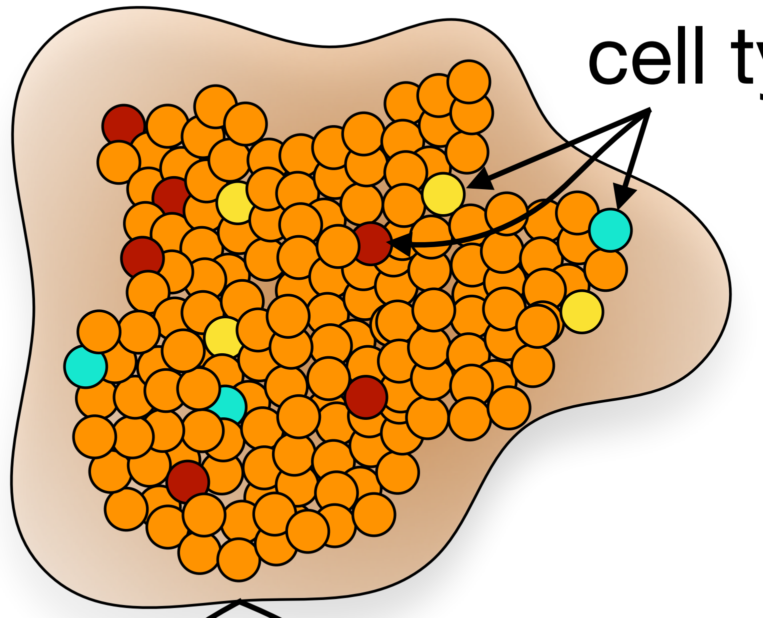


Bulk RNA-seq →Single-cell RNA-seq

A brave new world

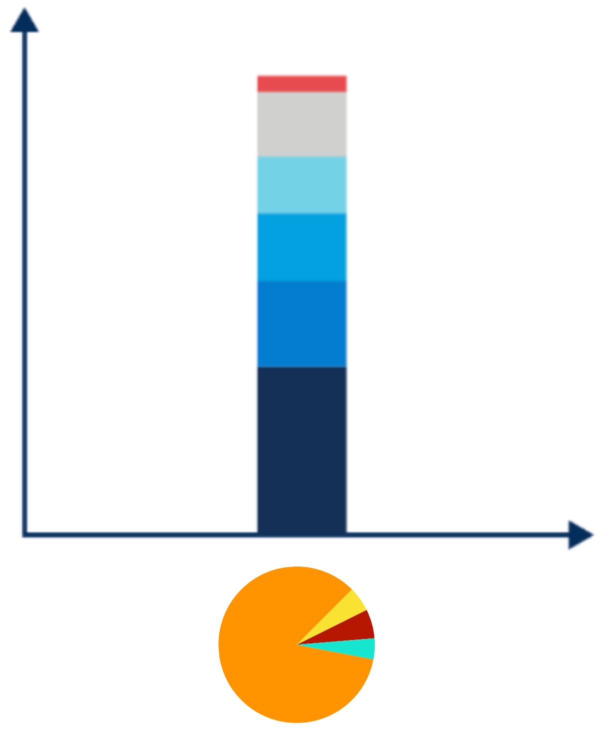
Tissue

rare
cell types



Bulk RNA-seq

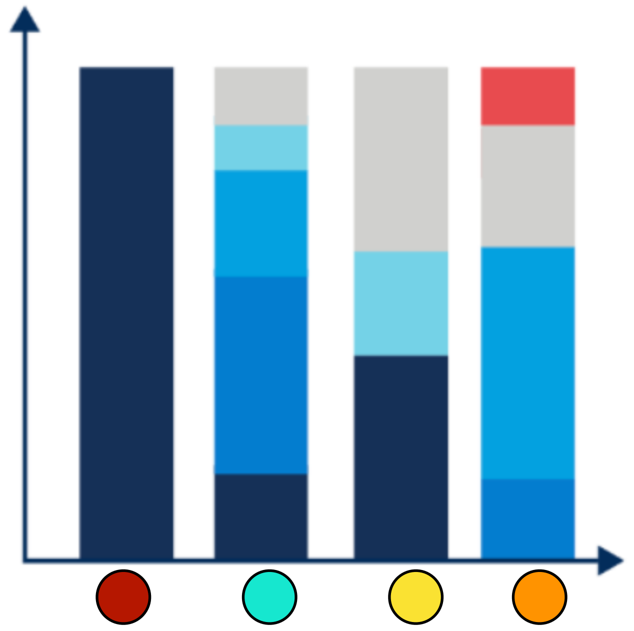
- Typically millions or 10s of millions of cells
- High-fidelity & high-sensitivity
- Measure transcript abundance at the population-level



These present **very** different challenges

Single-cell RNA-seq

- Typically tens of thousands of cells
- Low-coverage (reads / cell)
- Measure gene abundance at the single-cell level



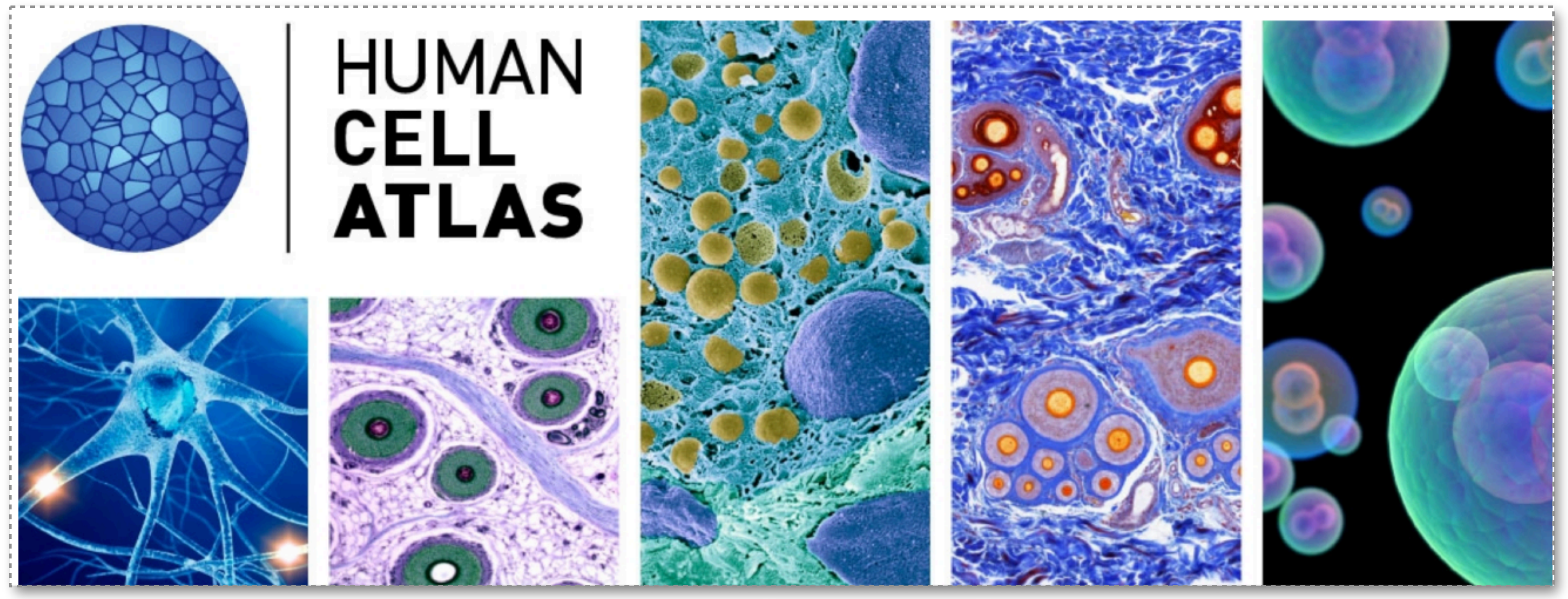
involves cell-type identification (supervised or unsupervised)

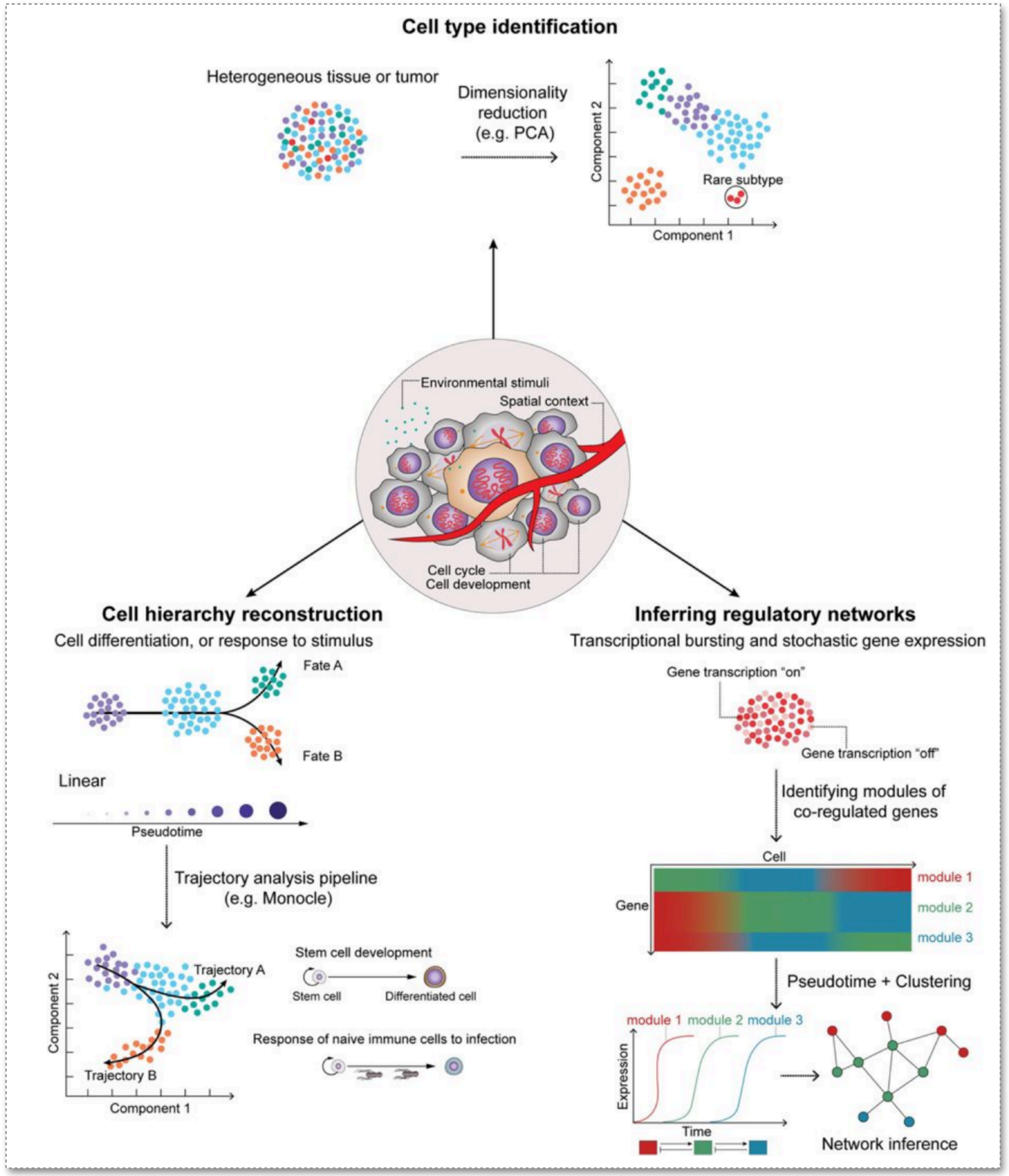
https://www.slideshare.net/QIAGENscience/why-scientist-analyze-single-cells (smoothie)
https://community.10xgenomics.com/t5/10x-Blog/Single-Cell-RNA-Seq-An-Introductory-Overview-and-Tools-for/ba-p/547 (bar chart images)

What are some exciting prospects of single-cell sequencing?

Explore biology at *unprecedented resolution.*
Some *transformative* applications:

- Study *treatment-resistant* cells in disease (cancer)
- Understand tissue / organism *development* (cell fate)
- Understand immune response at the cellular level
- Understand dynamic cellular processes
- Learn how expression is regulated (regulatory net.)
- Characterize new cell types & cell states (HCA)





Hwang, B., Lee, J. H., & Bang, D. (2018). Single-cell RNA sequencing technologies and bioinformatics pipelines. *Experimental & molecular medicine*, *50*(8), 96.
https://www.broadinstitute.org/research-highlights-human-cell-atlas

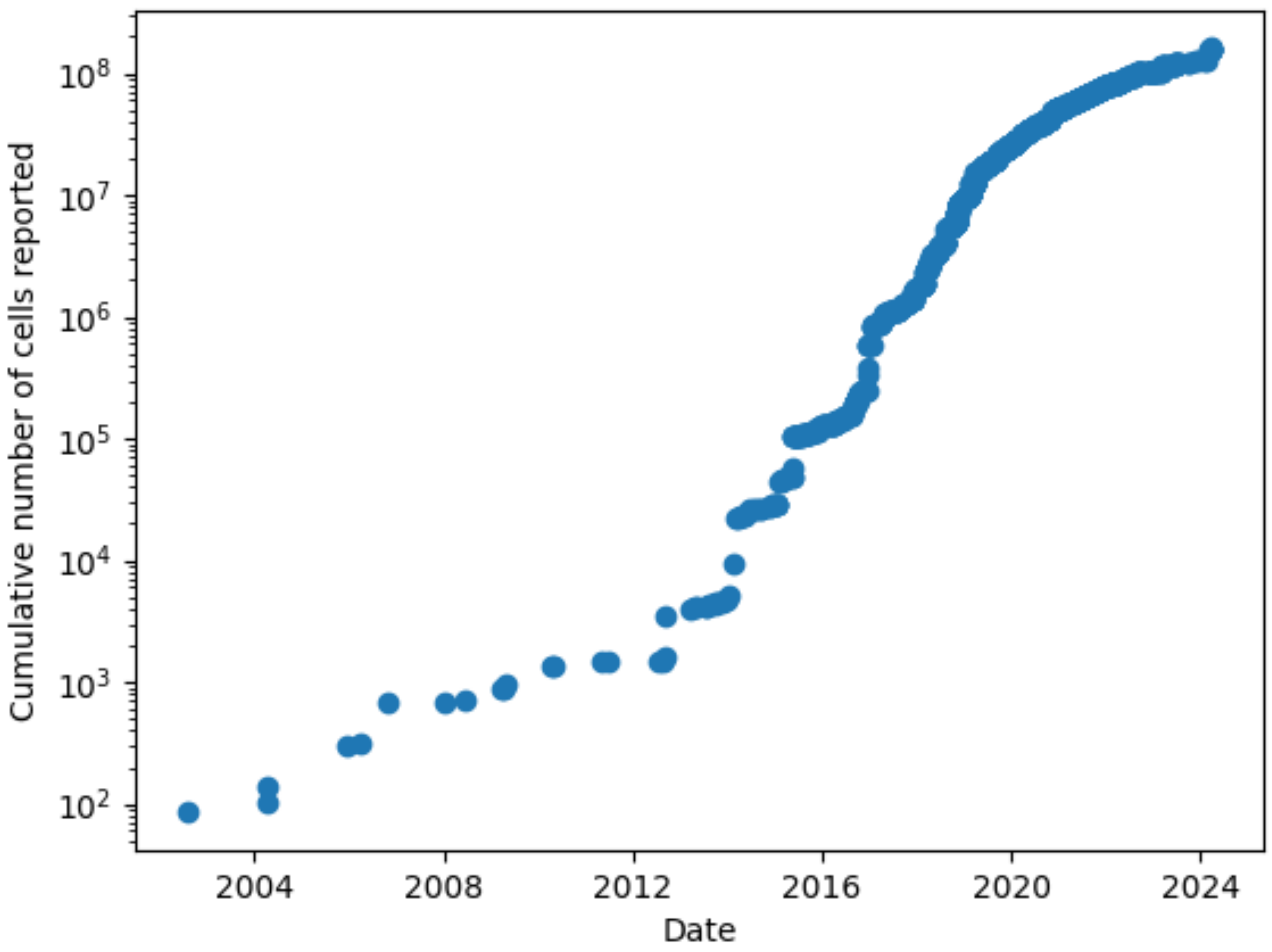
Alevin-fry
&
Simpleaf

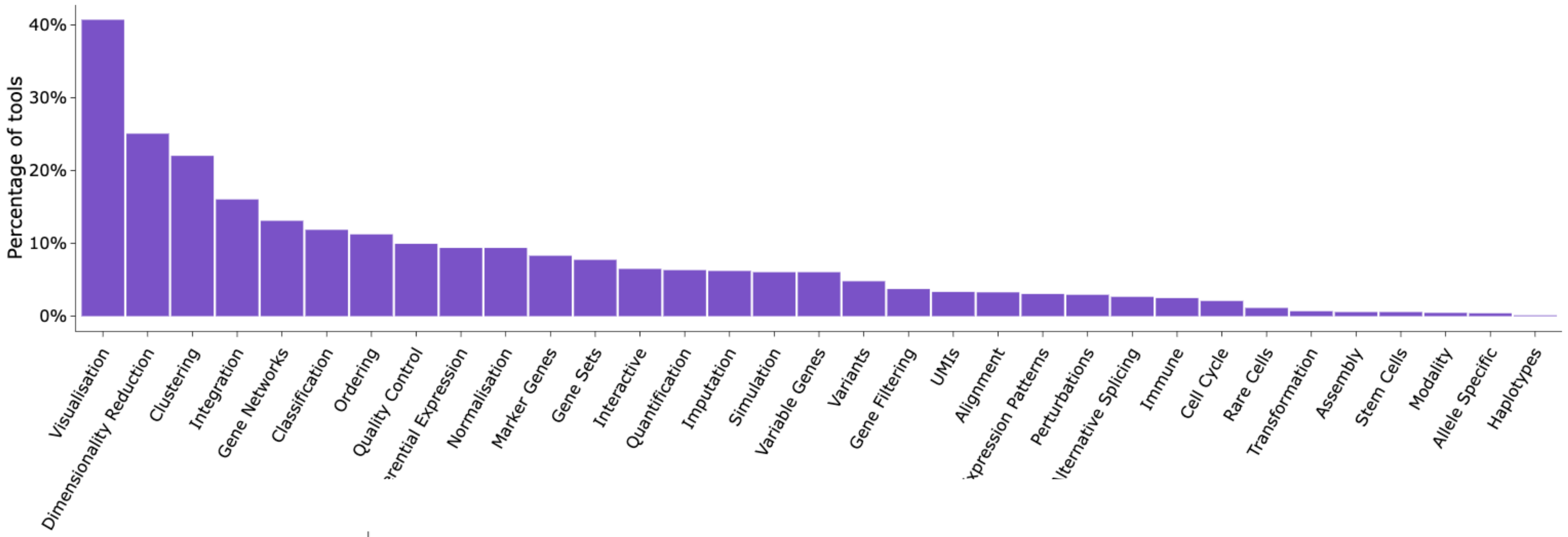
An end-to-end computational solution for single-cell RNA-seq pre-processing

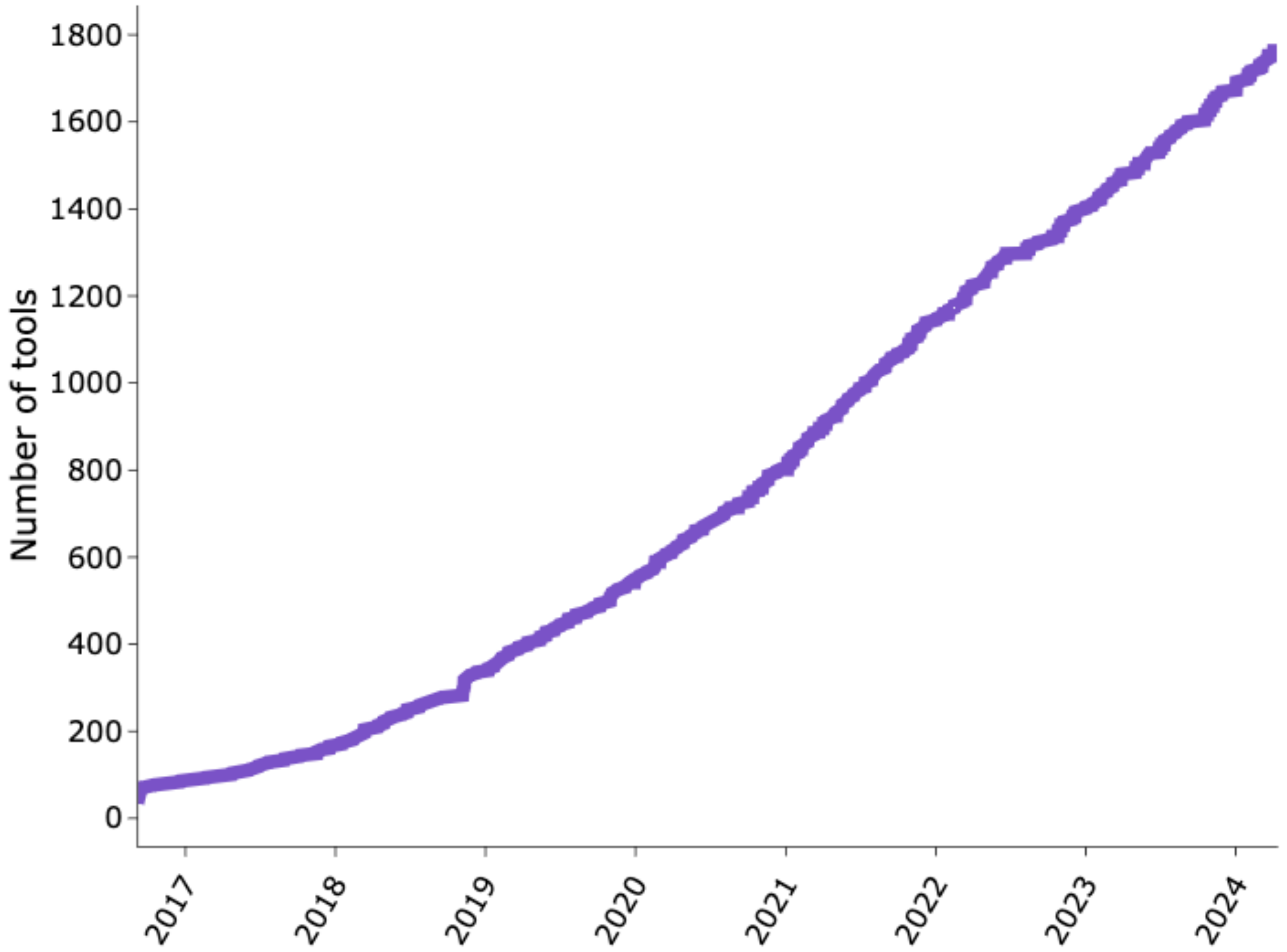
scRNA-seq
Splice Status Prediction
&
Off-Target Read Analysis

A comprehensive analysis of off-target reads and their potential impacts

Rapid growth in both data and methods (tools)







Data from: Svensson, Valentine, Eduardo da Veiga Beltrame, and Lior Pachter.
"A curated database reveals trends in single-cell transcriptomics." *Database* 2020 (2020).

Data from: Zappia L, Phipson B, Oshlack A. "Exploring the single-cell RNA-seq analysis landscape with the scRNA-tools database", PLOS Computational Biology (2018), DOI: 10.1371/journal.pcbi.1006245

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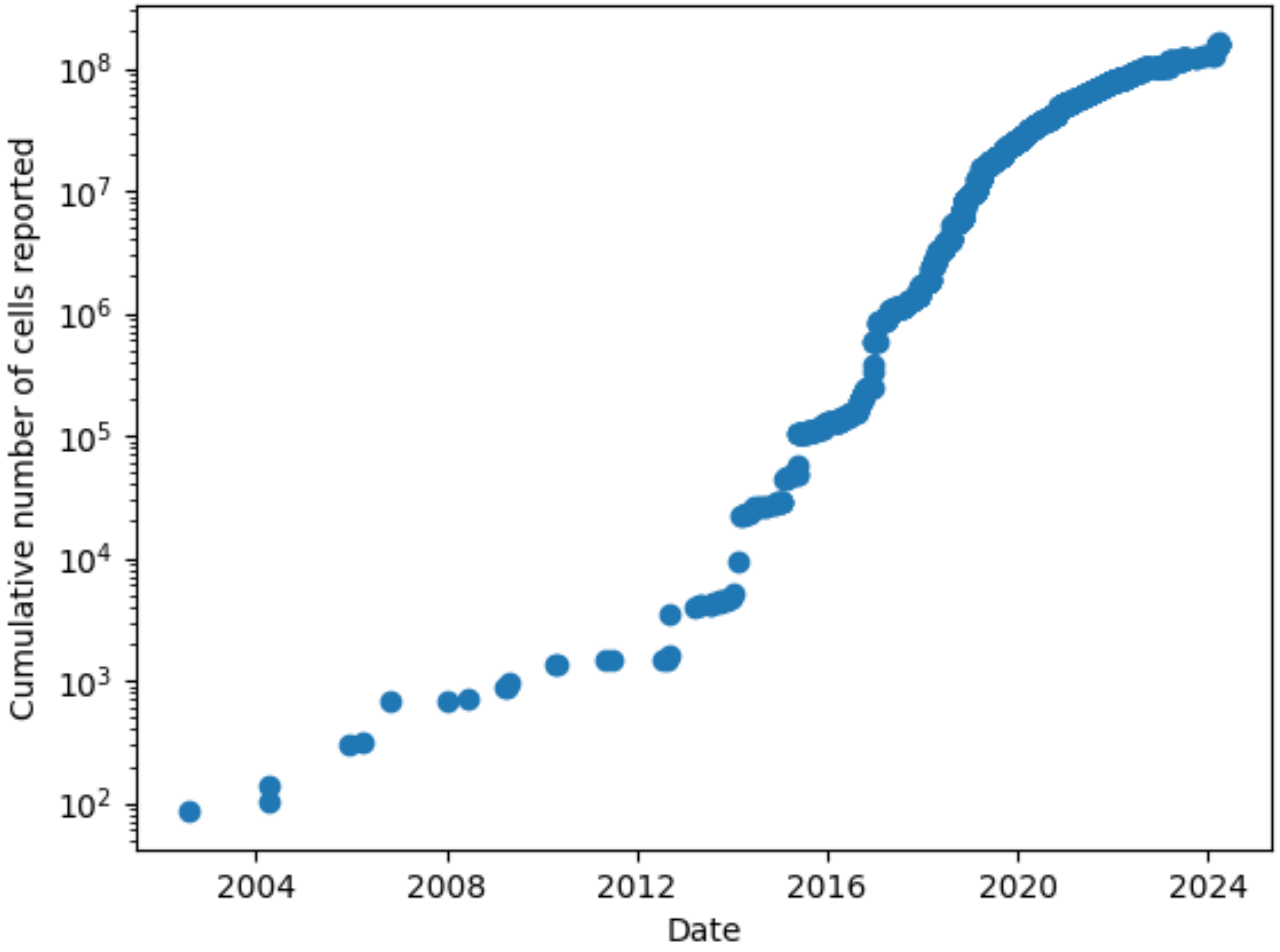

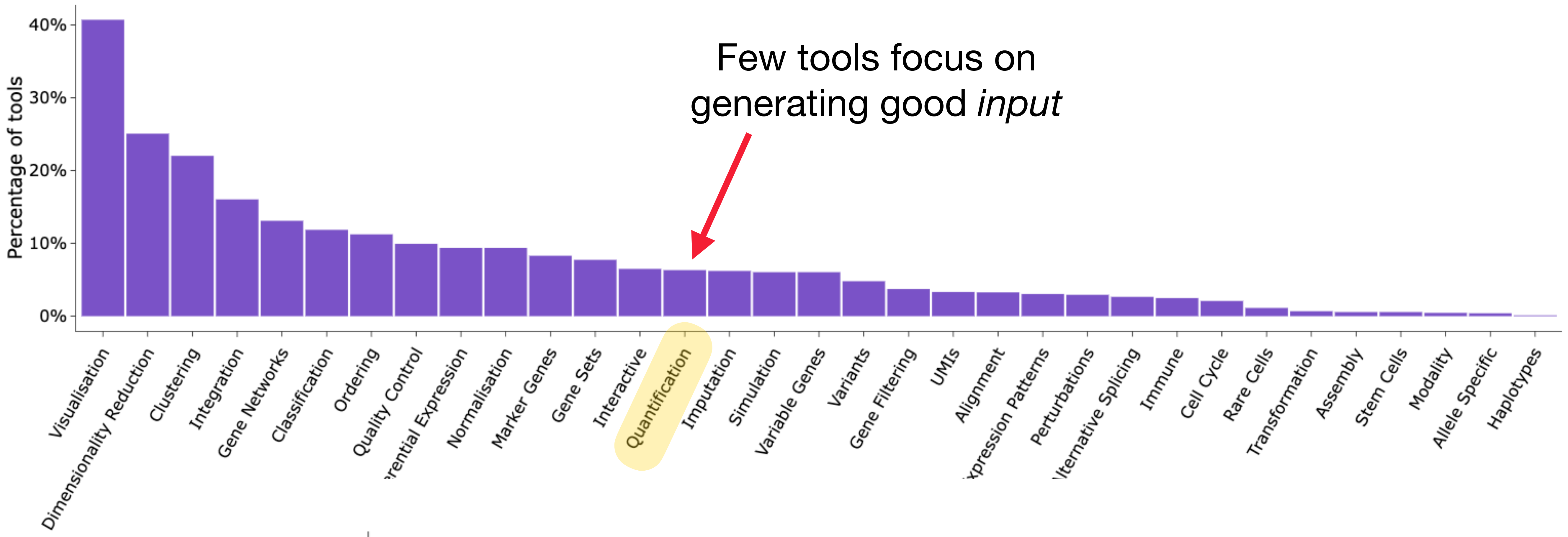

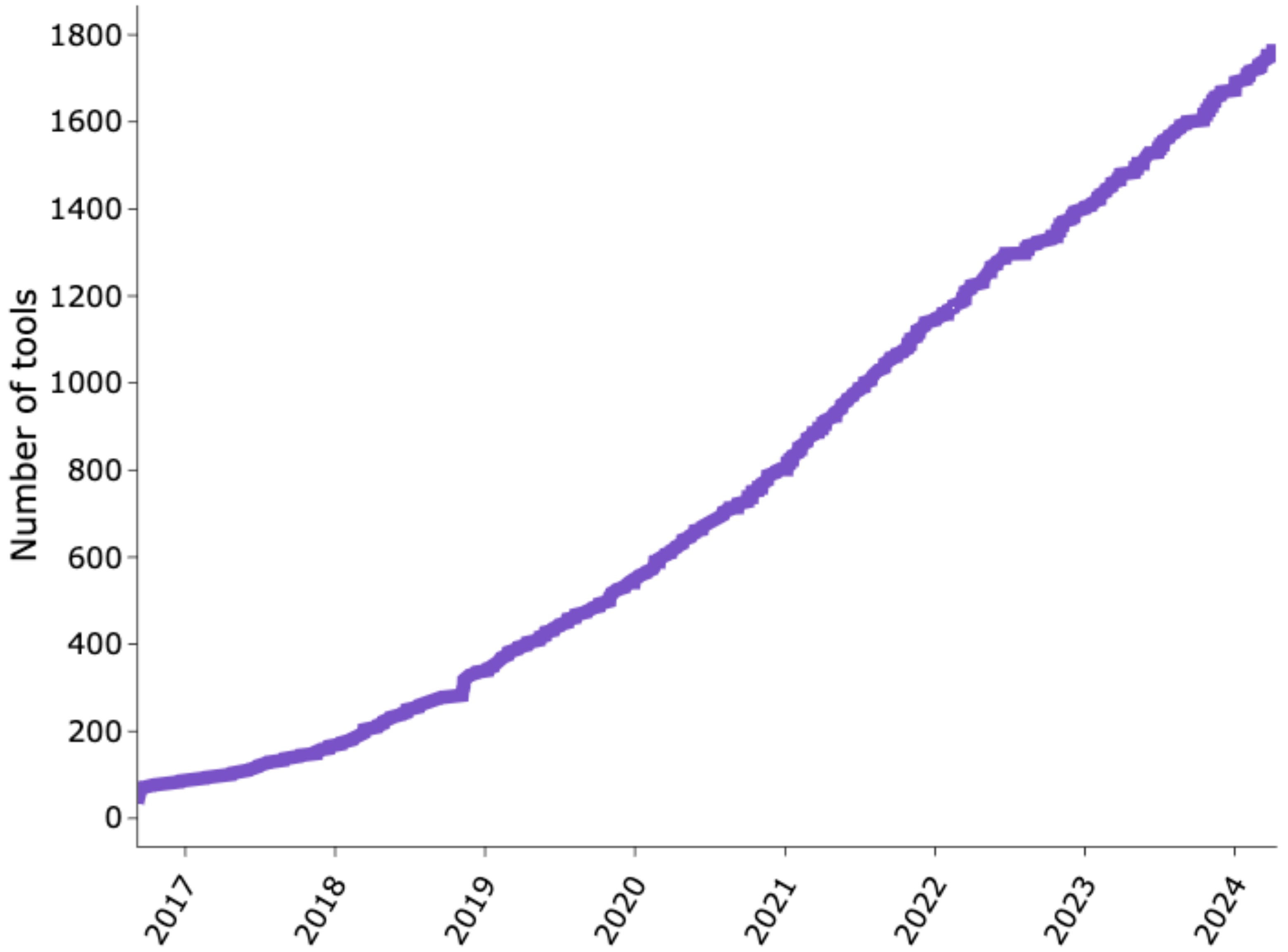

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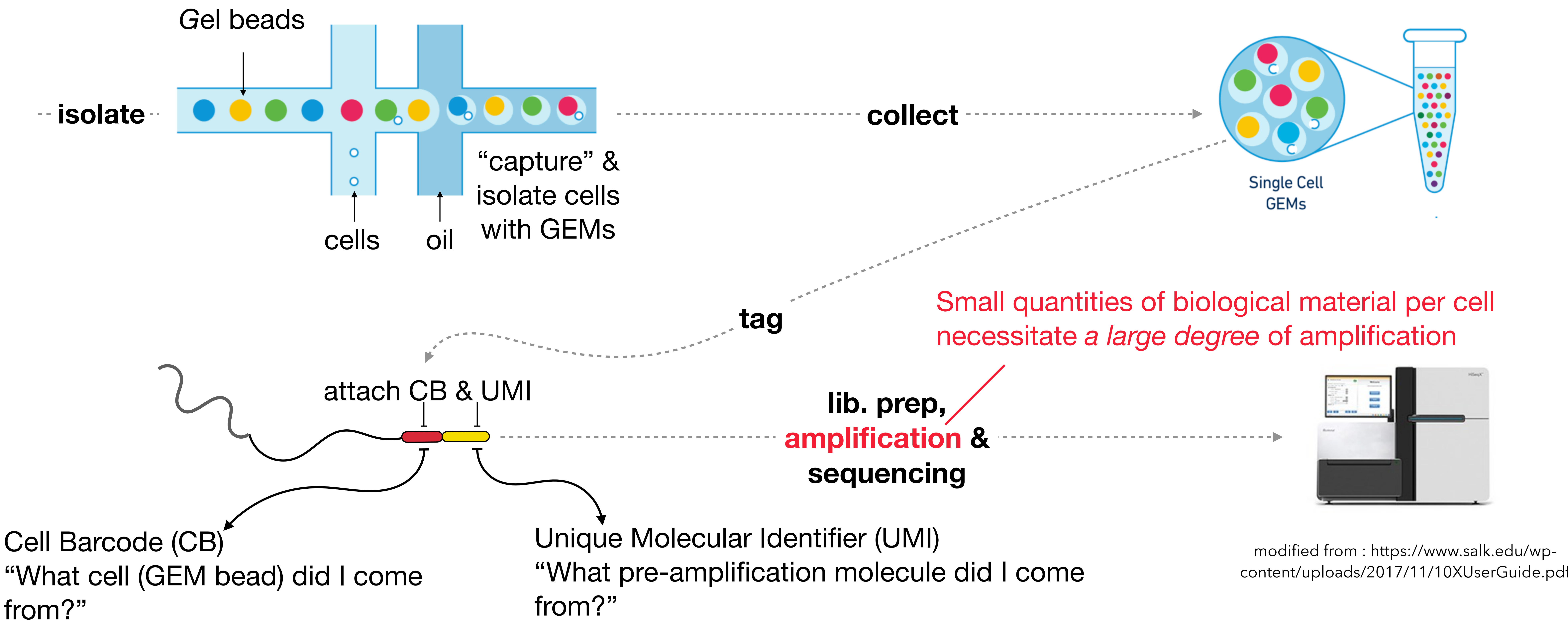
(Droplet-based) Single-cell RNA-seq

Many different protocols — here is a gist of droplet-based (microfluidic) techniques (the general approach used by 10x Genomics Chromium platform).

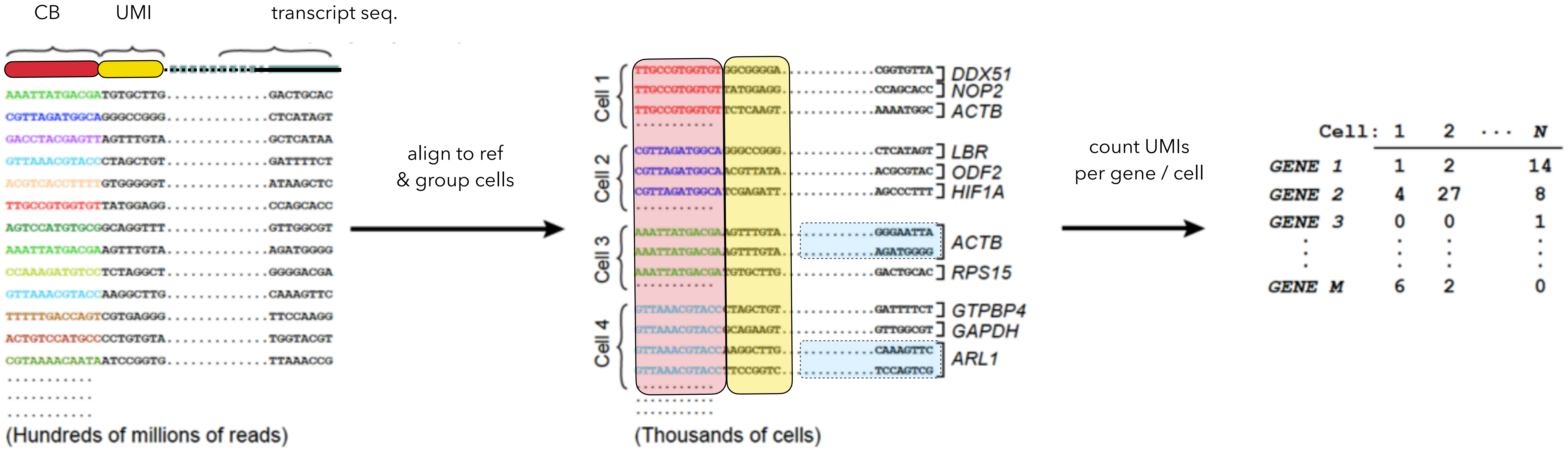


modified from : https://www.salk.edu/wp-content/uploads/2017/11/10XUserGuide.pdf

Partition by cell & group by UMI

In the case of perfect alignment & sequencing, and no ambiguity, using the CB & UMI to estimate gene expression we simply:

1. Group sequenced reads by their CB (read → cell)
2. Collapse all reads with the same UMI (read in cell → molecule)
3. Count!

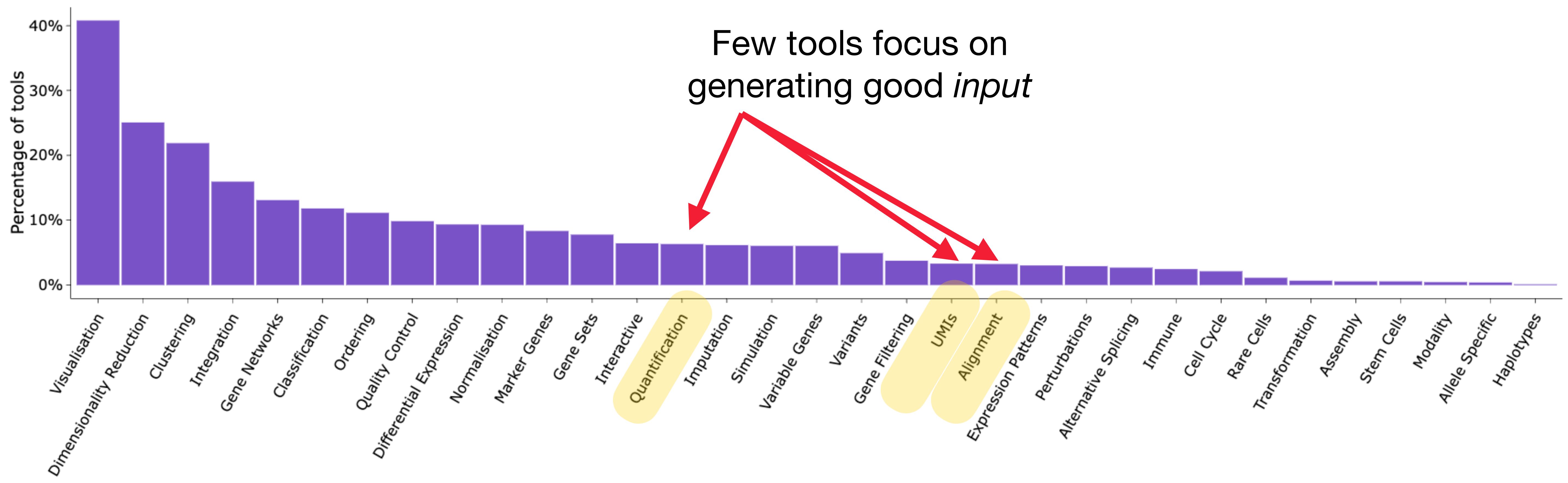


In practice both mapping & sequencing are imperfect and the process is more complicated

http://mccarrolllab.org/wp-content/uploads/2015/05/figure-2D.png

Rapid growth in both data and methods (tools)

Methods



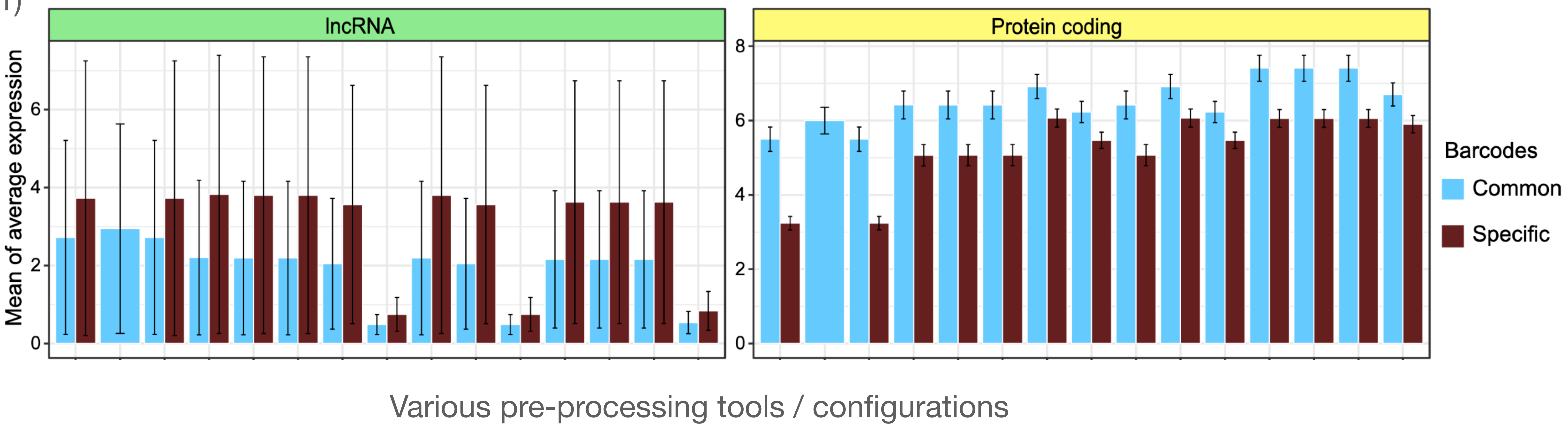
But: Pre-processing is not a “solved” problem, and the choices we make *matter*!

🗑 in → 🗑 out

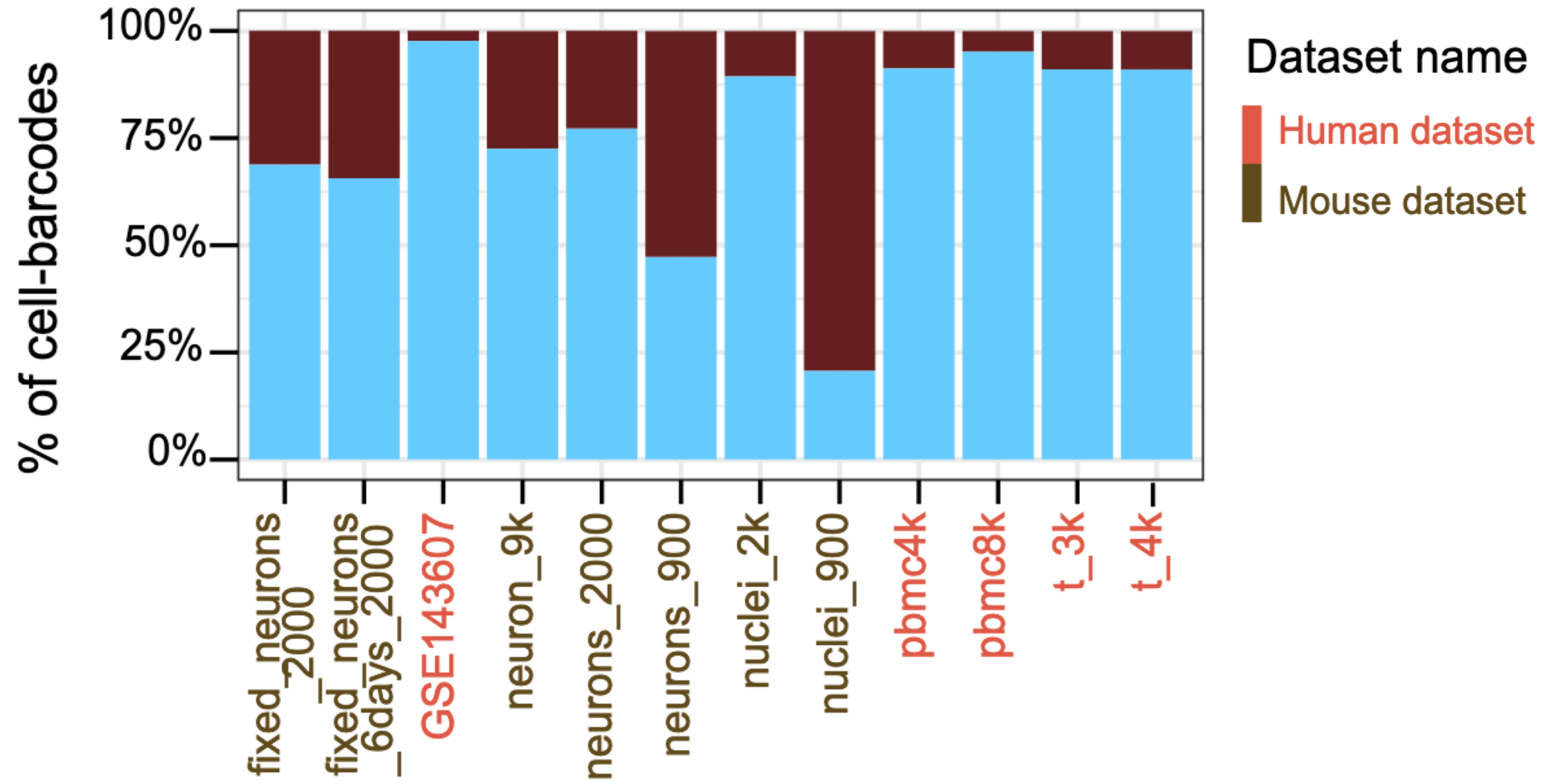
Ultimately, the quality of your analysis will depend on the quality of your input!

What kind of differences could pre-processing make?

1)



2)



> Bar plot showing the average gene expression of lncRNA (left) and protein coding genes (right) for human 8k PBMCs from a Healthy Donor.

1) There can be a *tremendous* degree of variation between which barcodes (cells) are called, and the corresponding expression assigned to genes.

2) This variation itself can vary substantially between dataset and organism.

What kind of differences could pre-processing make?

This variation is just between different versions of the same software; Cell Ranger!

Impacts of Cell Ranger versions on Chromium gene expression data. Imad Abugessaisa, Akira Hasegawa, Scott Walker, Shintaro Katayama, Juha Kere, Takeya Kasukawa
bioRxiv 2024.08.10.607413; doi: https://doi.org/10.1101/2024.08.10.607413

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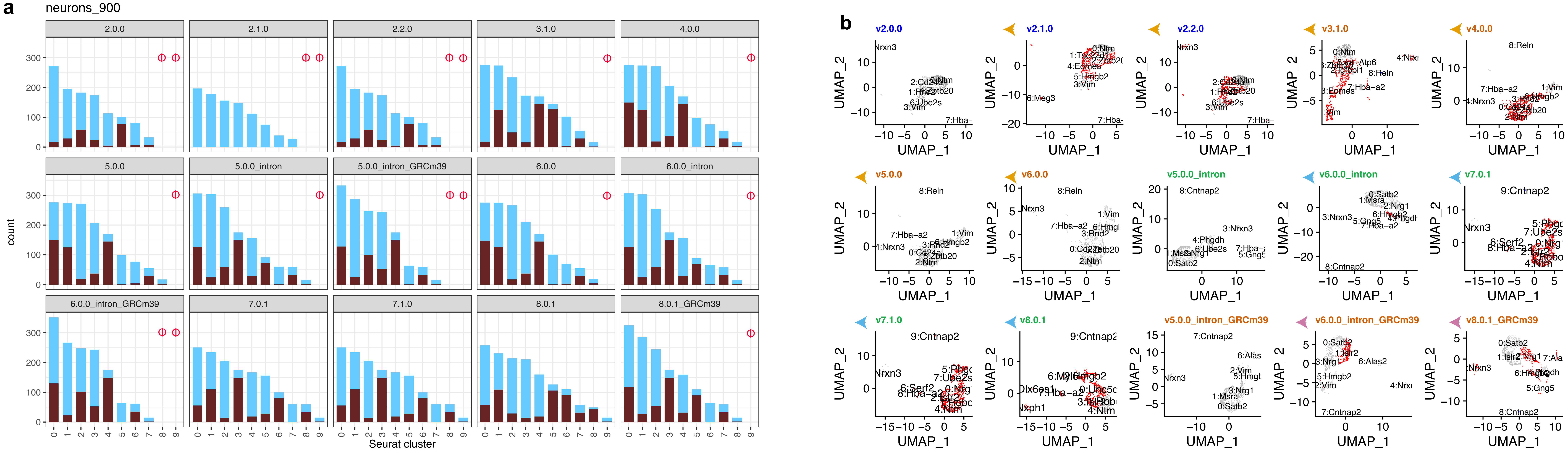
> "The choice of CR version affects scores for quality, average gene expression, clustering results, and top cluster marker genes for each dataset. Our study indicates a demonstrable, quantitative effect on downstream analysis from choice of CR version, resulting in widely different Chromium single cell gene expression data for different CR versions."

These differences are also not uniform and ambient, but induce different clusters, markers, etc.

In other words; they can *fundamentally* affect downstream analysis!



Supplementary figure 5

Preprocessing choice can determine what you can even compute

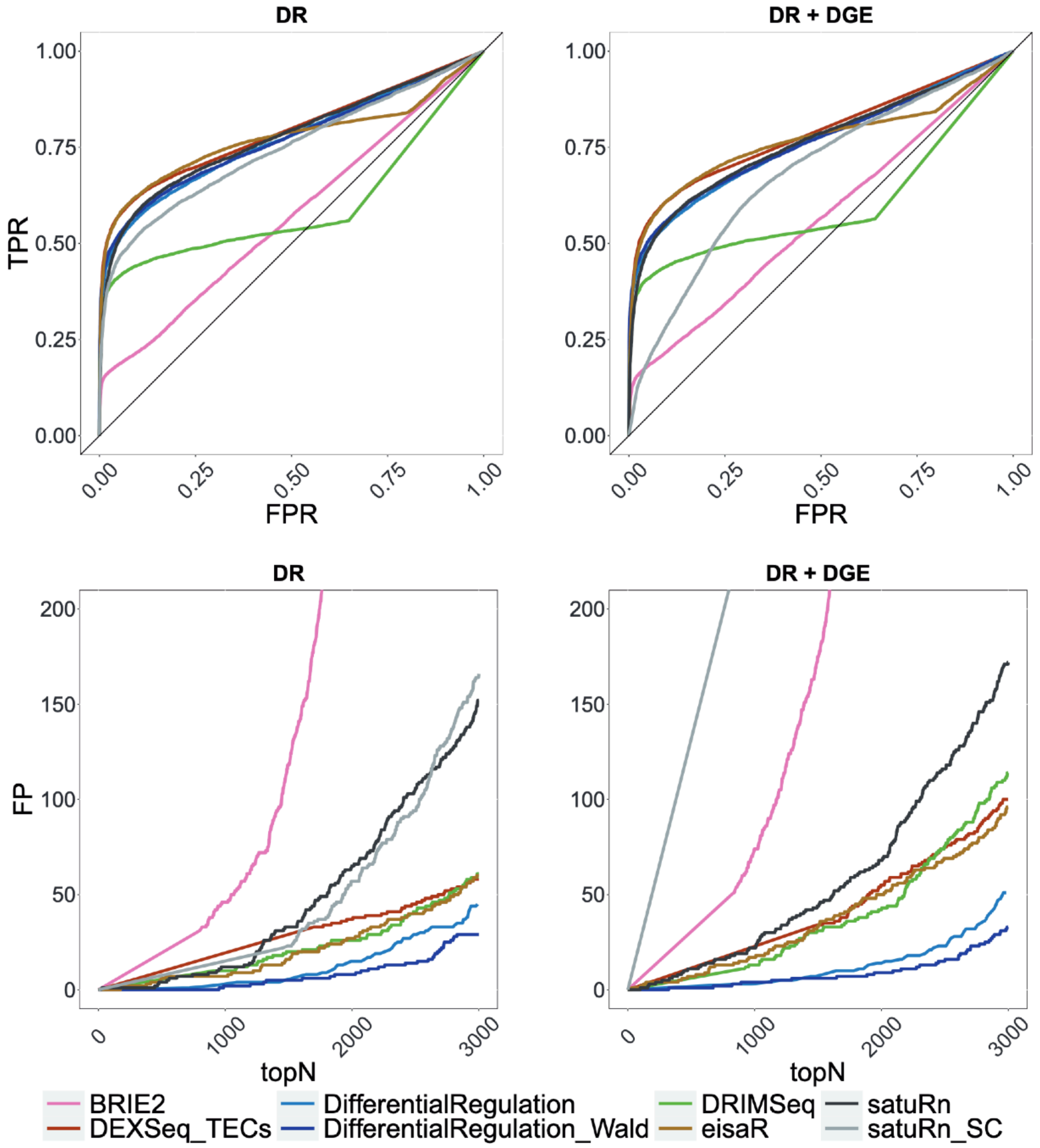
Biostatistics, 2024, **00**, 1–15
https://doi.org/10.1093/biostatistics/kxae017
Article

OXFORD

DifferentialRegulation: a Bayesian hierarchical approach to identify differentially regulated genes

Simone Tiberi[1,2,*], Joël Meili[2,+], Peiying Cai[2,+], Charlotte Soneson[3,+], Dongze He[4,5], Hirak Sarkar[6], Alejandra Avalos-Pacheco[7,8], Rob Patro[5,9], Mark D. Robinson[2]

> "A Bayesian hierarchical method to discover changes between experimental conditions with respect to the relative abundance of unspliced mRNA (over the total mRNA). We model the quantification uncertainty via a latent variable approach, where reads are allocated to their gene/transcript of origin, and to the respective splice version"



- Quantification uncertainty can be accounted for in the model, as alevin-fry can output (splicing-aware) transcript equivalence class (TEC) counts.
- TEC defined by the set of transcripts to which reads map
- Each TEC associated with # of UMIs having this label

So what are the challenges?

A *non-exhaustive* list

Resolving the cell, gene and splicing status of origin for a fragment is not always simple : **methods need to be smart**

- Fragments can arise from **unexpected** locations and should be accounted for
- Cellular barcodes (& UMIs) are subject to **sequencing** error just like "biological" reads
- Reads are not always clearly attributable to a single gene (e.g. sequence similar genes)

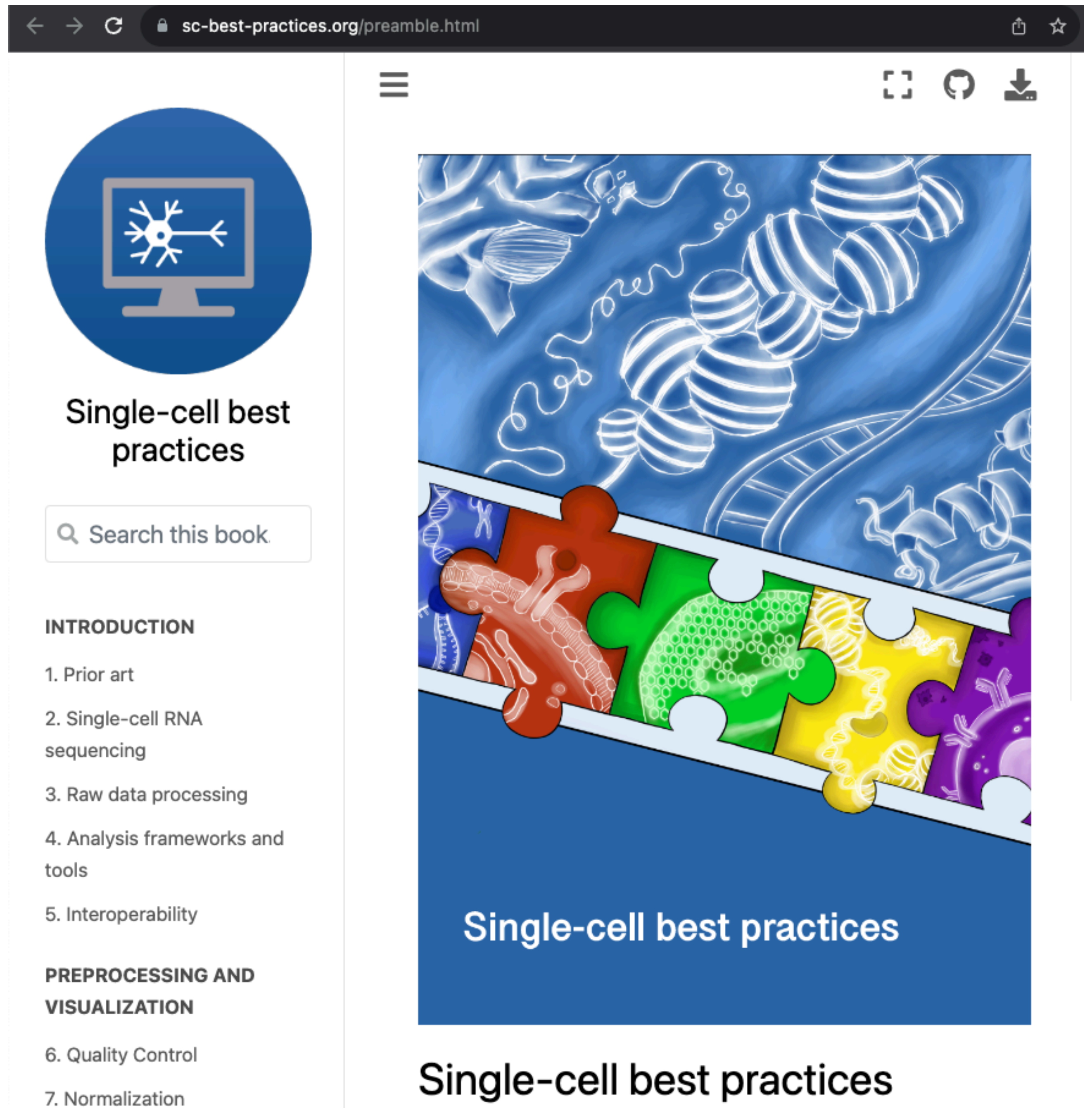
There is a *lot* of data — e.g. PBMC 10k dataset (~10k cells) has more than 600 millions (paired-end) fragments, which is 44GB (*compressed*) : **methods need to scale just to be usable**

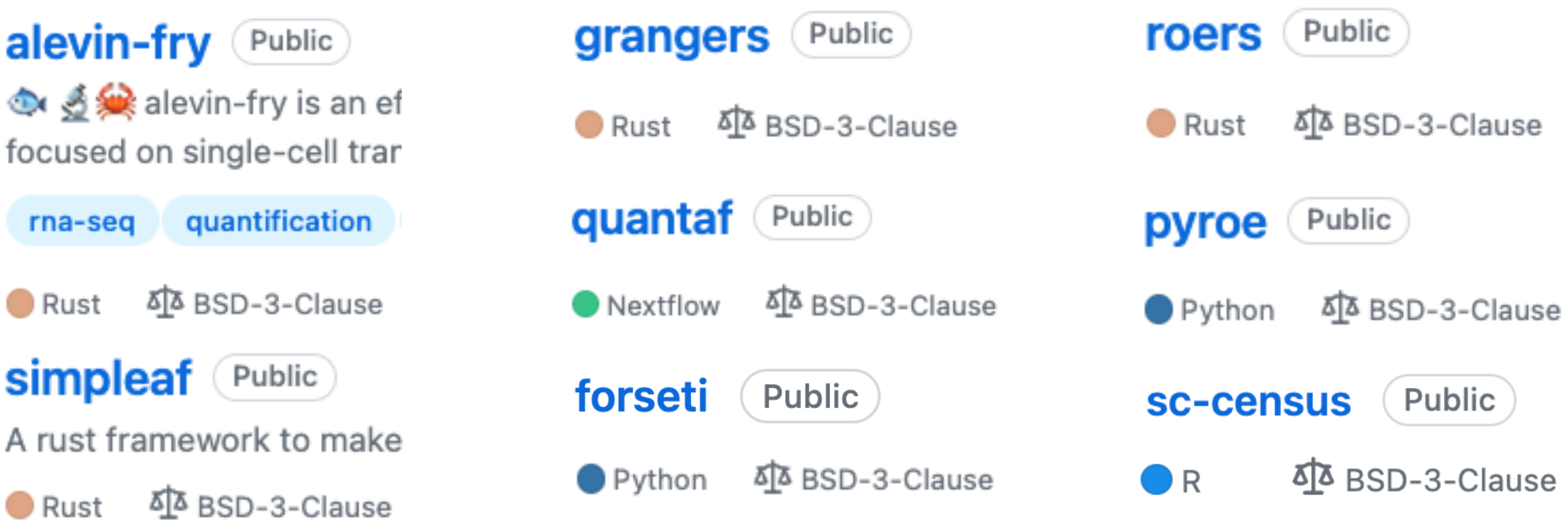
Scalability needed in (at least) 2 dimensions

- Time : we'd like to process data **quickly** and scale well with **many threads**
- Memory : we'd like to be able to process data with **moderate & stable memory** requirements

An efficient, accurate, and open method & tooling ecosystem

- Single-cell analyses can be complex, and optimal methods and best practices are still evolving
- We are building scalable and accurate algorithms, implemented in modular and composable tools for analyzing such data
- It is critical that methods are open & transparent and, to whatever extent possible, track best practices (concerns regarding **CellRanger**)





Alevin-fry

- The alevin-fry framework

ARTICLES
https://doi.org/10.1038/s41592-022-01408-3
nature methods
Check for updates

Alevin-fry unlocks rapid, accurate and memory-frugal quantification of single-cell RNA-seq data

Dongze He[1], Mohsen Zakeri[2], Hirak Sarkar[3], Charlotte Soneson[4,5], Avi Srivastava[6] and Rob Patro[2]

Bioinformatics, 2023, **39(10)**, btad614
https://doi.org/10.1093/bioinformatics/btad614
Advance Access Publication Date: 6 October 2023
Applications Note
OXFORD

Sequence analysis

`simpleaf`: a simple, flexible, and scalable framework for single-cell data processing using alevin-fry

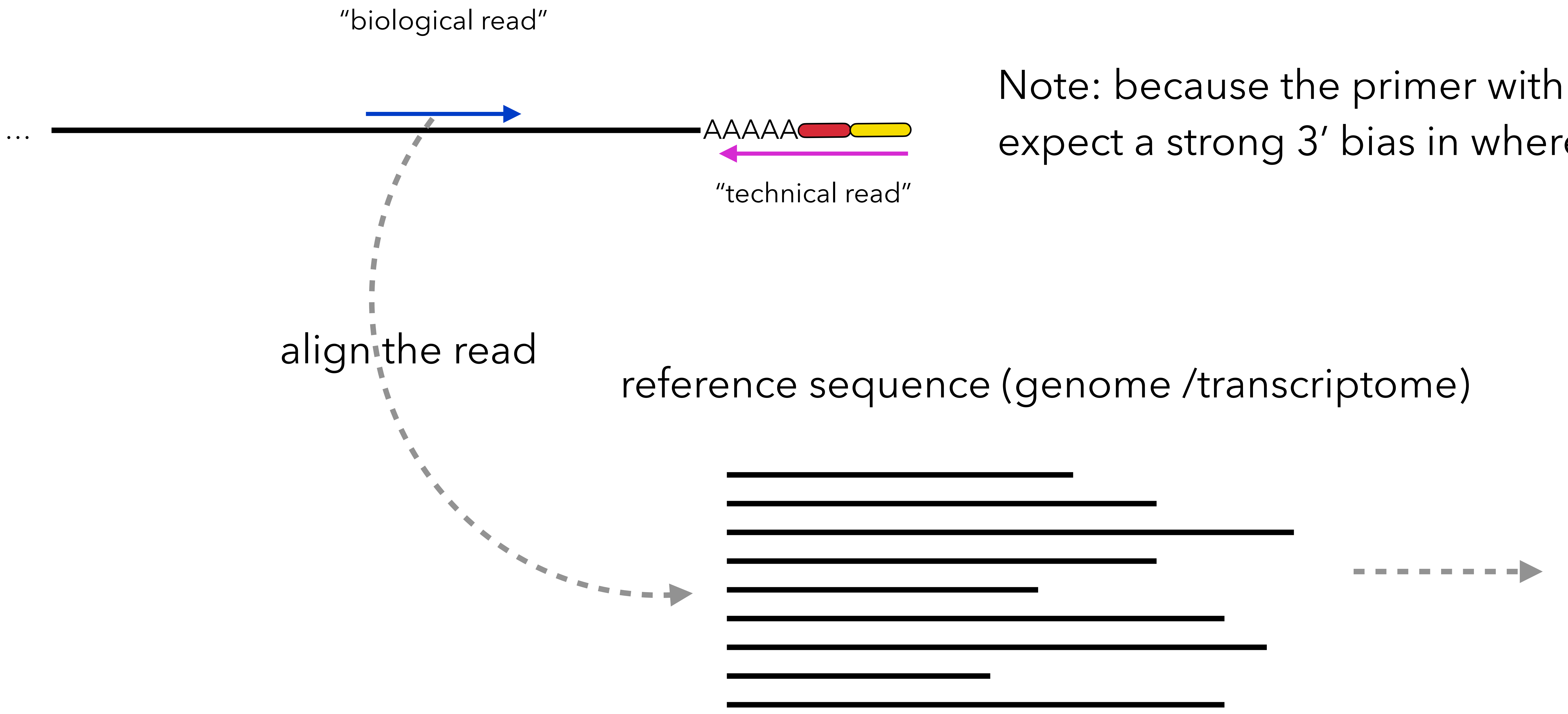
Dongze He[1,*] and Rob Patro[2,*]

nature reviews genetics
https://doi.org/10.1038/s41576-023-00586-w
Expert recommendation
Check for updates

Best practices for single-cell analysis across modalities

Lukas Heumos[1,2,3,28], Anna C. Schaar[1,4,5,28], Christopher Lance[1,6], Anastasia Litinetskaya[1,4], Felix Drost[1,3], Luke Zappia[1,4], Malte D. Lücken[1,7], Daniel C. Strobl[1,3,8,9], Juan Henao[1], Fabiola Curion[1,4], Single-cell Best Practices Consortium*, Herbert B. Schiller[2] & Fabian J. Theis[1,3,4,5]

Determining origin of "biological" reads



Note: because the primer with cell barcode & UMI binds to polyA, we expect a strong 3′ bias in where the biological reads are sampled

{transcript 1, position 2454, forward,
transcript 2, position 2278, forward,
transcript 7, position 1346, rev-comp**}**

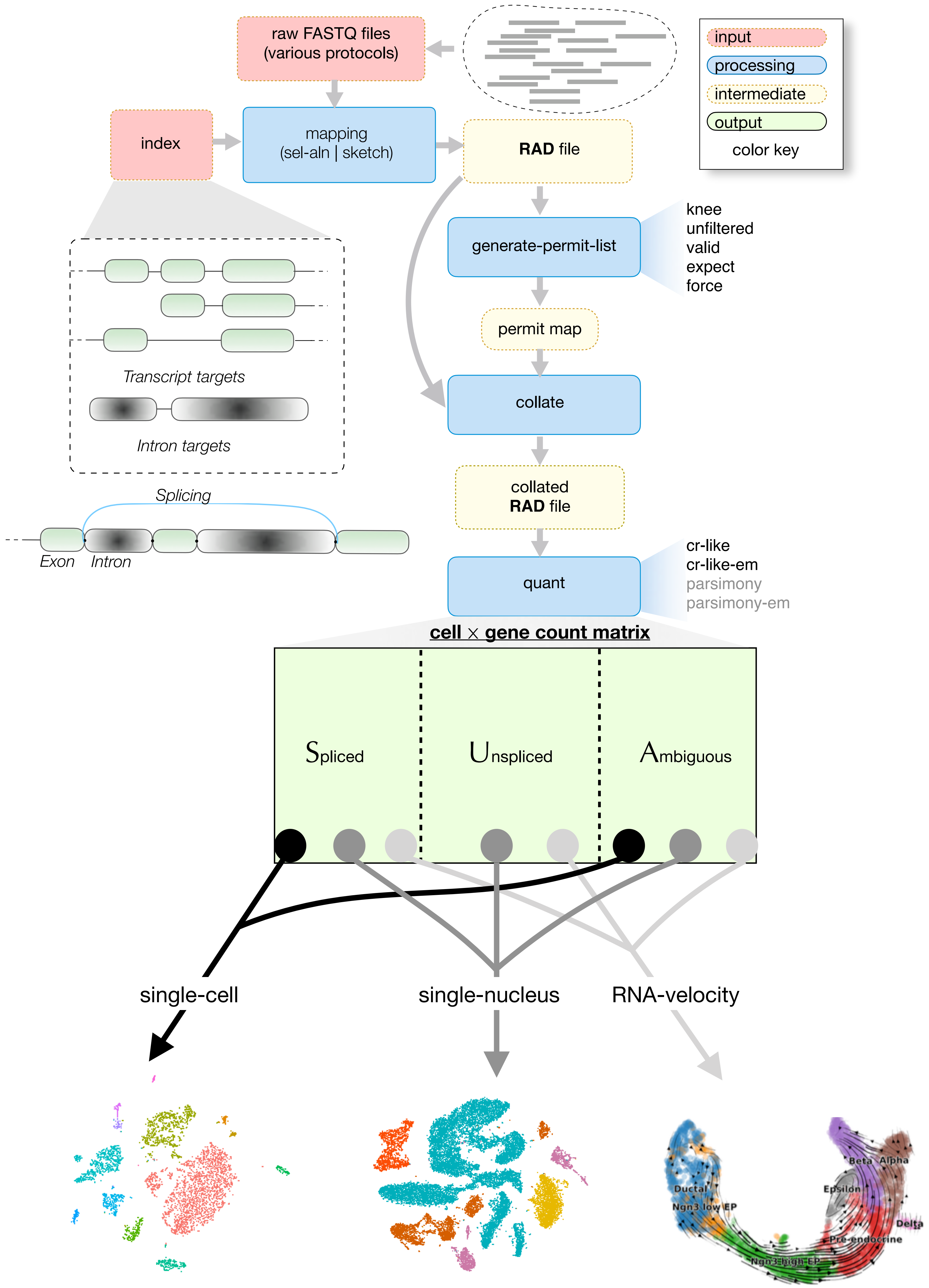
There are a host of different ways this alignment problem can be solved:

- Alignment against the annotated genome using a spliced aligner (e.g. STAR)[1]
- Selective-alignment against the transcriptome using a tool like salmon/alevin[2]
- Pseudoalignment against the transcriptome using a tool like kallisto[3]

Will discuss the implications of this choice later.

1. Kaminow, Benjamin, Dinar Yunusov, and Alexander Dobin. "STARsolo: accurate, fast and versatile mapping/quantification of single-cell and single-nucleus RNA-seq data." *bioRxiv* (2021).
2. Avi Srivastava, Laraib Malik, Hirak Sarkar, Mohsen Zakeri, Fatemeh Almodaresi, Charlotte Soneson, Michael I Love, Carl Kingsford, and Rob Patro. Alignment and mapping methodology influence transcript abundance estimation. *Genome Biology*, **21**(1):1–29, 2020
3. Nicolas L Bray, Harold Pimentel, Páll Melsted, and Lior Pachter. Near-optimal probabilistic RNA-seq quantification. *Nature Biotechnology*, **34**(5):525, 2016

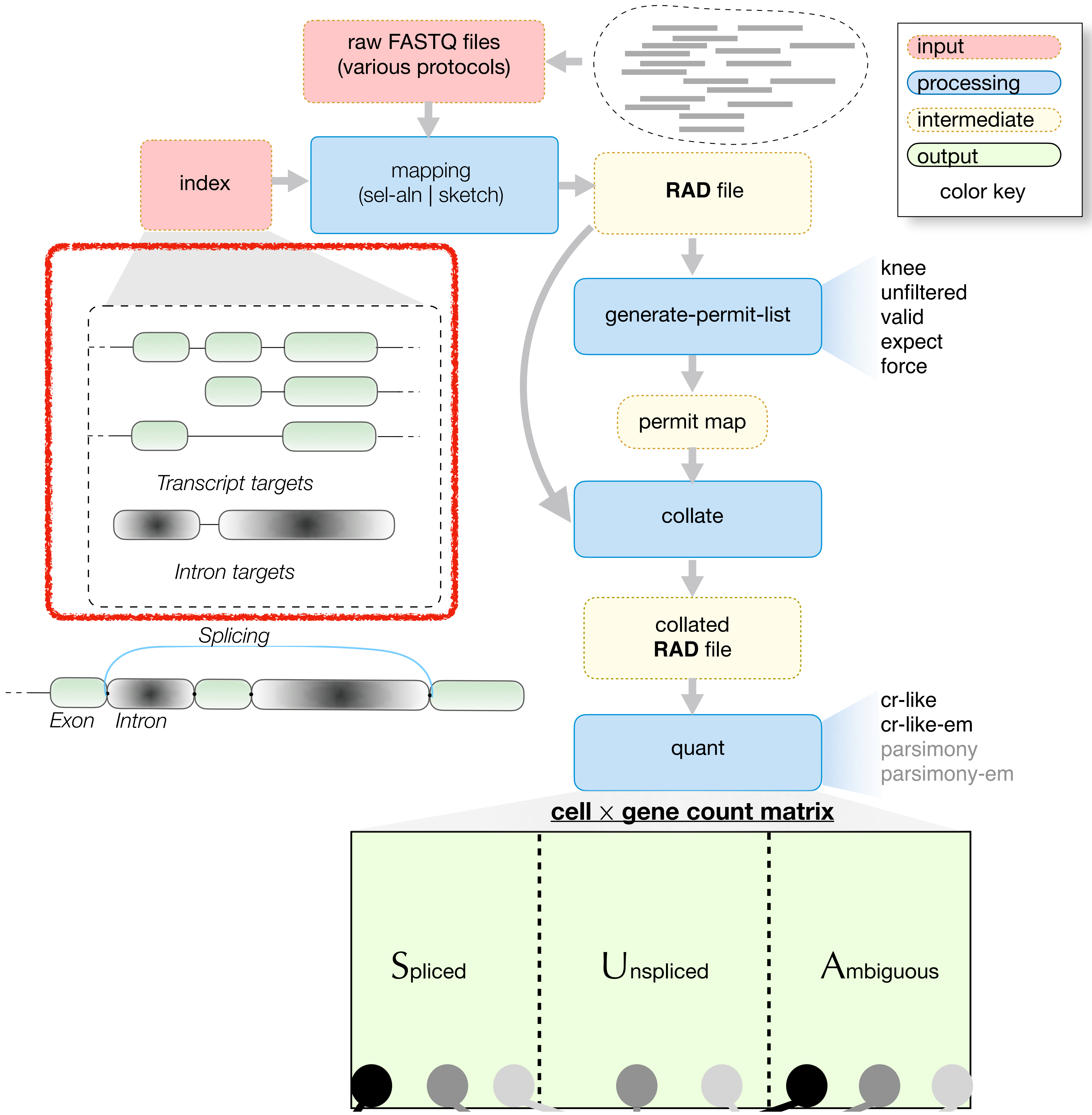
The alevin-fry framework



Alevin-fry allows accurate processing of single-cell sequencing data quickly and with low memory footprint.

- Input: FASTQ files.
- Output: Three cell-by-gene count matrices representing the count of each gene in each cell with the splicing status of spliced, unspliced and ambiguous, respectively.
- The three count matrices can be combined to perform different types of data analyses.

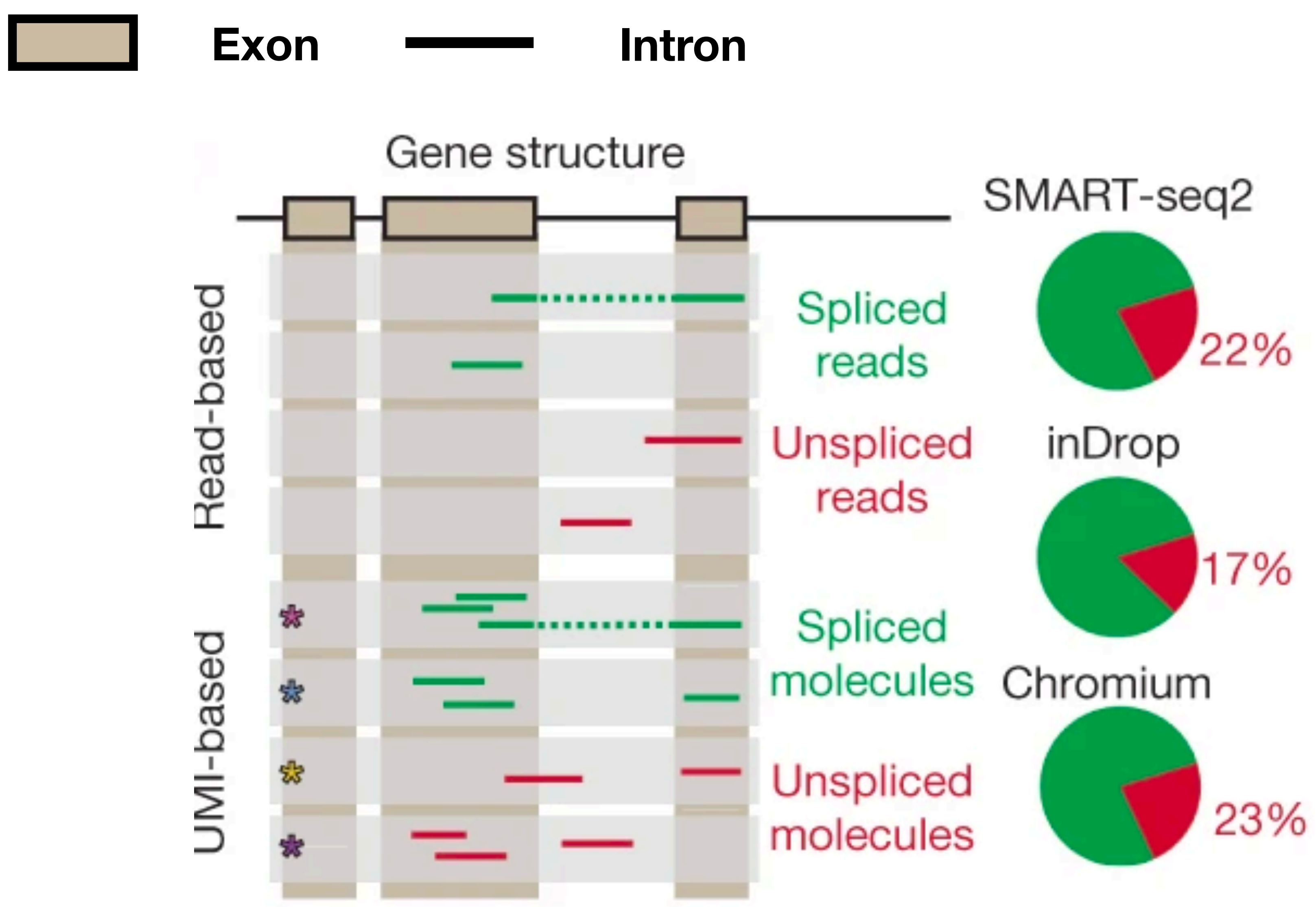
Alevin-fry accepts augmented transcriptome references

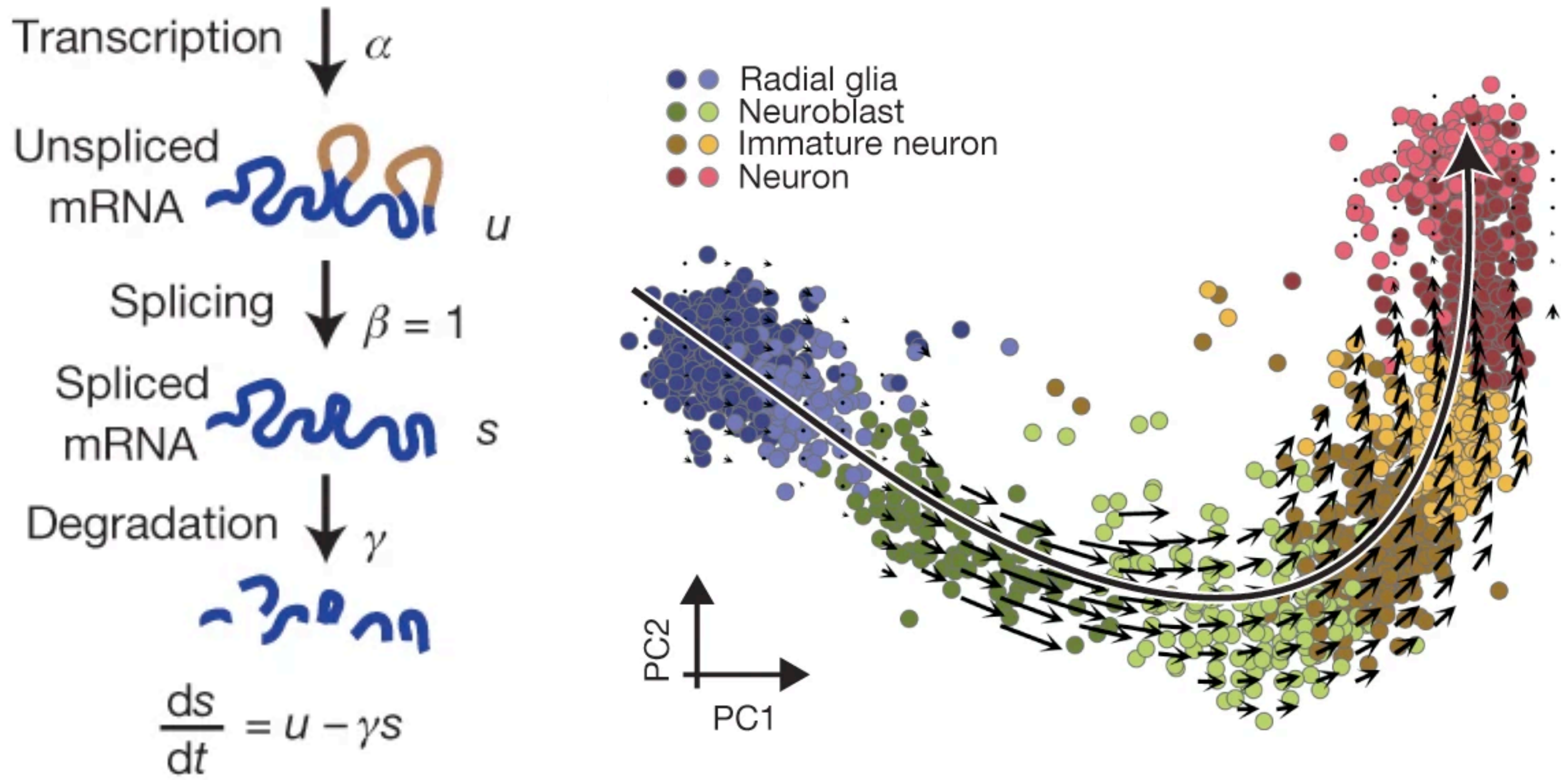


Alevin-fry makes an augmented transcriptome reference.

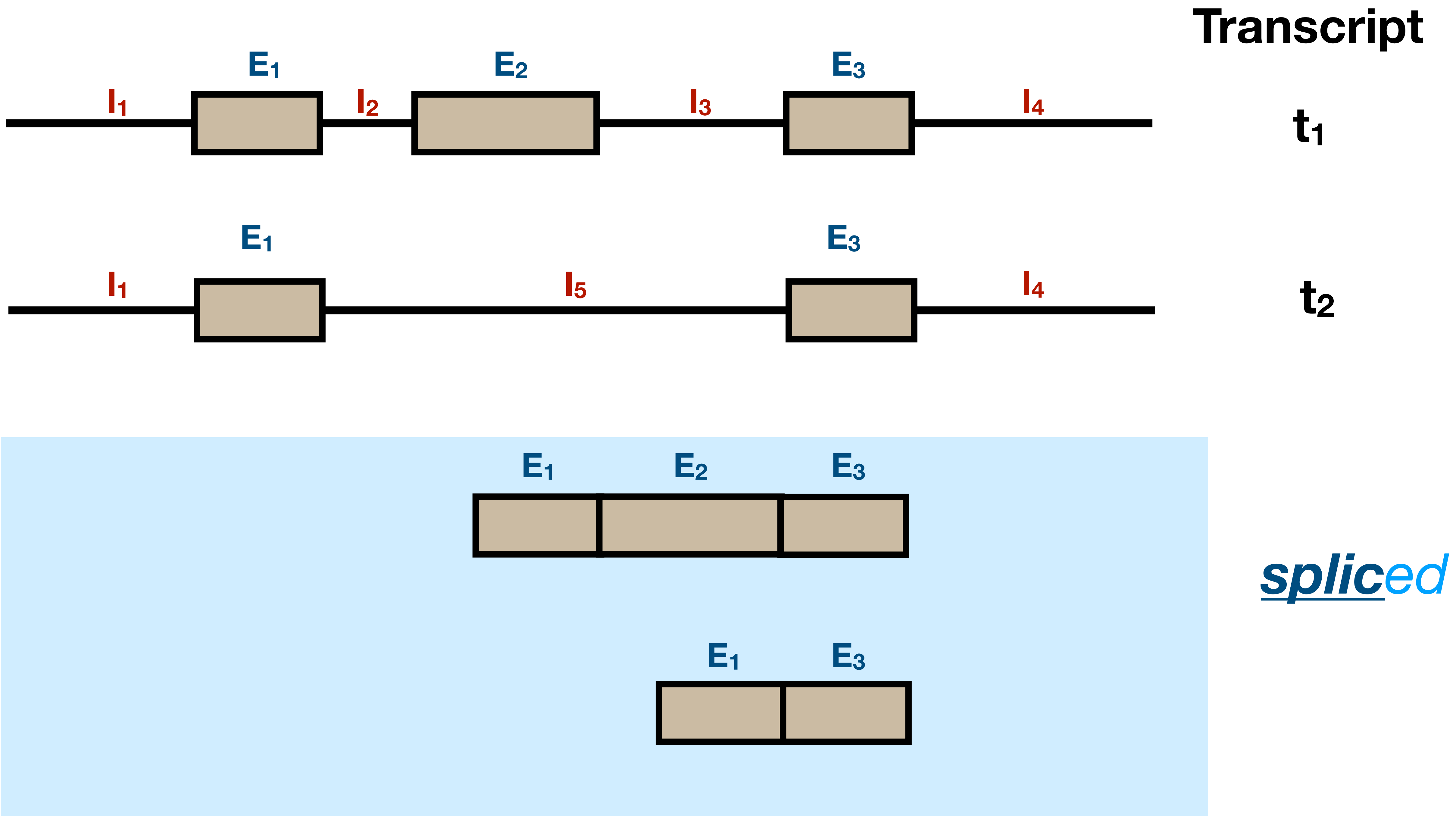
- Input:
  - A genome FASTA file
  - A gene annotation GTF file
- Output:
  - An augmented transcriptome reference including spliced transcripts and other augmented sequences.

Alevin-fry accepts augmented transcriptome references





RNA velocity



splici **- An augmented transcriptomic reference**

La Manno, G., Soldatov, R., Zeisel, A. et al. RNA velocity of single cells. Nature 560, 494–498 (2018). https://doi.org/10.1038/s41586-018-0414-6

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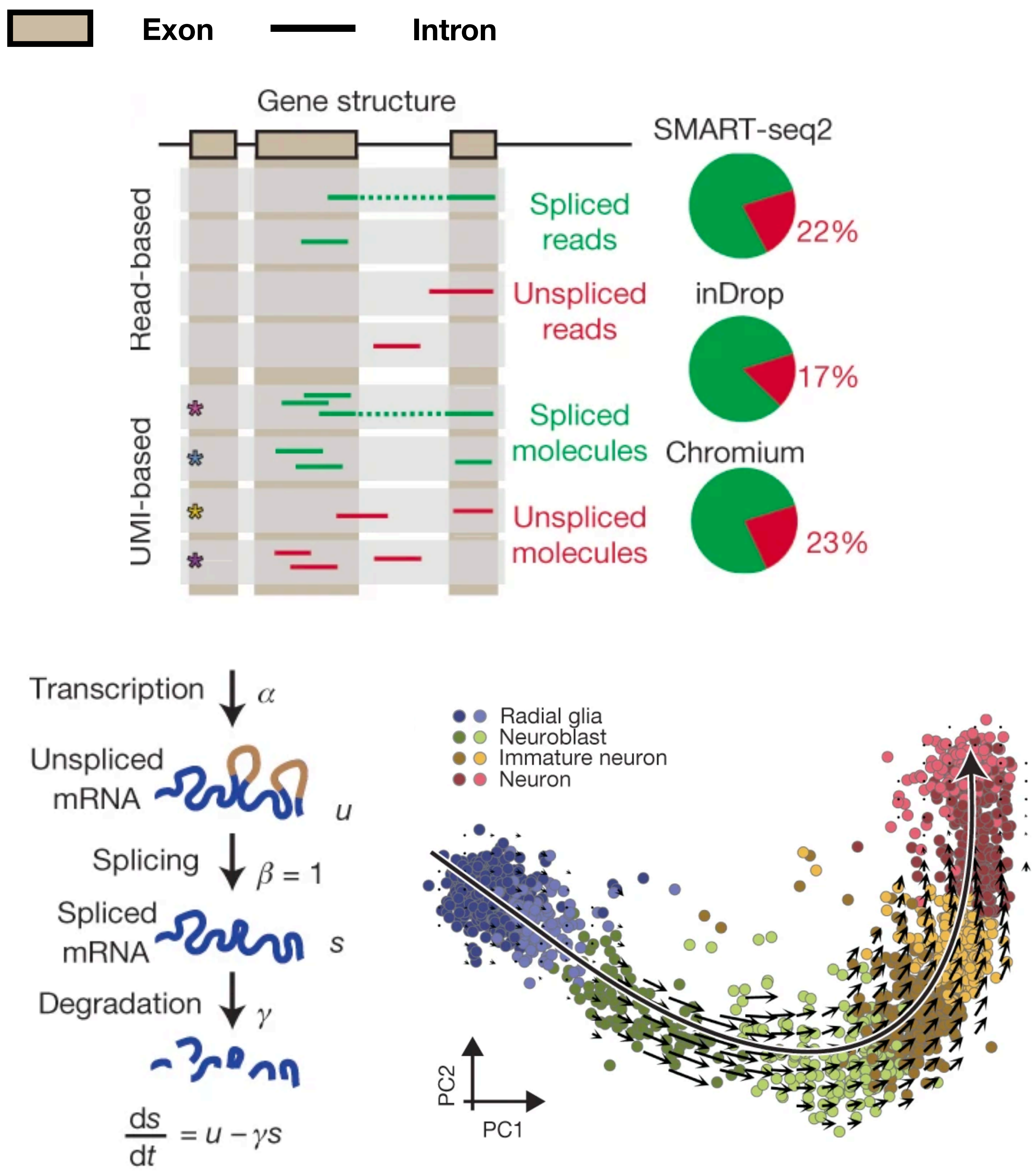

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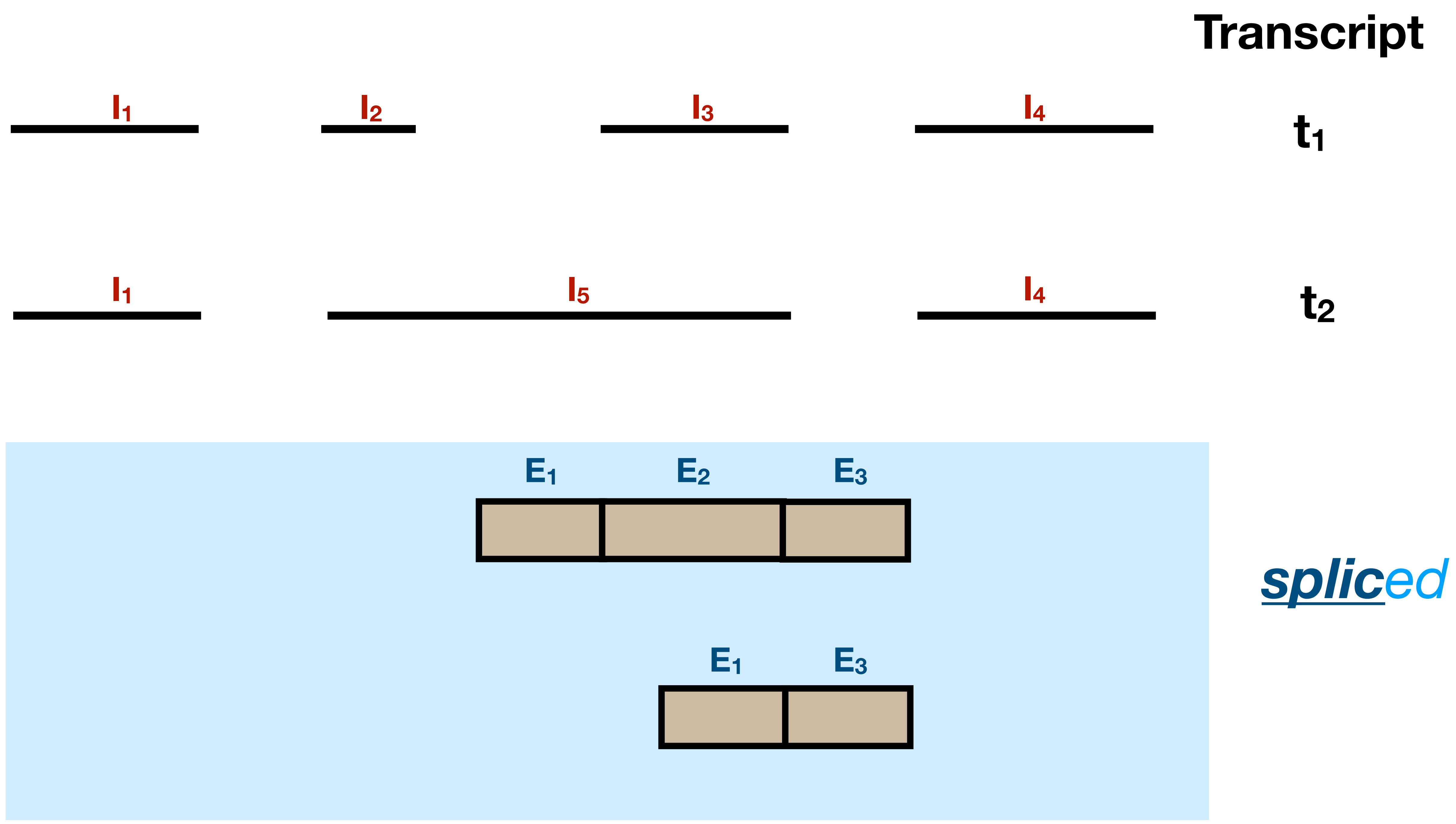

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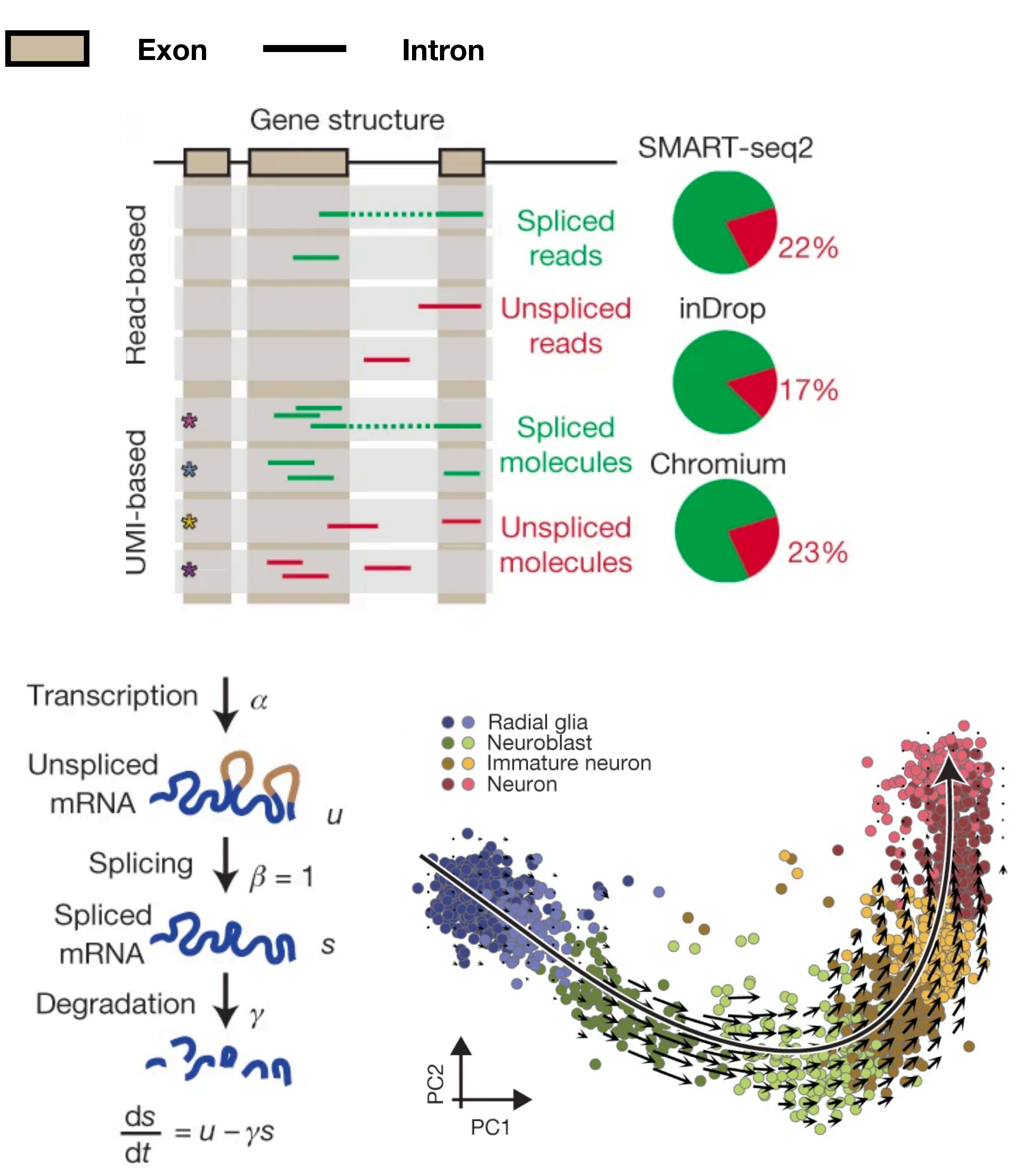

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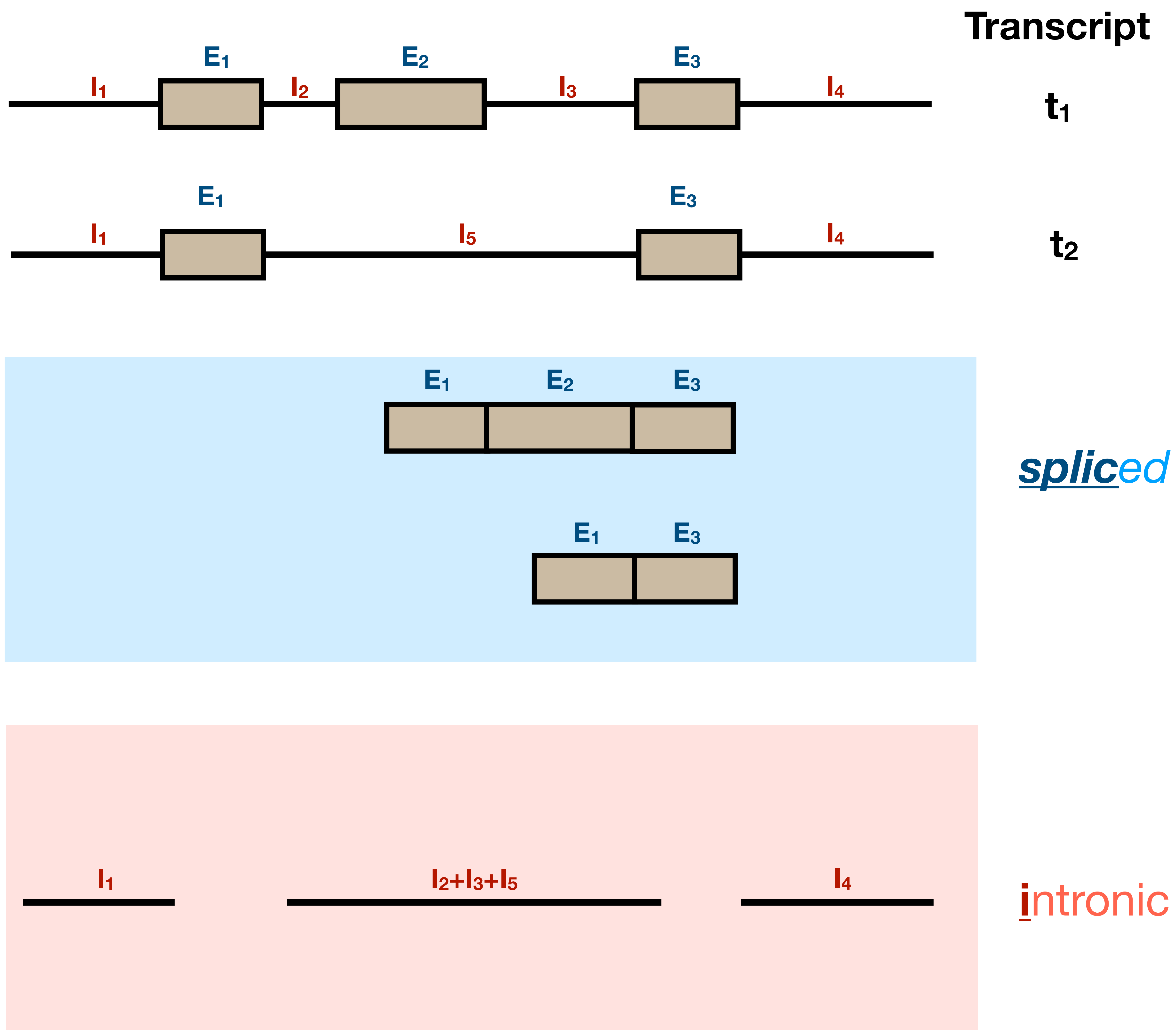

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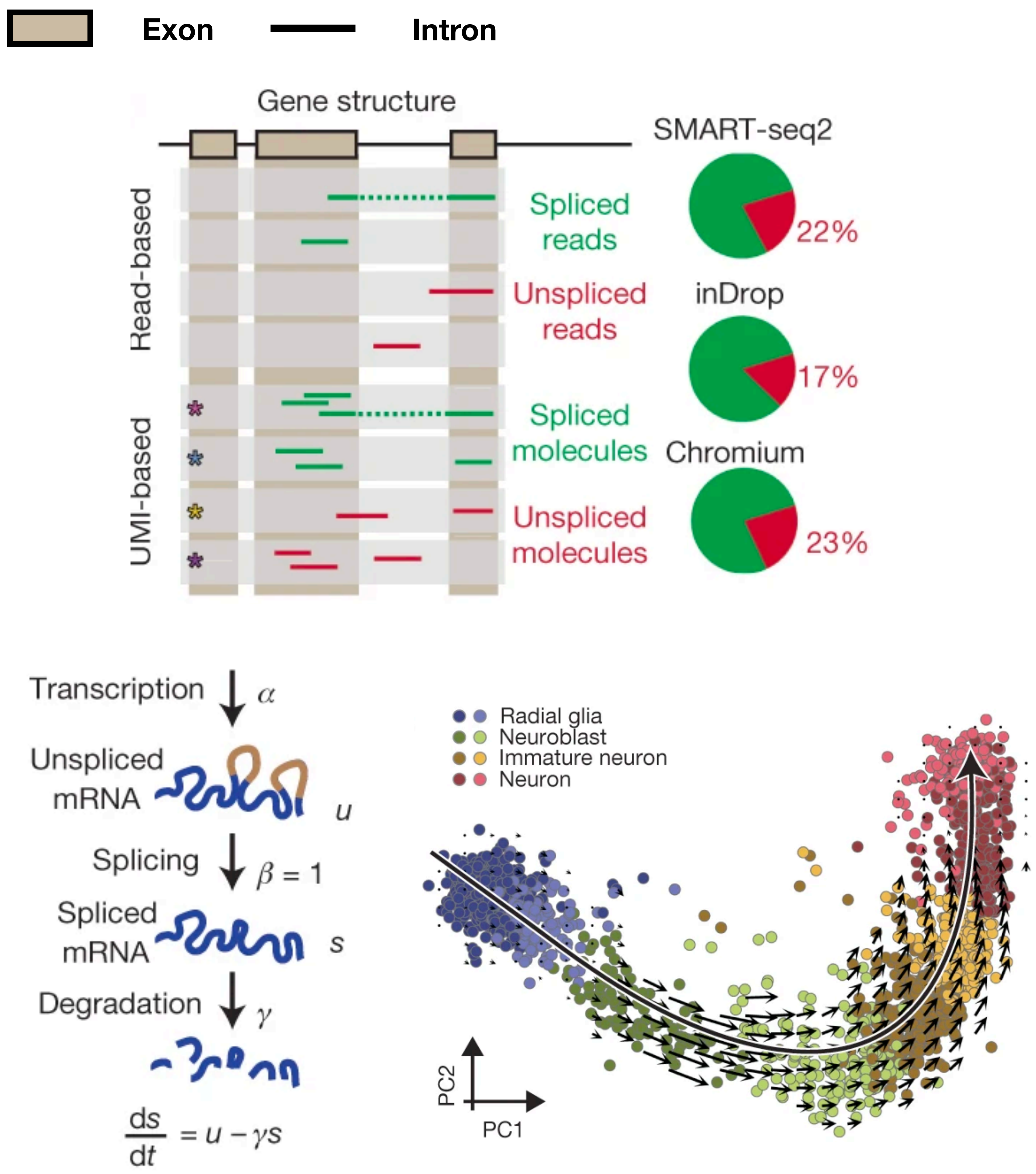

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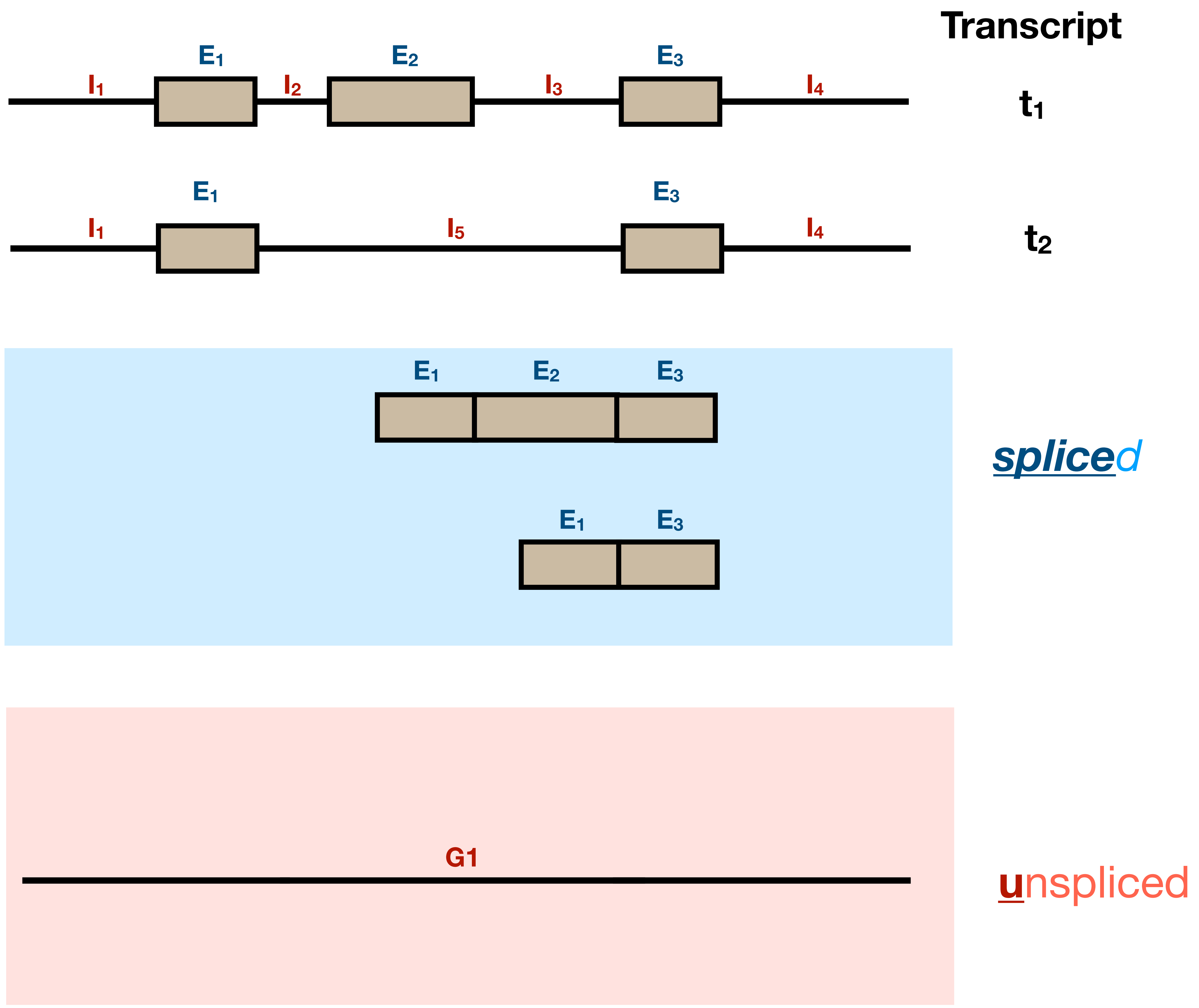
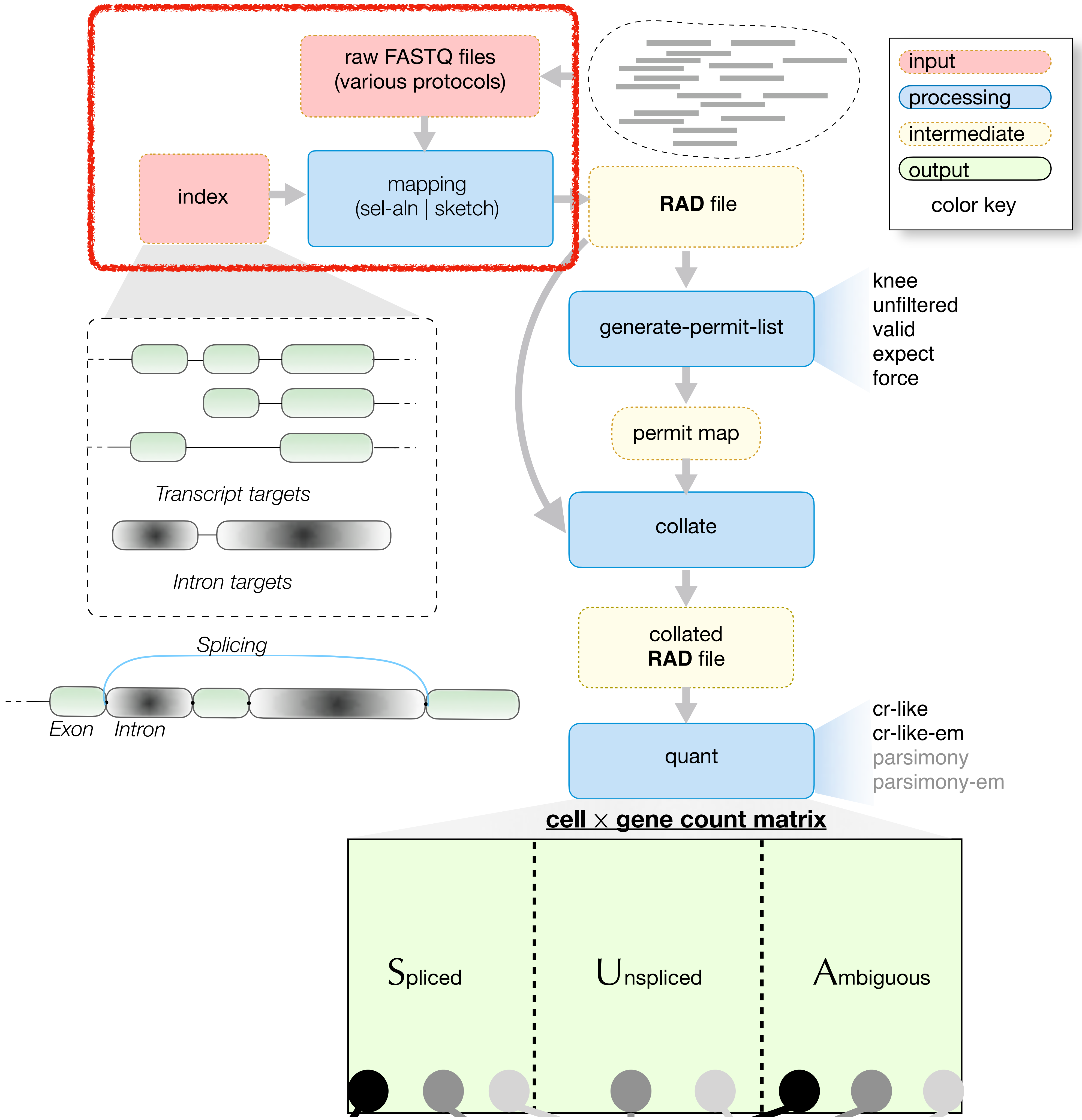


***spliceu* - An augmented transcriptome reference**

La Manno, G., Soldatov, R., Zeisel, A. et al. RNA velocity of single cells. Nature 560, 494–498 (2018). https://doi.org/10.1038/s41586-018-0414-6

Alevin-fry provides efficient indexing and mapping

Alevin-fry uses transcriptome-based indices and mapping algorithms.



Alevin-fry builds an reference index over the transcriptome reference and maps sequencing reads against it.

- Input:
  - An augmented reference
  - The sequencing reads
- Output:
  - The mapping records of reads

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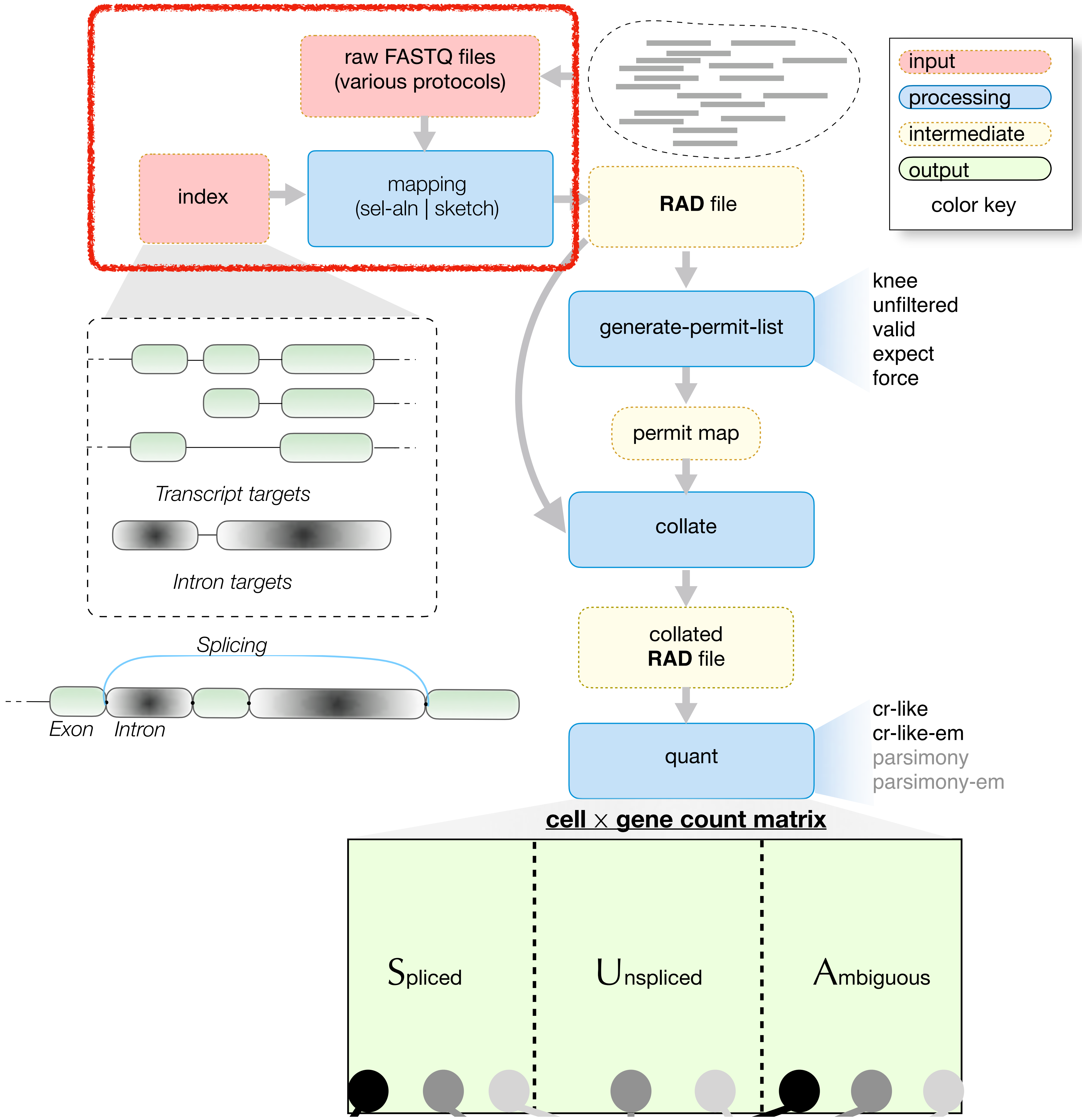

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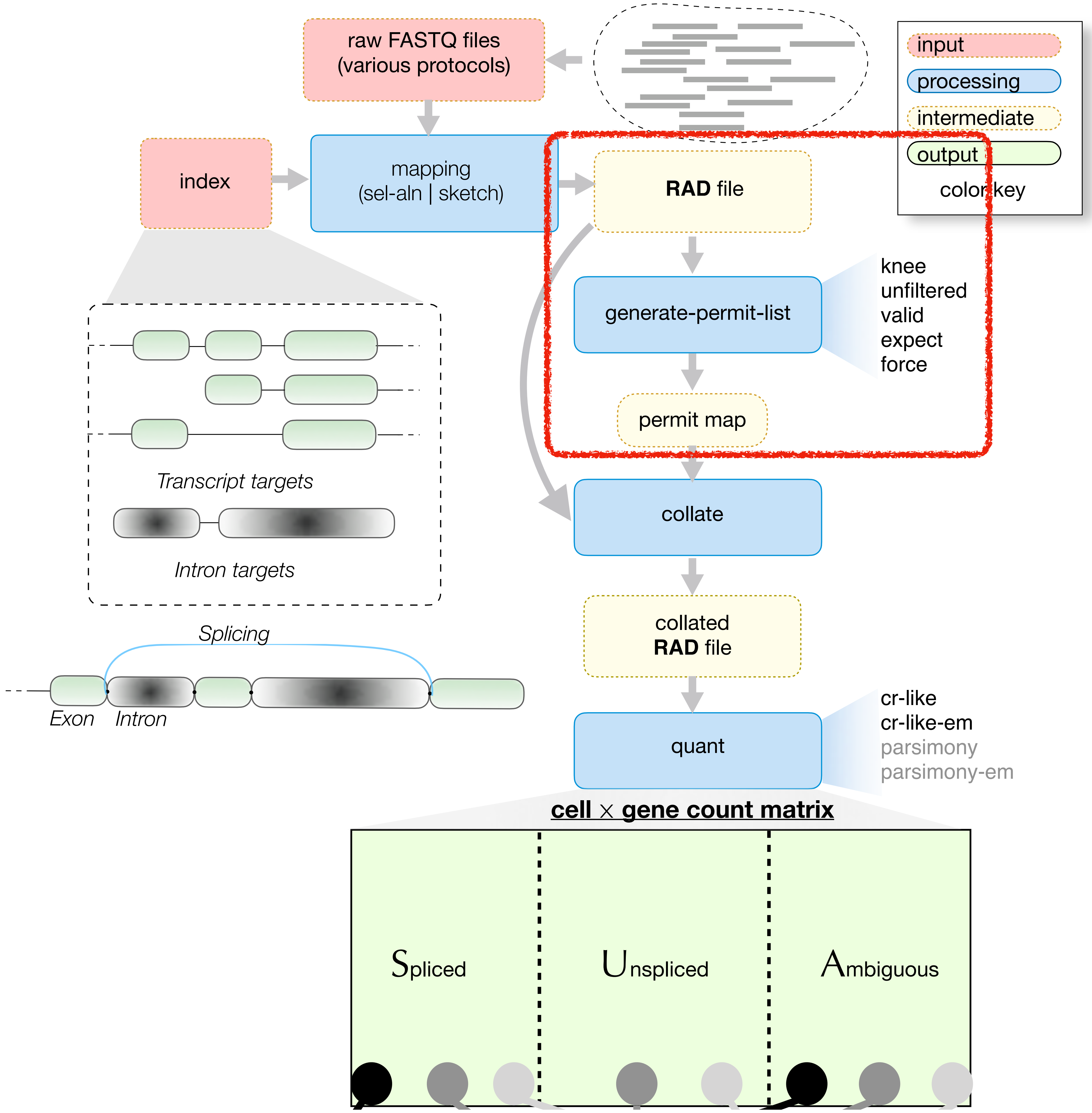
The piscem index — a very space-efficient index over the colored, compacted reference de Bruijn graph.

The piscem index for 10X Human 2020-A reference is **1.5 GB**.

It allows read mapping with less than **3 GB memory usage** regardless of the sample size.

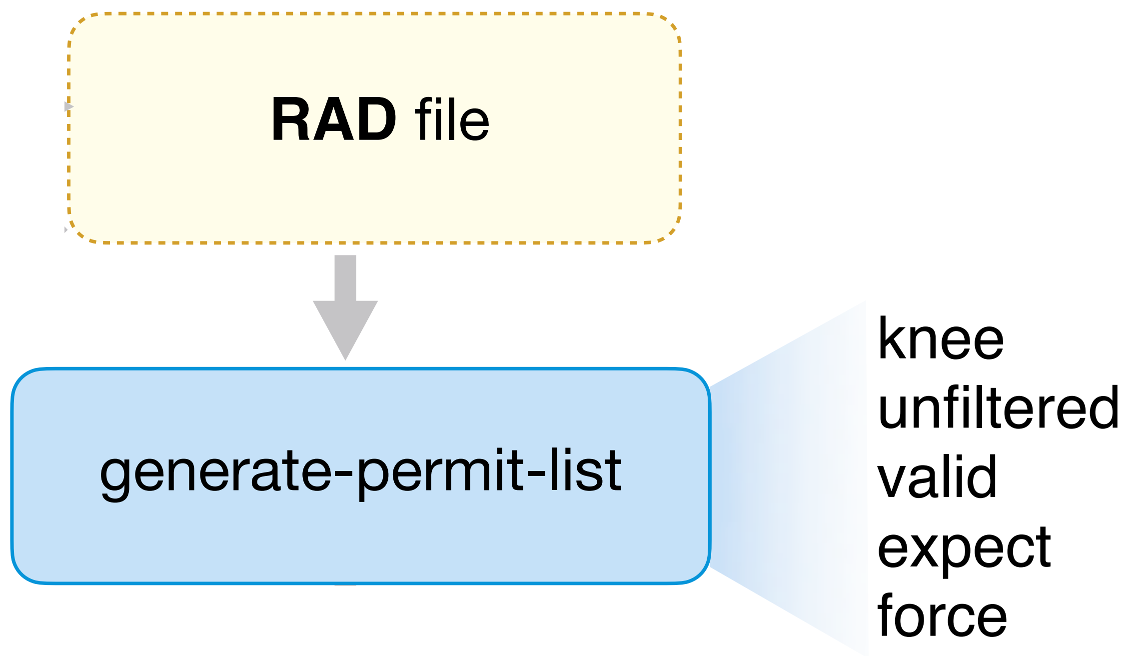
Determining which cells are present & correcting their barcodes



Alevin-fry corrects cell barcodes with errors.

- Input:
  - Observed cell barcodes
- Output:
  - Corrected cell barcodes

Determining which cells are present & correcting their barcodes

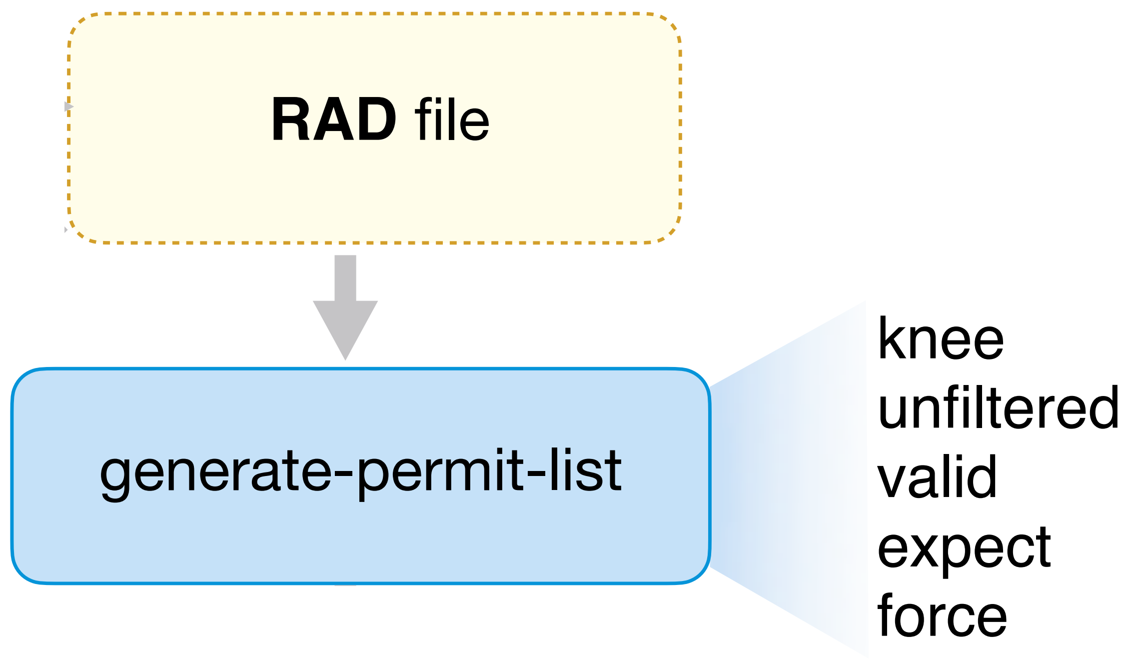


Alevin-fry corrects cell barcodes according to a list of permitted cells.

AAAAAAACAAAAAAAA Observed CB #1

AAAAAAAAAAAAAAAA Permitted CB #1

Determining which cells are present & correcting their barcodes



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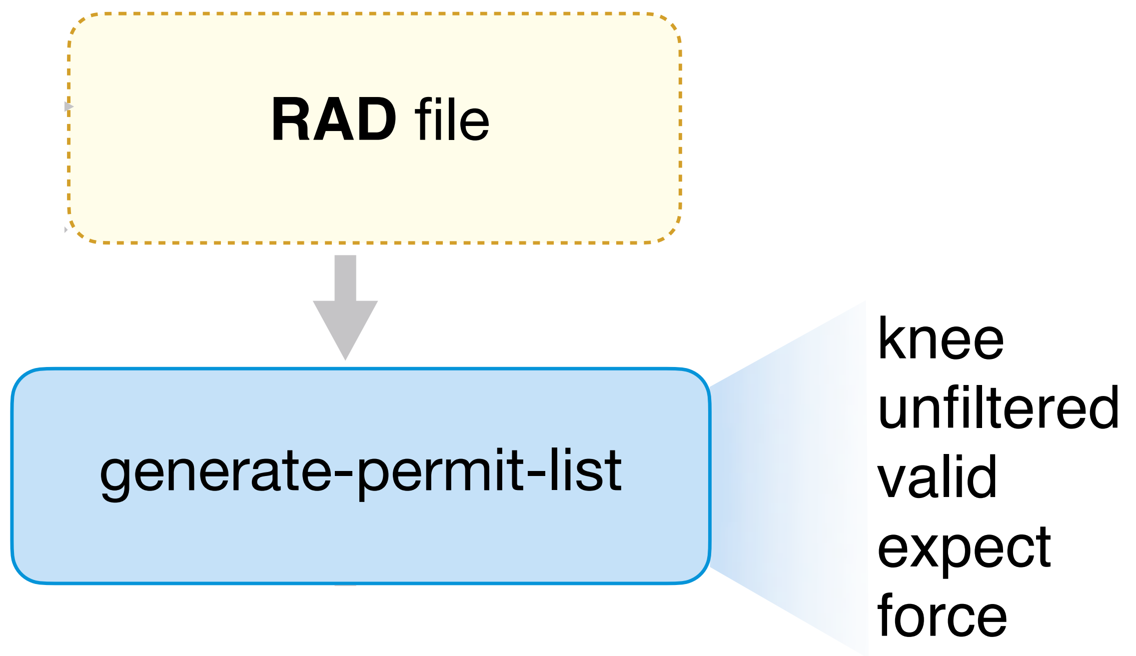
✓ AAAAAAAAAAAAAAAA Observed CB #1

AAAAAAAAAAAAAAAA Permitted CB #1

Determining which cells are present & correcting their barcodes



Alevin-fry corrects cell barcodes according to a list of permitted cells.

	AAAAAAAAAAAAAAAA	Permitted CB #1
?	AAAAAAAAAAAAAAGA	Observed CB #2
	AAAAAAAAAAAAAAGG	Permitted CB #2

Actual algorithm is 2-pass:

- First, scan list for exactly-matching barcodes, and keeps only codes with > k reads (e.g. k = 10)
- Second, correct against this "truly present" list. Reduces the amount of ambiguity, but ambiguity is still possible. Ambiguous corrections are skipped.

Determining which cells are present & correcting their barcodes

RAD file

generate-permit-list

knee
unfiltered
valid
expect
force

Alevin-fry corrects cell barcodes according to a list of permitted cells.

min read count (τ)	# barcodes $\geq \tau$	# recoverable reads
1	$1,816,810$	$2,609,604$
10	$115,396$	$4,676,792$
50	$46,185$	$4,867,972$
100	$15,377$	$4,882,645$

AAAAAAAAAAAAAAAA Permitted CB #1

? AAAAAAAAAAAAAAGA Observed CB #2

AAAAAAAAAAAAAAGG Permitted CB #2

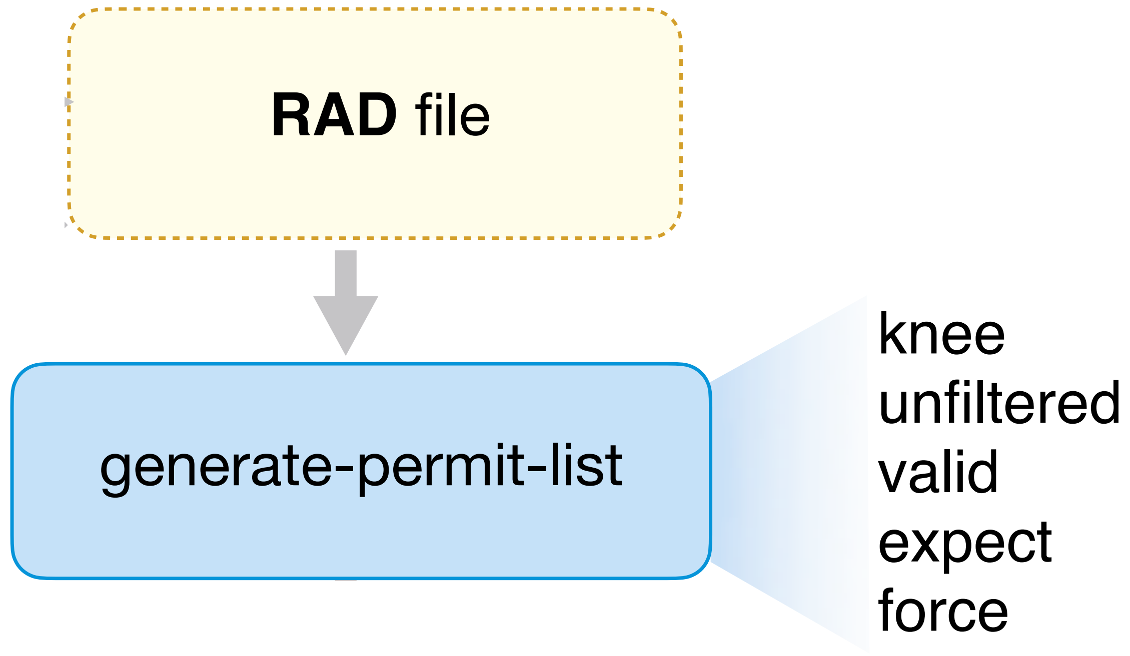
Actual algorithm is 2-pass:

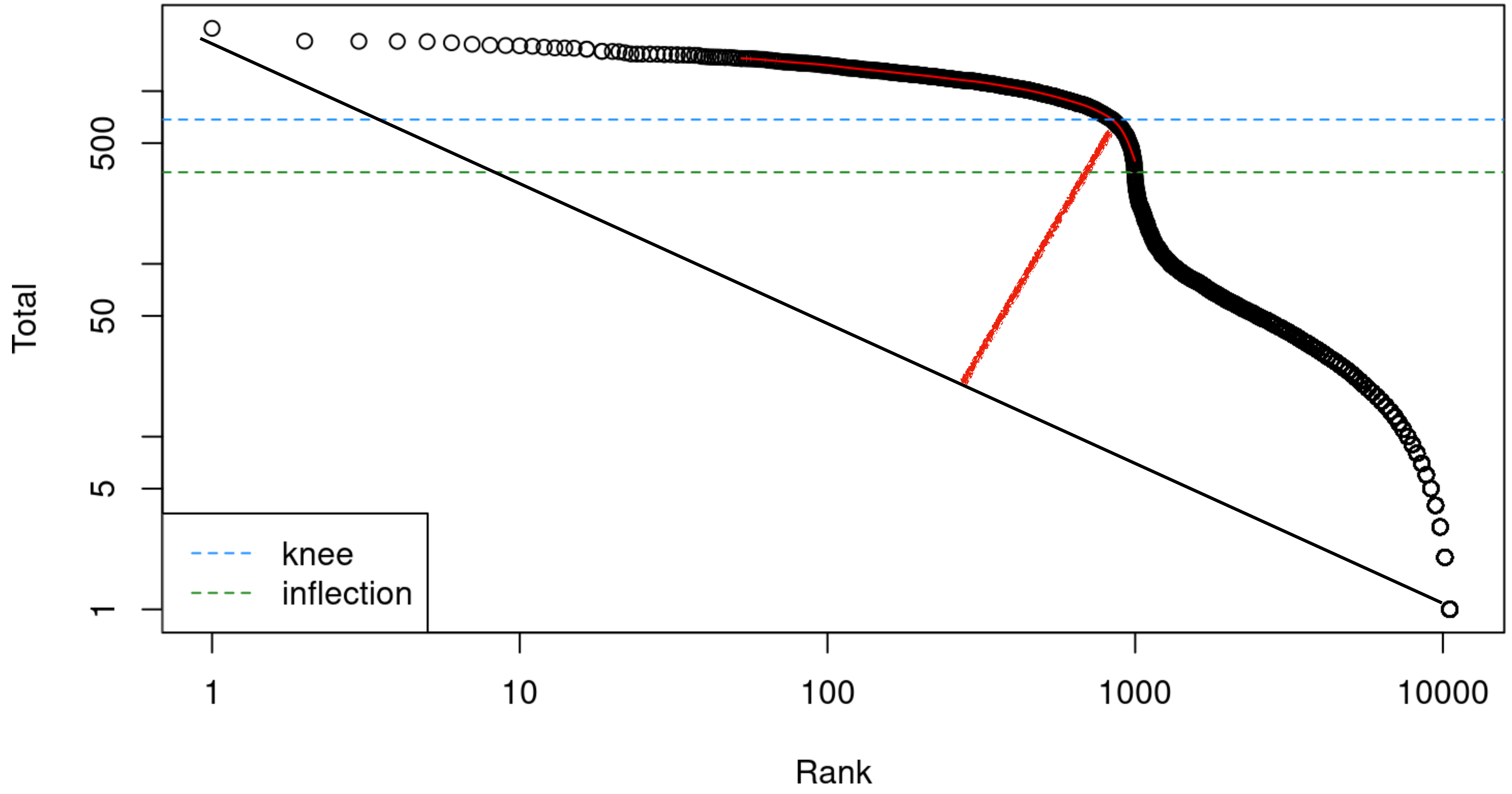
- First, scan list for exactly-matching barcodes, and keeps only codes with > k reads (e.g. k = 10)
- Second, correct against this “truly present” list. Reduces the amount of ambiguity, but ambiguity is still possible. Ambiguous corrections are skipped.

Determining which cells are present & correcting their barcodes

Provides other options if no permit-list is available

There are many methods for correcting barcodes: For example, the “robust multi-step knee-finding method”



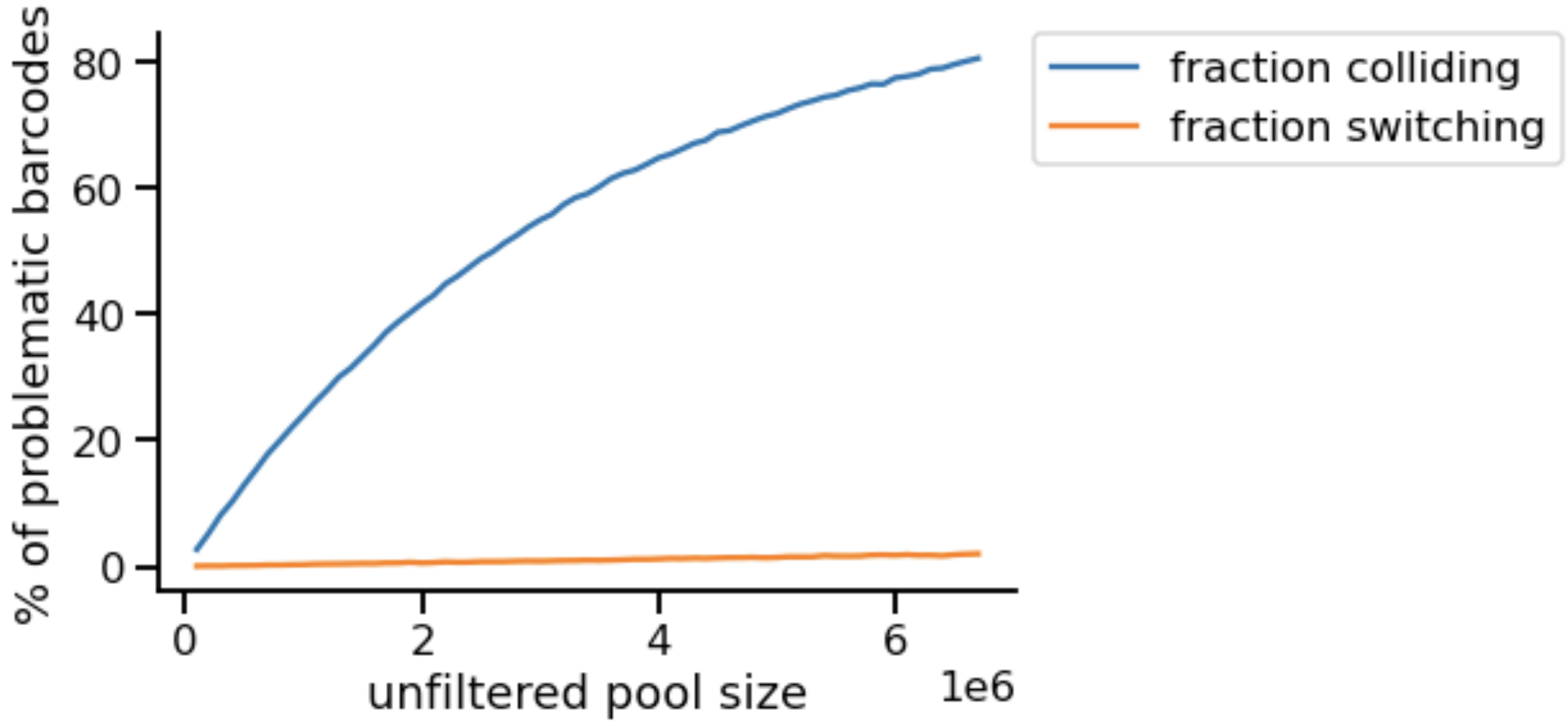


Uses such an un-supervised method to determine a list of “true” barcodes, and then corrects against this list.

An aside about permit-list “design”

Correcting against a known list can pose some interesting challenges:

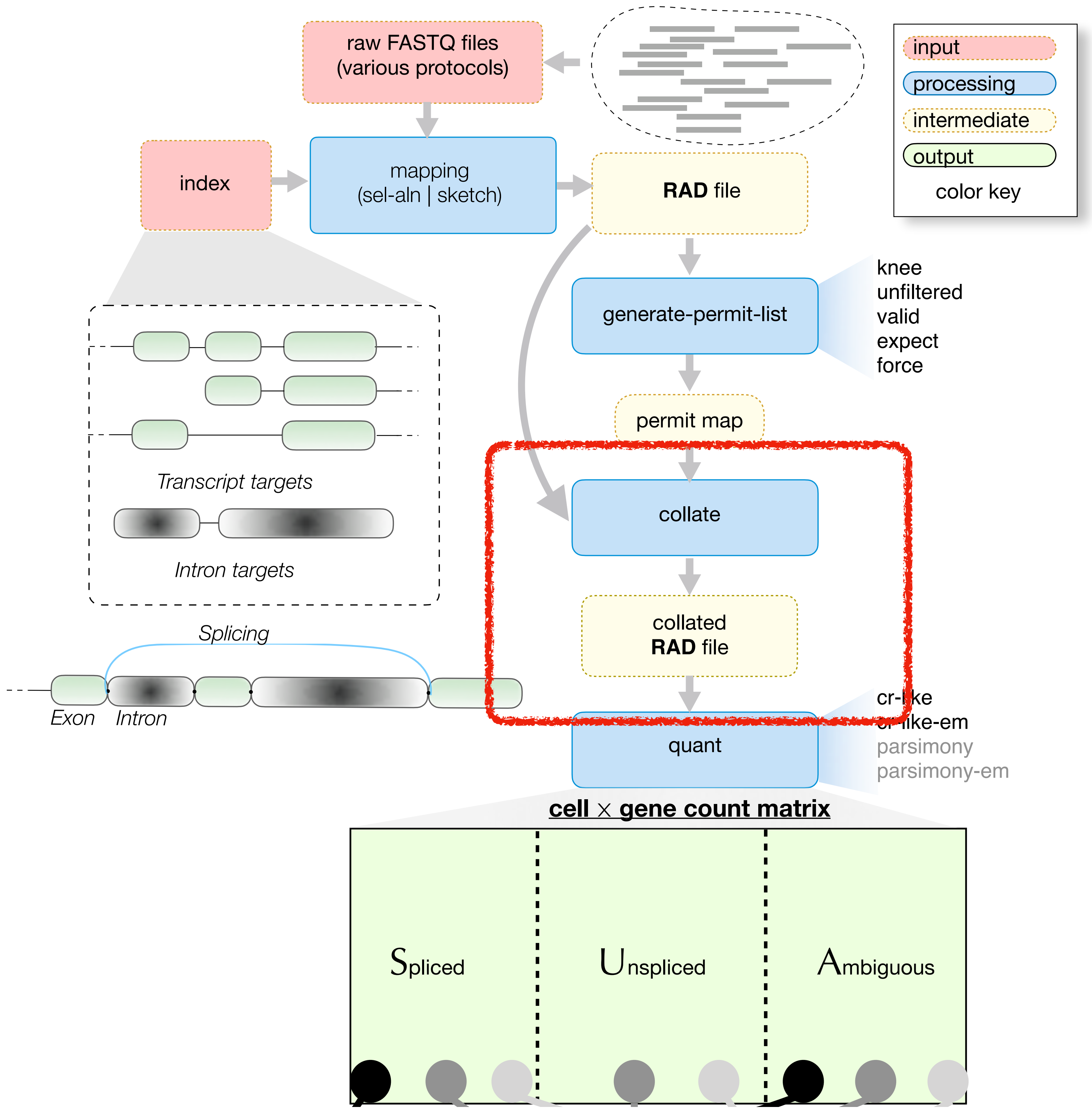
If we take a 10X CB and mutate at a single position to an non-10X CB, what percentage of the time does the new barcode have >= 2 permitted neighbors, at Hamming distance 1, in the 10X CB permit list?



The fraction of ambiguous corrections we can expect (y-axis) based on the number of barcodes from the permit list that appear in the unfiltered pool (x-axis).

Note: Proposals have been made to use error correcting codes for BC design. That may provide much better performance. Why aren’t we doing this?

Collating reads by cell barcode: Semi-sort to the rescue



Use a 2-pass "scatter-and-gather" algorithm to collate in limited memory.

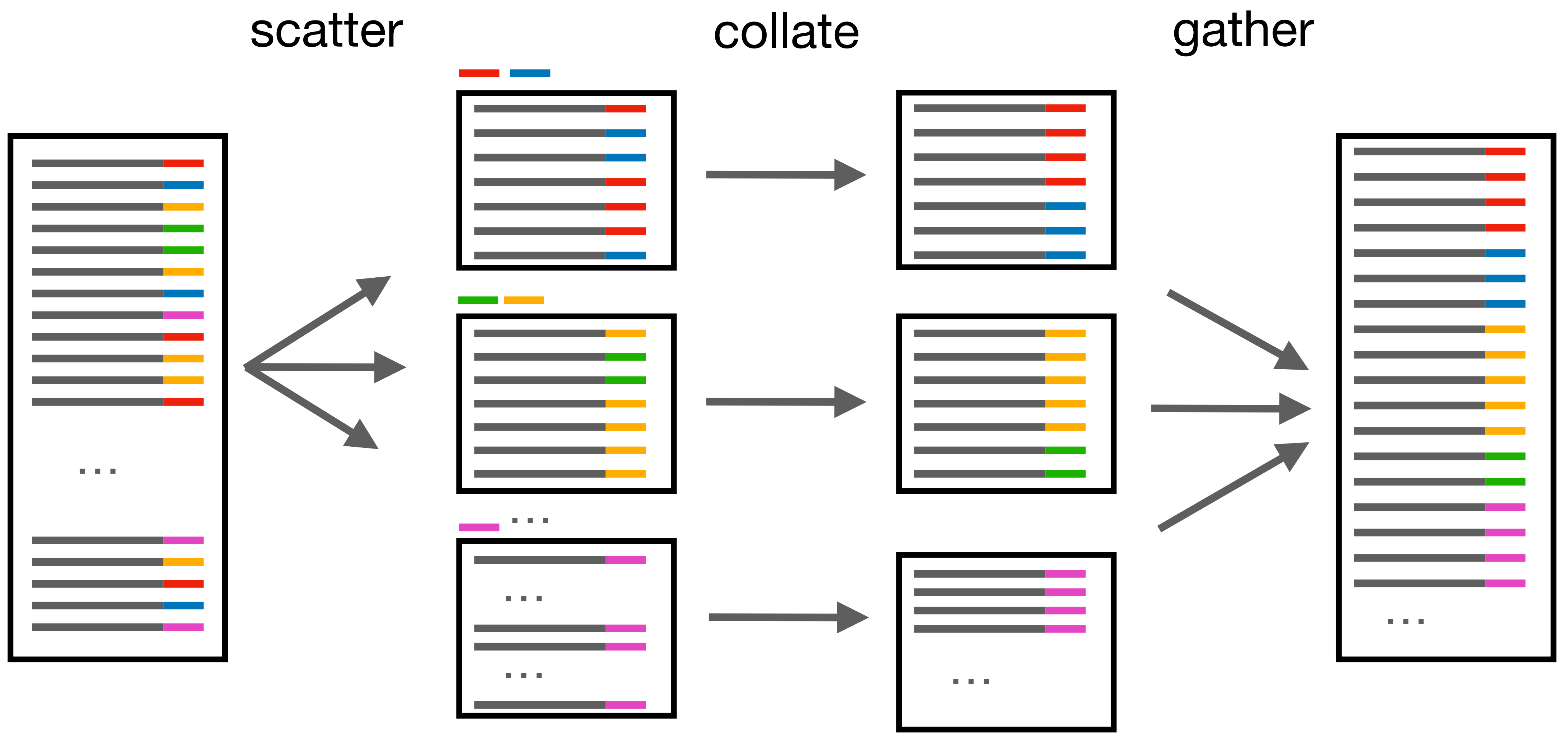
Pass 1 (scatter) :

In descending order of frequency,

- assign each corrected CB to a "bucket" temp file.
- CBs are assigned to each bucket until it is full (greedy packing).
- CBs never split across buckets.
- parse the records of each CB (in parallel) and write them to the same temp file.

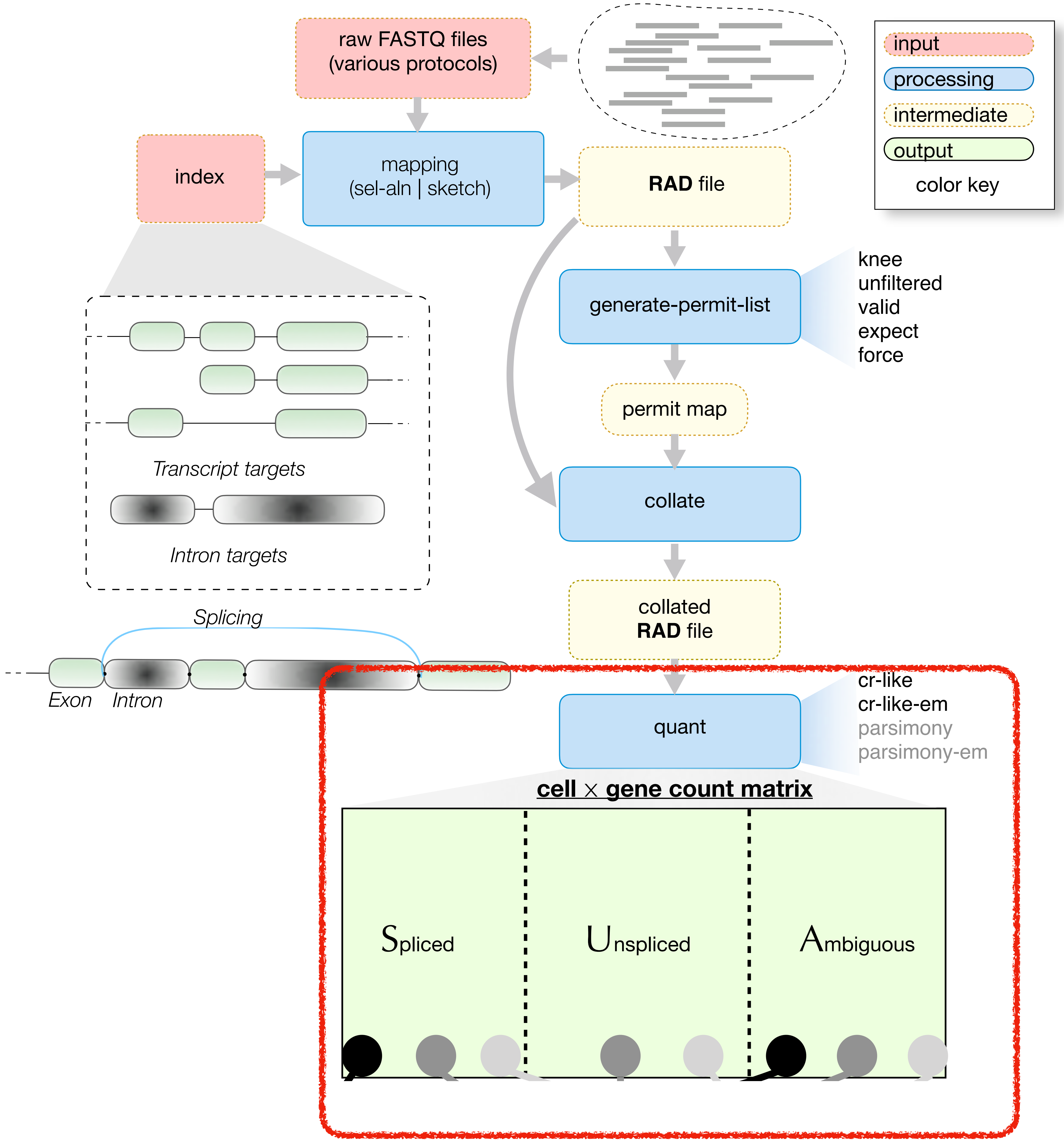
Pass 2 (gather) :

- read each temp file (a bucket) into memory (*In parallel*)
- collate the record within the bucket in memory by hashing their CB to a list of records
- append each collated bucket to the global output file



The resulting output file is globally collated; cells can be processed one at a time.

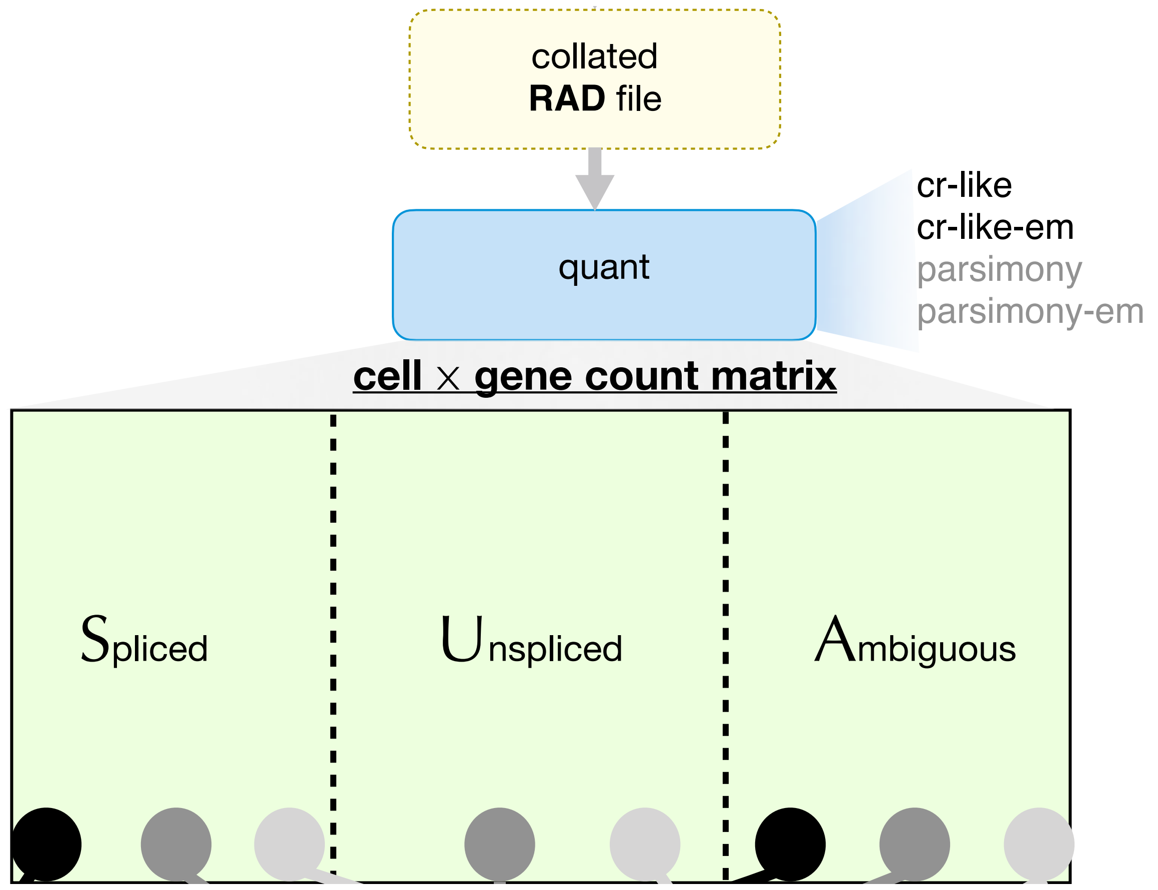
Resolve UMIs to assign counts to genes



Alevin-fry resolves UMIs to assign counts.

- Input:
  - The CB corrected and collated mappings
- Output:
  - Three count matrices containing the spliced, unspliced and ambiguous gene counts.

Resolve UMIs to assign counts to genes



Since cells are independent, we can consider them separately (for now).

Goal: Characterize the number of *distinct pre-PCR* molecules arising from each splicing status of each gene within the cell.

Challenges: UMIs can be duplicated, reads can be mappable to sequence that exists in both introns / exons depending on the isoform, reads can map to multiple genes.

Many strategies exist to resolve UMIs, depending on one's assumptions; describe alevin-fry's **parsimony** here.

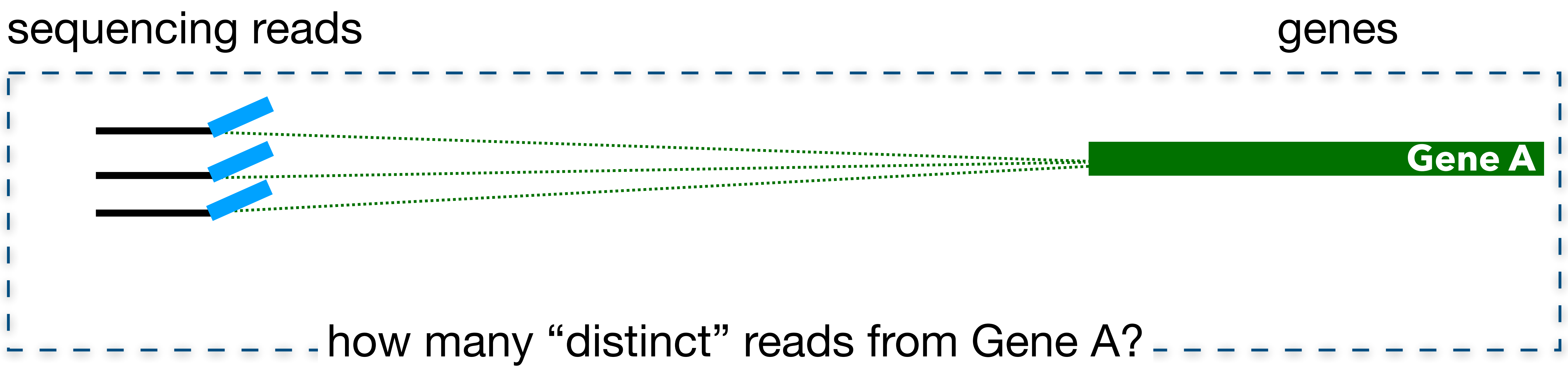
Srivastava *et al. Genome Biology* (2019) 20:65
https://doi.org/10.1186/s13059-019-1670-y
Genome Biology
METHOD Open Access

Alevin efficiently estimates accurate gene abundances from dscRNA-seq data

Check for updates

Avi Srivastava[1], Laraib Malik[1], Tom Smith[2], Ian Sudbery[3] and Rob Patro[1*]

Multi-mapping is what makes this difficult



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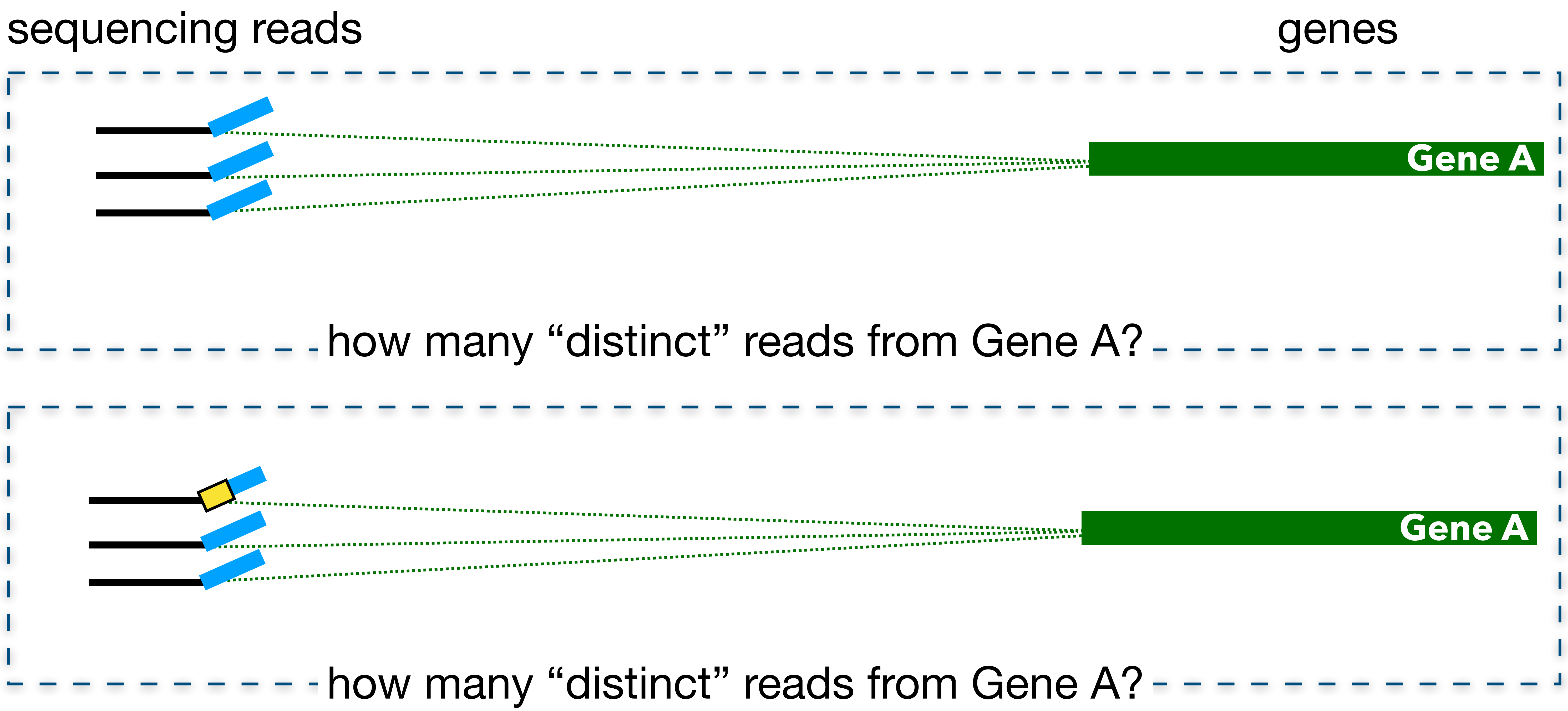

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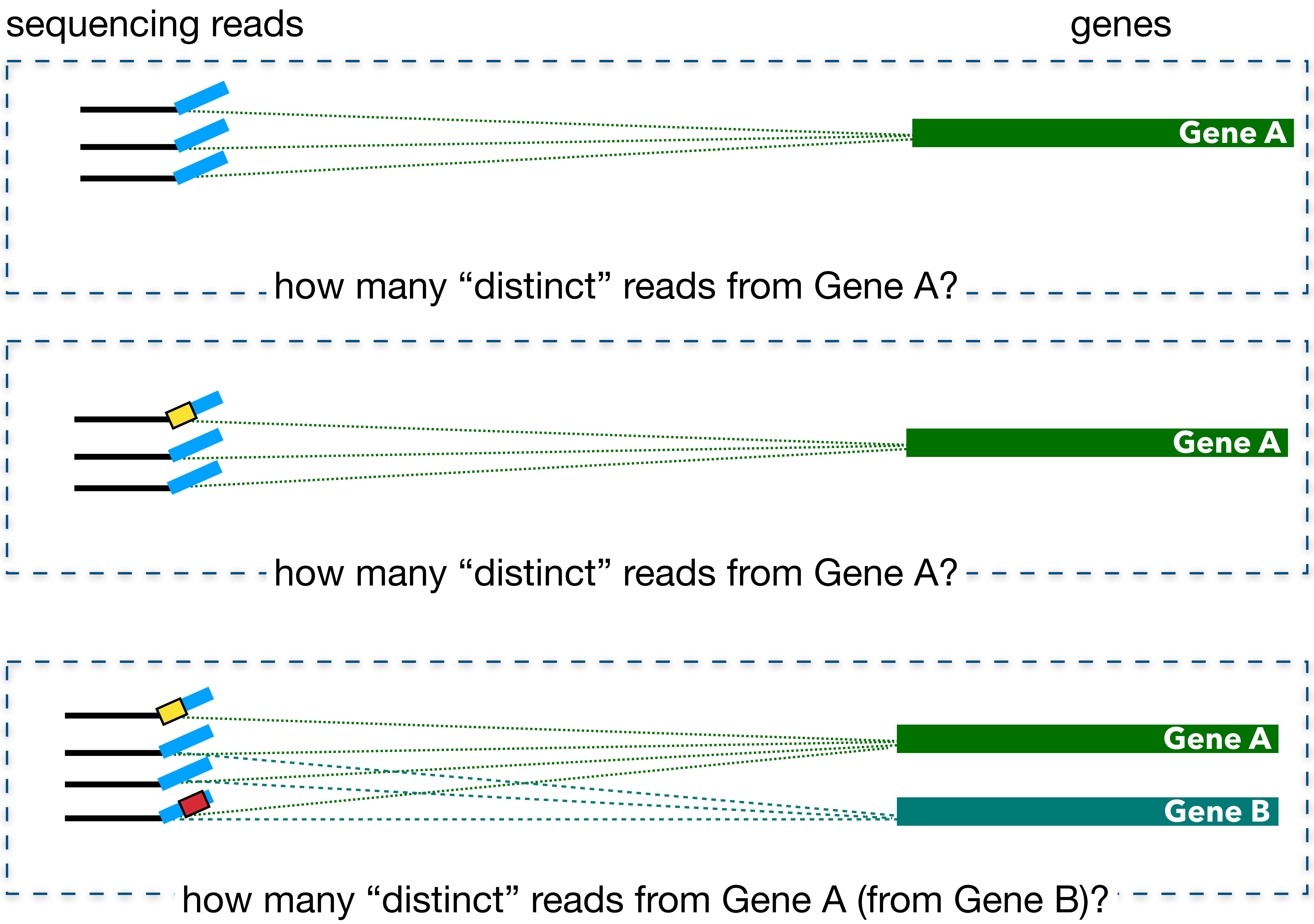


The answer for one depends on the other!

Multi-mapping is what makes this difficult

There are *many* reasons single-cell quantification is challenging!

most of these stem from the difficulty of cell isolation & small quantities of genetic material:

- 1 UMI ≠ 1 pre-PCR molecule (often); because of small material per-cell, typical protocols involve many rounds of PCR; UMI tags are subject to PCR and sequencing error. There is also “dropout” and bias.

Parsimonious UMI Graph (PUG) resolution

Represent the UMIs / transcripts relationship as a graph $G = (V, E)$.

Each **vertex** ($v \in V$) is a tuple:

- $\mathbf{eq_v}$: the set of transcripts to which the read maps
- $\mathbf{s_v}$: the UMI sequence tagging the read

Each v has count, $c(v)$ — number of "**equivalent**" reads.
"**equivalent**" means aligns to same transcripts and has same UMI.

There is an **edge** ($e = \{u,v\} \in E$) if, for some chosen edit distance τ :

- $s_u = s_v$ and $|eq_u \cap eq_v| > 0$ (*bidirected* edge)
- $d(s_u, s_v) < \tau$, $c(u) \sim c(v)$ and $|eq_u \cap eq_v| > 0$ (*bidirected* edge)
- $d(s_u, s_v) < \tau$, $c(u) > 2c(v)+1$ and $|eq_u \cap eq_v| > 0$ (*directed* $u \rightarrow v$)

Smith, T. S., Heger, A., & Sudbery, I. (2017). UMI-tools: Modelling sequencing errors in Unique Molecular Identifiers to improve quantification accuracy. *Genome research*

A parsimony-guided approach

We will attempt to *explain* the PUG using the *minimum number* of pre-PCR molecules.
More formally:

Given: UMI-resolution graph $G = (V, E)$

Find: a *minimum cardinality* cover by *monochromatic arborescences.*

We seek parsimony

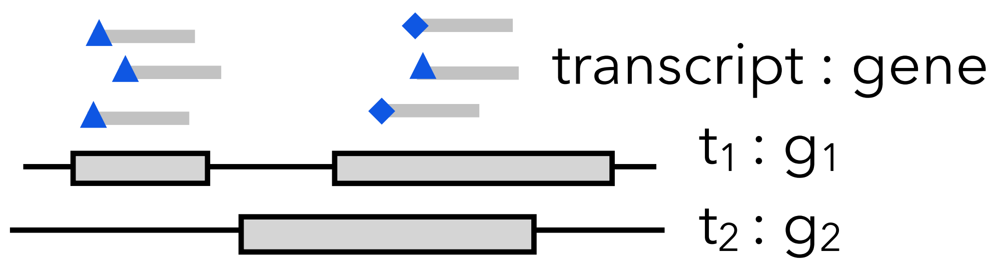
Each component can be ascribed to the reads sequenced from a *single initial molecule* (before amplification).

Each *monochromatic arborescence* is a set of vertices that can all be described as "reads" coming from the same, original transcript.

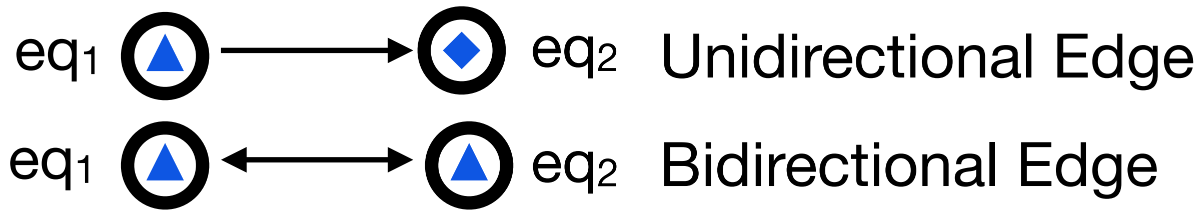
The decision problem is *NP-complete* (reduction from DOMINATING SET); but experimental instances are usually *simple* and we adopt a *greedy heuristic.* (solutions are known to be optimal in > 80% of cases)

UMI Resolution

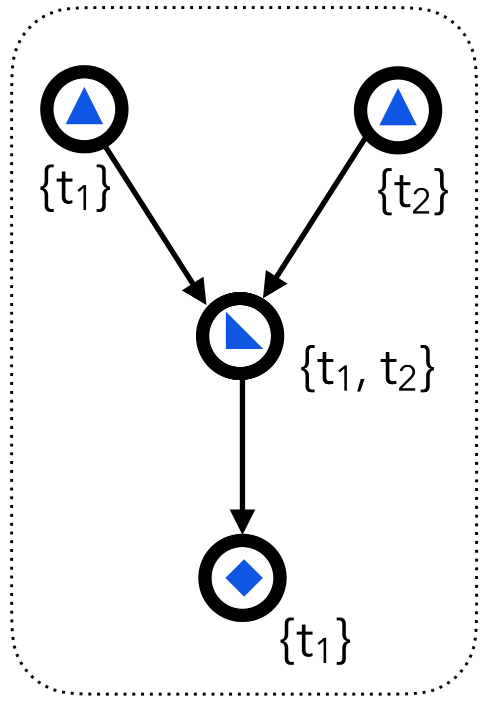
read mappings + UMIs
define equivalence classes (vertices)



UMI seq & freq define the edge set

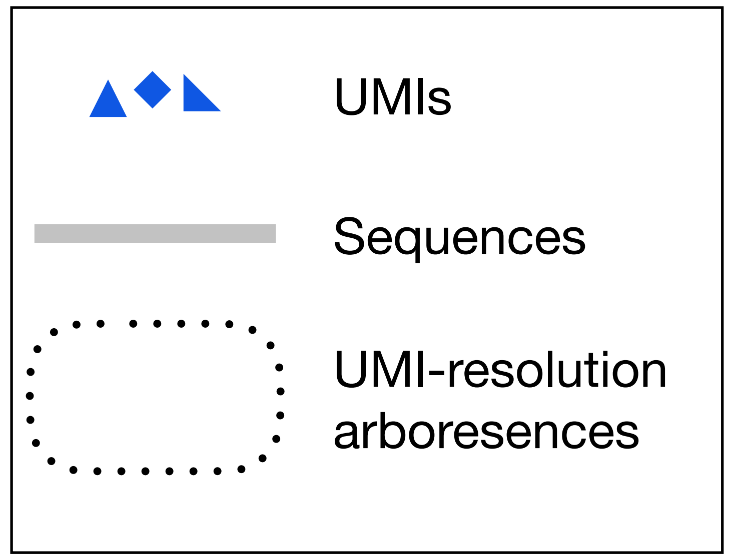


where $d(▲, ◆) \leq \tau$ and $|eq_1 \cap eq_2| > 0$



Algorithm seeks
parsimonious arboresence

legend



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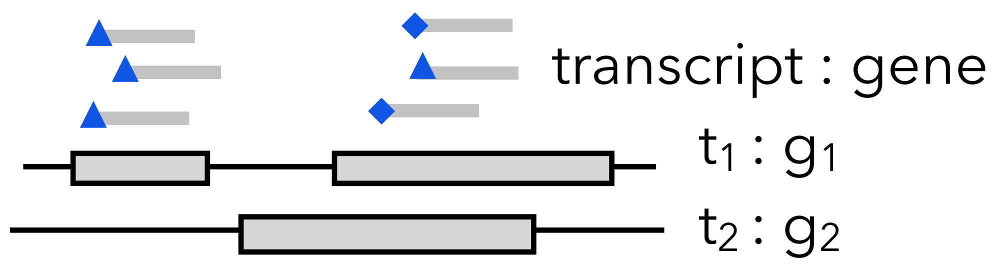
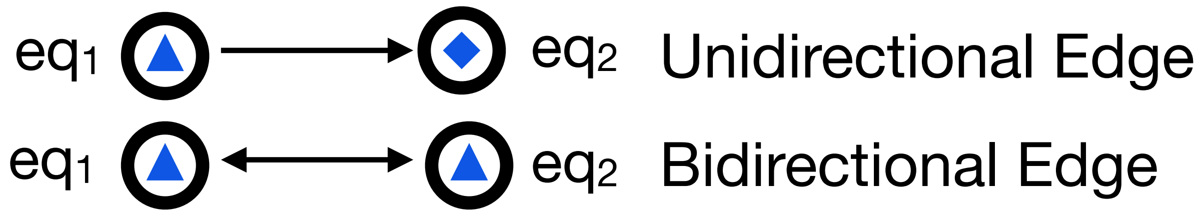
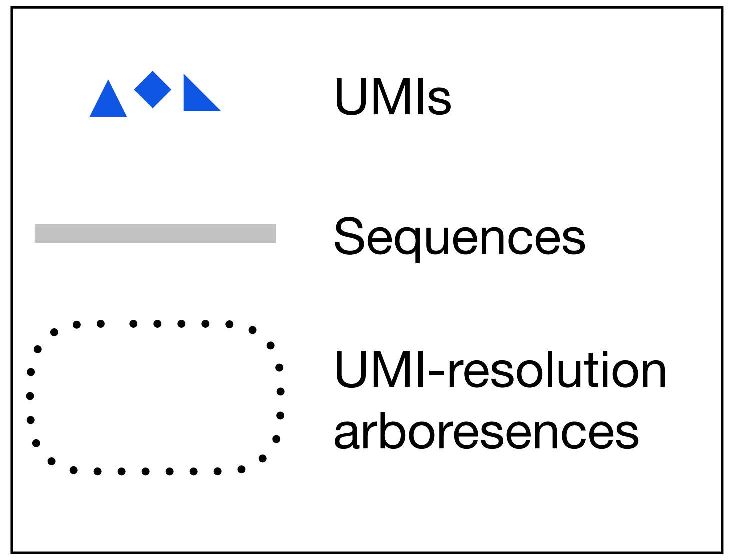


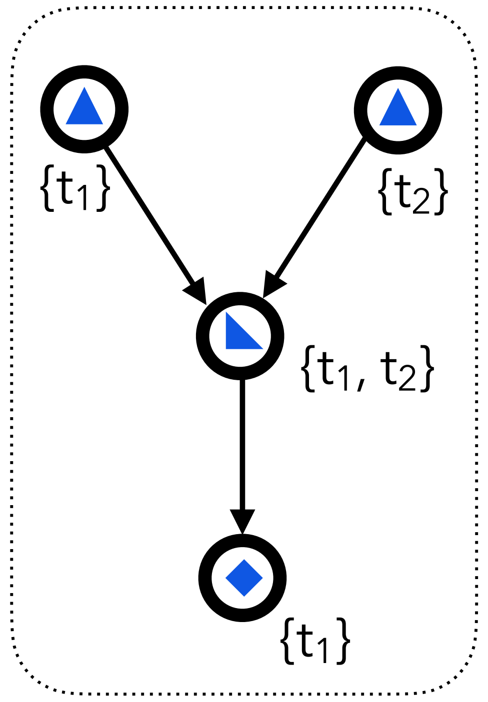
UMI seq & freq define the edge set



where d(▲, ◆) $\leq \tau$ and $|eq_1 \cap eq_2| > 0$

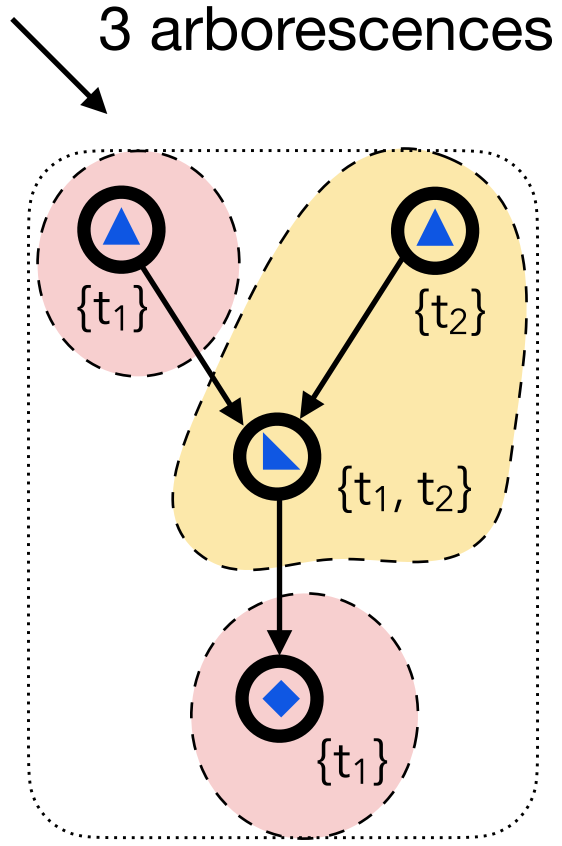
legend





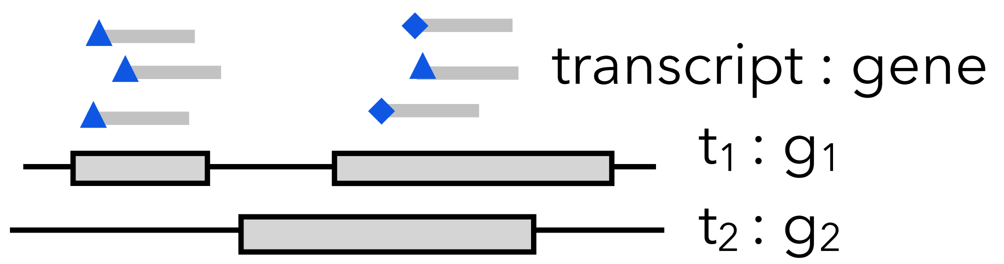
Algorithm seeks
parsimonious arboresence

3 arborescences

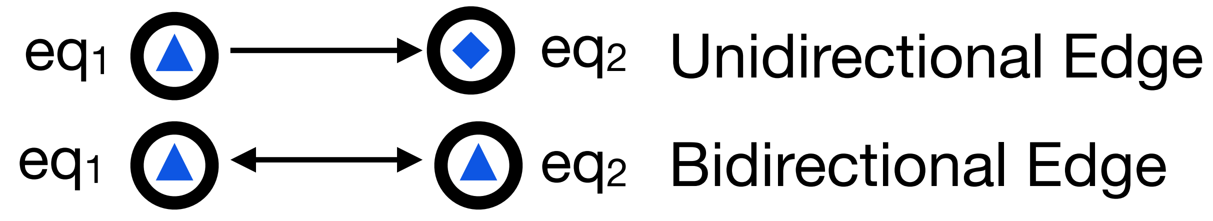


UMI Resolution

read mappings + UMIs
define equivalence classes (vertices)

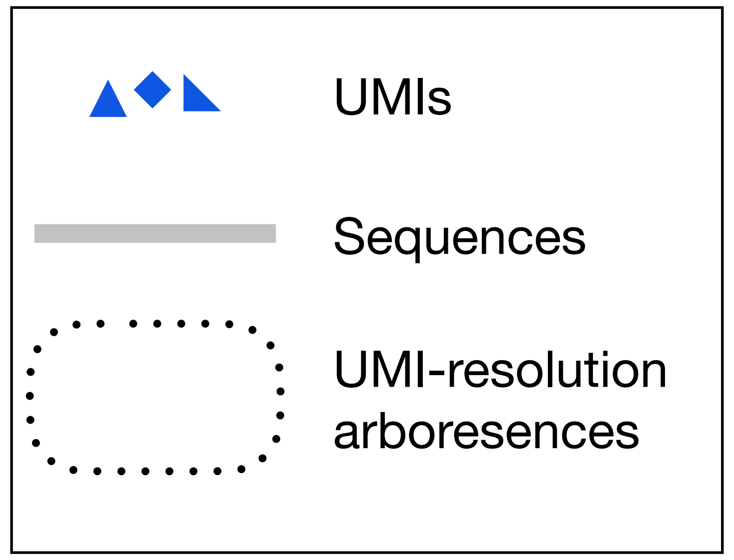


UMI seq & freq define the edge set



where d(▲, ◆) $\leq \tau$ and | eq$_1 \cap$ eq$_2$ | > 0

legend



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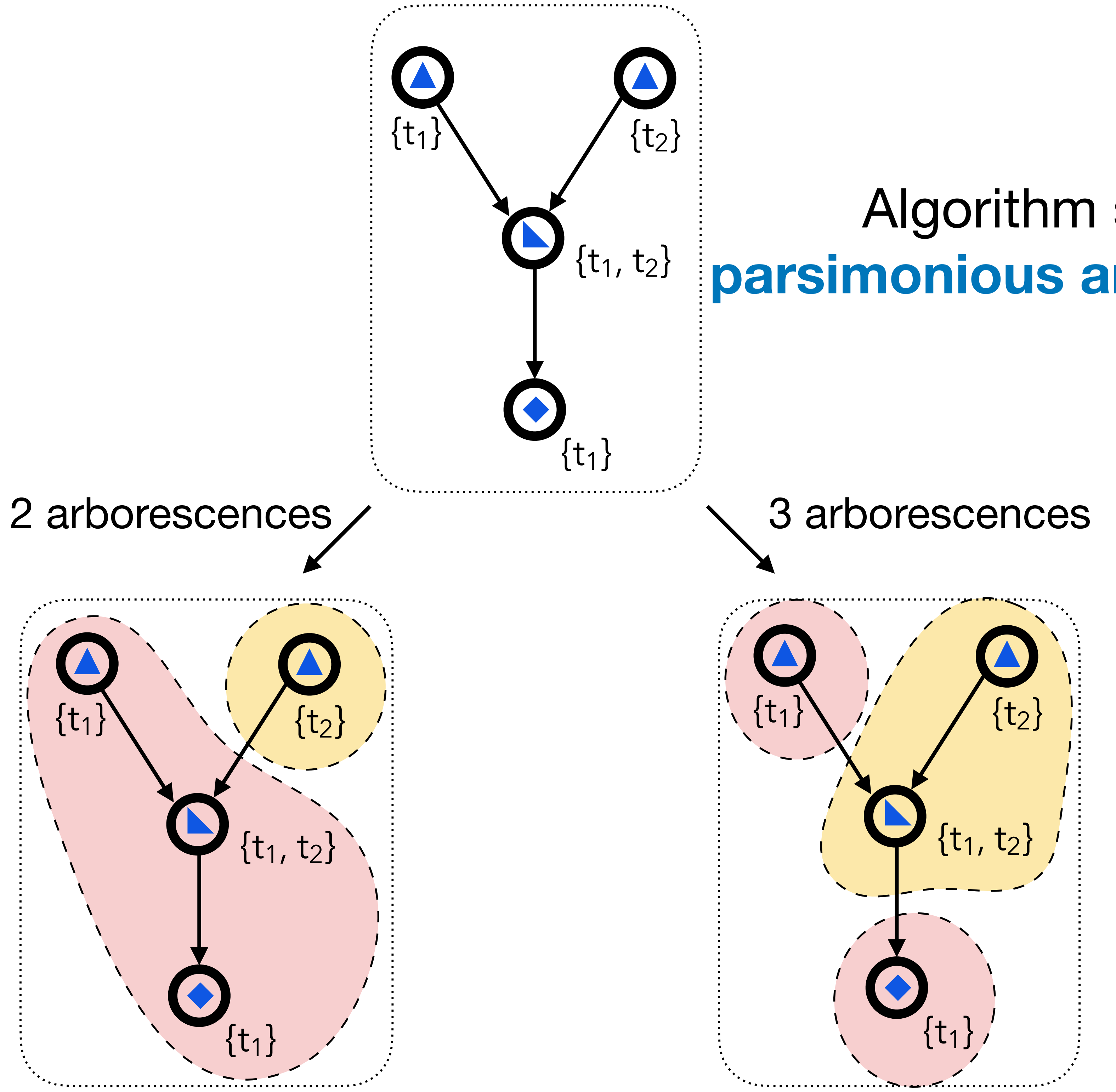
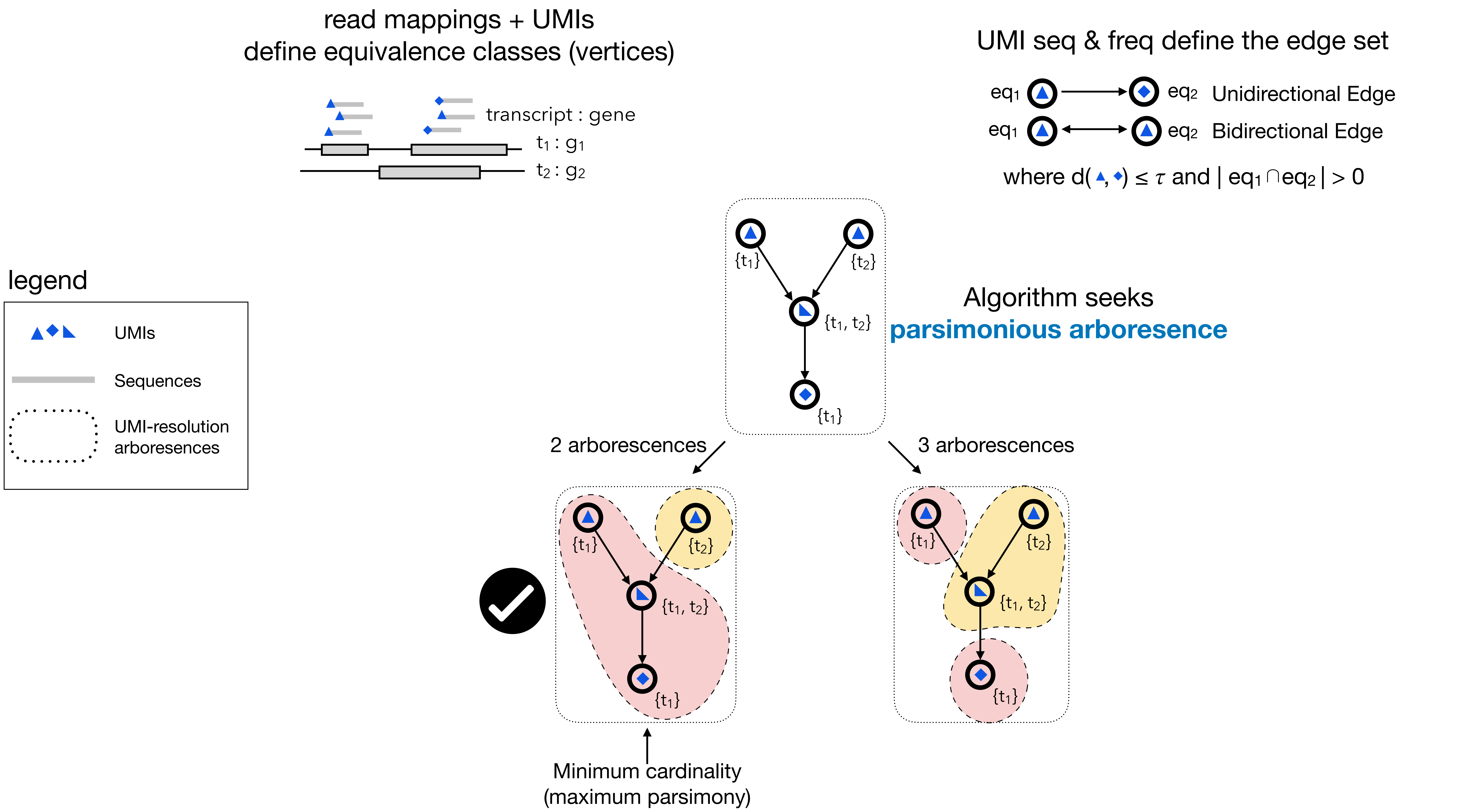
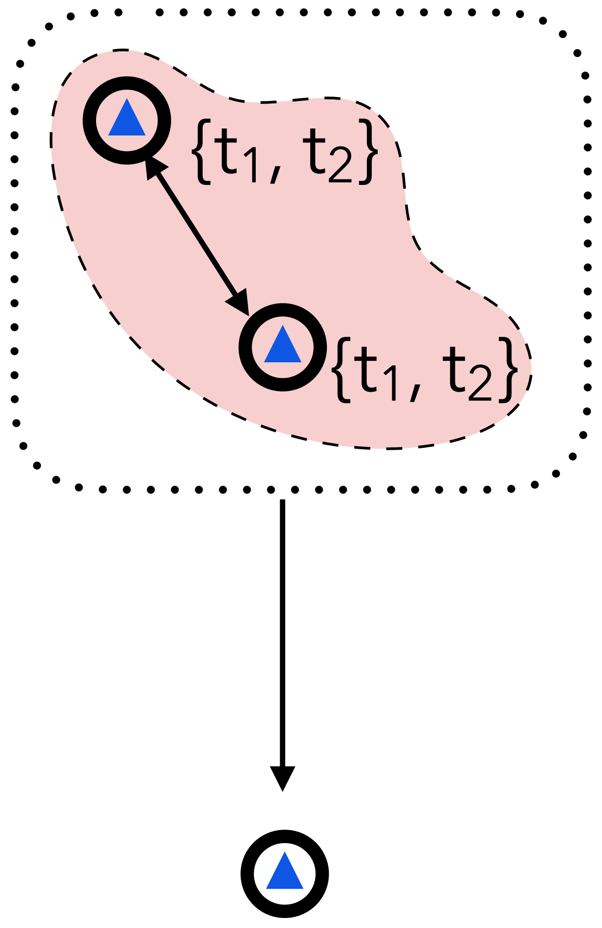


UMI Resolution



UMI Resolution

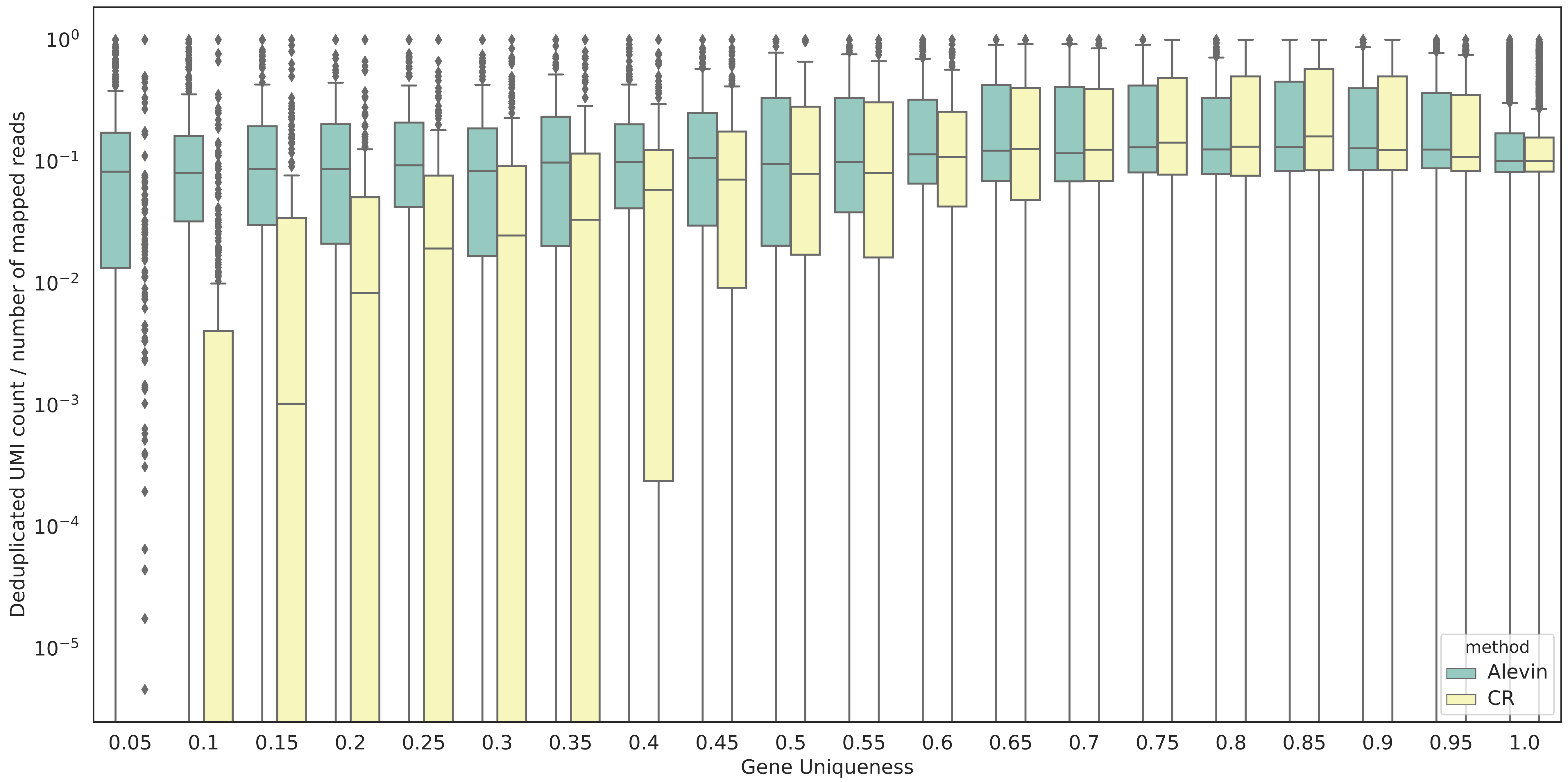
Parsimony does not always provide a unique solution, even if we can solve it optimally.



Equally-parsimonious under t1 **or** t2

- Parsimony cannot resolve the gene of origin here
- We treat UMI as gene-ambiguous & resolve via EM-algorithm

Discarding gene-ambiguous reads does not affect all genes equally

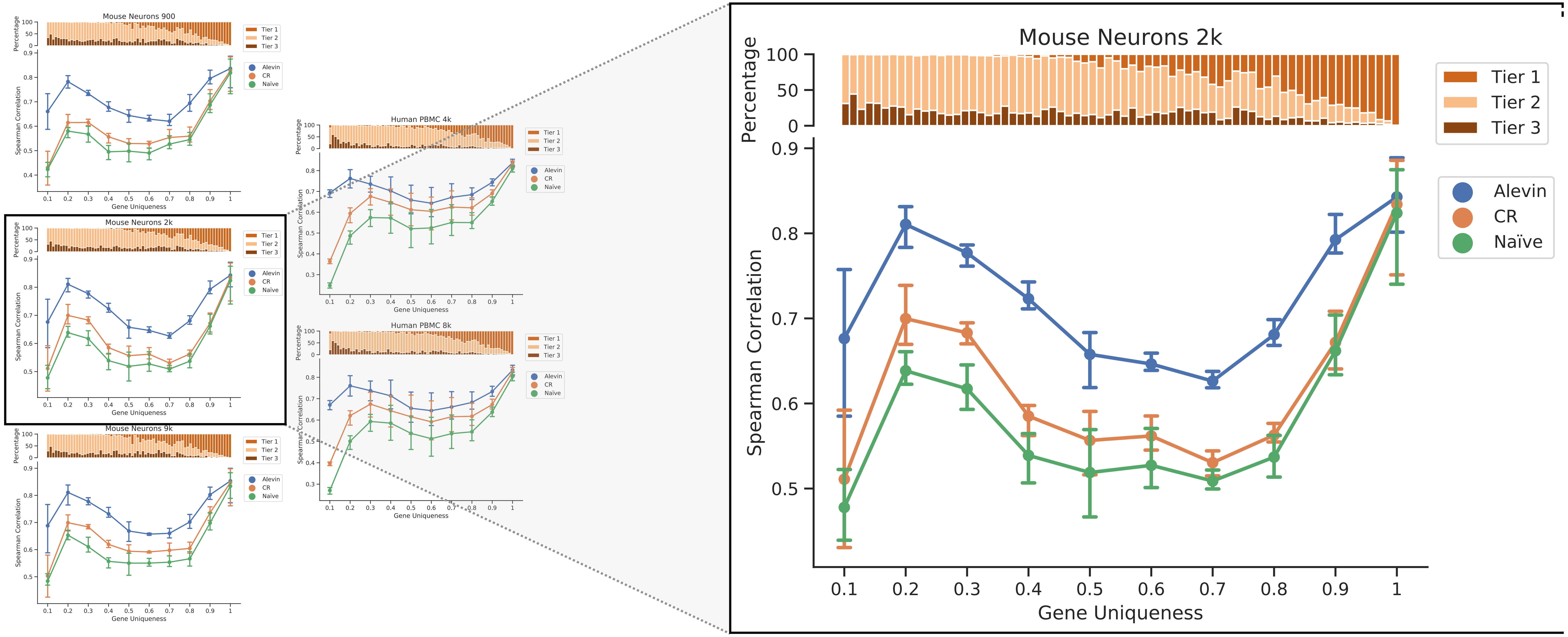


Bin number	Human	Mouse
1	2538	3984
2	1094	1398
3	813	996
4	852	944
5	908	1137
6	1044	1178
7	1339	1392
8	1943	1615
9	3360	2226
10	44397	38509

Stratify genes by sequence uniqueness and look at distribution of # of de-duplicated UMIs per input read (PBMC 4k). Large effect for genes with lots of sequence homology (including important paralog families). Thus, the discarding of gene multi-mapping reads leads to systematic bias.

Accounting for gene-ambigous improves correlation with bulk RNA-seq

Trend across 5 public 10x datasets [+]



For each scRNA-seq dataset, we obtained multiple bulk samples from the same tissue. We compare gene-level quants in bulk (quantified with RSEM*) to average expressions across single cells.

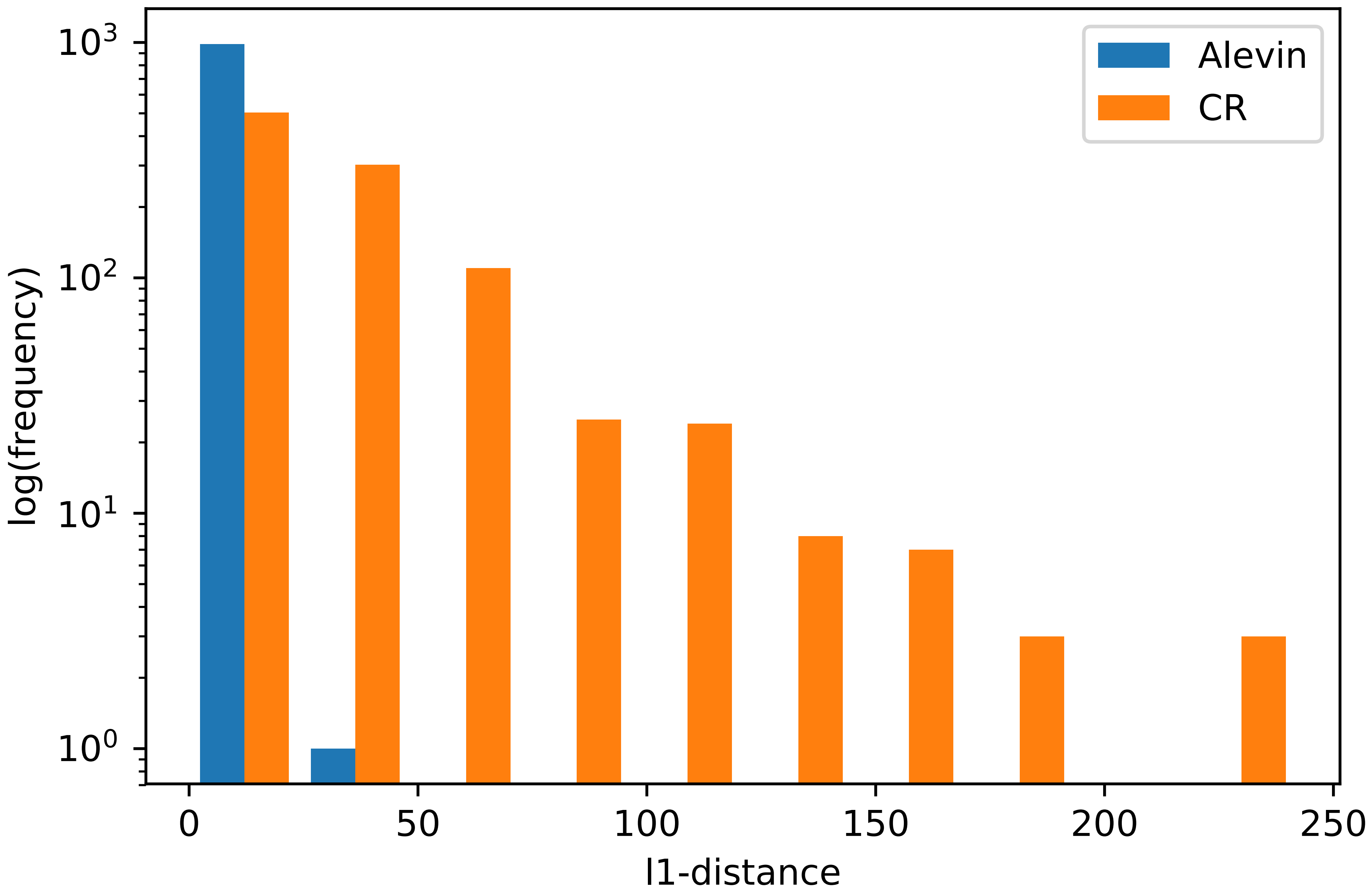
*Li, B., & Dewey, C. N. (2011). RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. *BMC bioinformatics*, *12*(1), 323.

+Zheng, Grace XY, et al. "Massively parallel digital transcriptional profiling of single cells." Nature communications 8 (2017): 14049.

Robustness to *in silico* homology

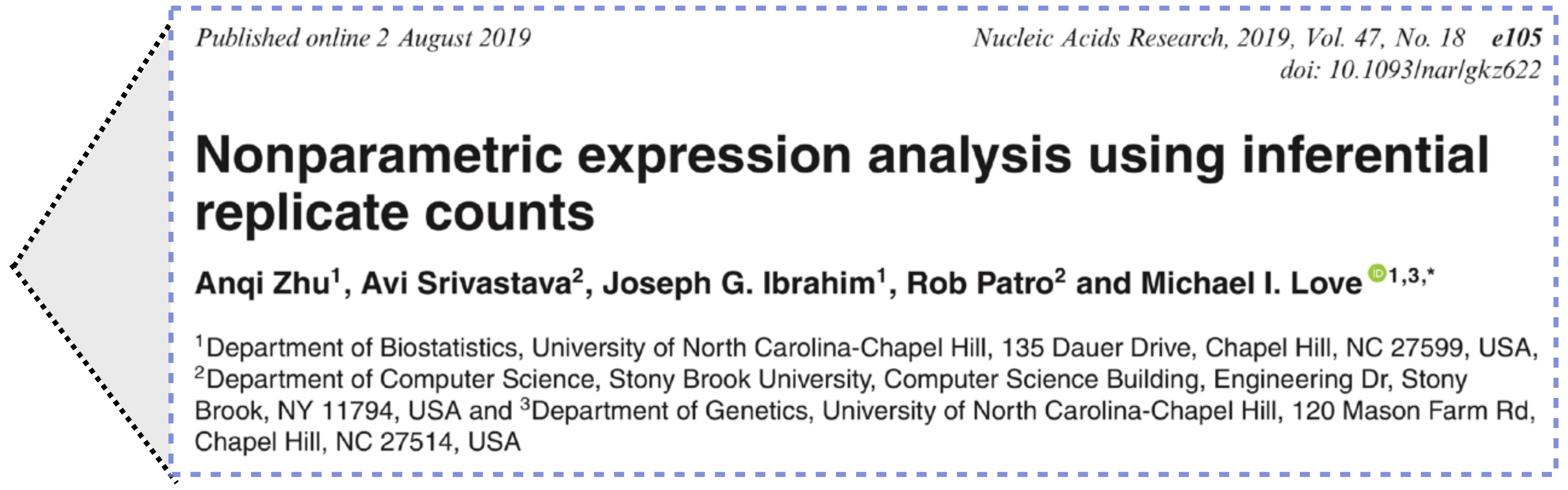
Compare quantifications of a mouse neuron sample:

aligned and quantified against **mouse reference** vs. **mouse + human reference**



> "A cow is … basically a human" (paraphrased)
> – Bruce Futcher (yeast biologist, extraordinaire)

Also: alevin can provide <u>bootstrap</u> uncertainty estimates of gene counts to be used in downstream analysis



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doi: 10.1093/nar/gkz622

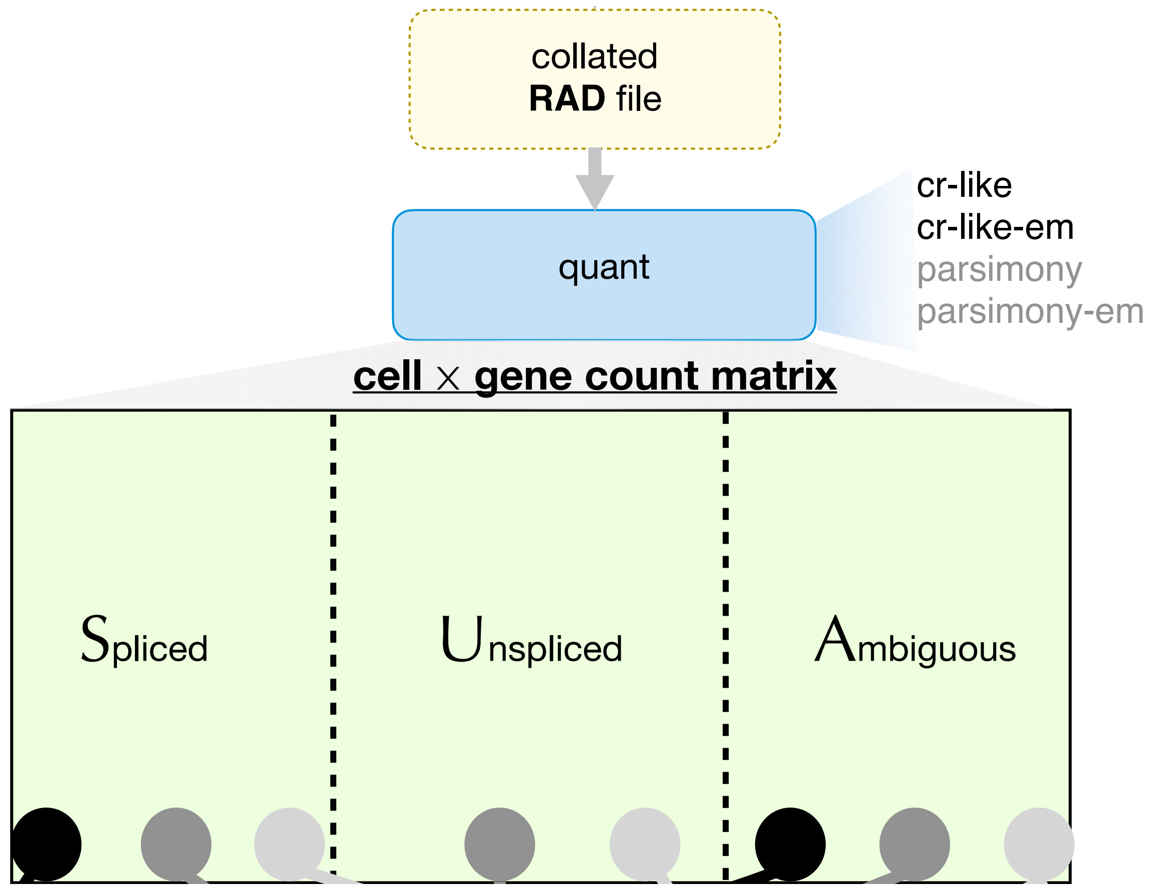
Nonparametric expression analysis using inferential replicate counts

Anqi Zhu[1], Avi Srivastava[2], Joseph G. Ibrahim[1], Rob Patro[2] and Michael I. Love[1,3,*]

[1]Department of Biostatistics, University of North Carolina-Chapel Hill, 135 Dauer Drive, Chapel Hill, NC 27599, USA, [2]Department of Computer Science, Stony Brook University, Computer Science Building, Engineering Dr, Stony Brook, NY 11794, USA and [3]Department of Genetics, University of North Carolina-Chapel Hill, 120 Mason Farm Rd, Chapel Hill, NC 27514, USA

Resolve UMIs to assign counts to genes

Characterize *the number of distinct pre-PCR* molecules arising from *each splicing status* of *each gene* within *each cell*.

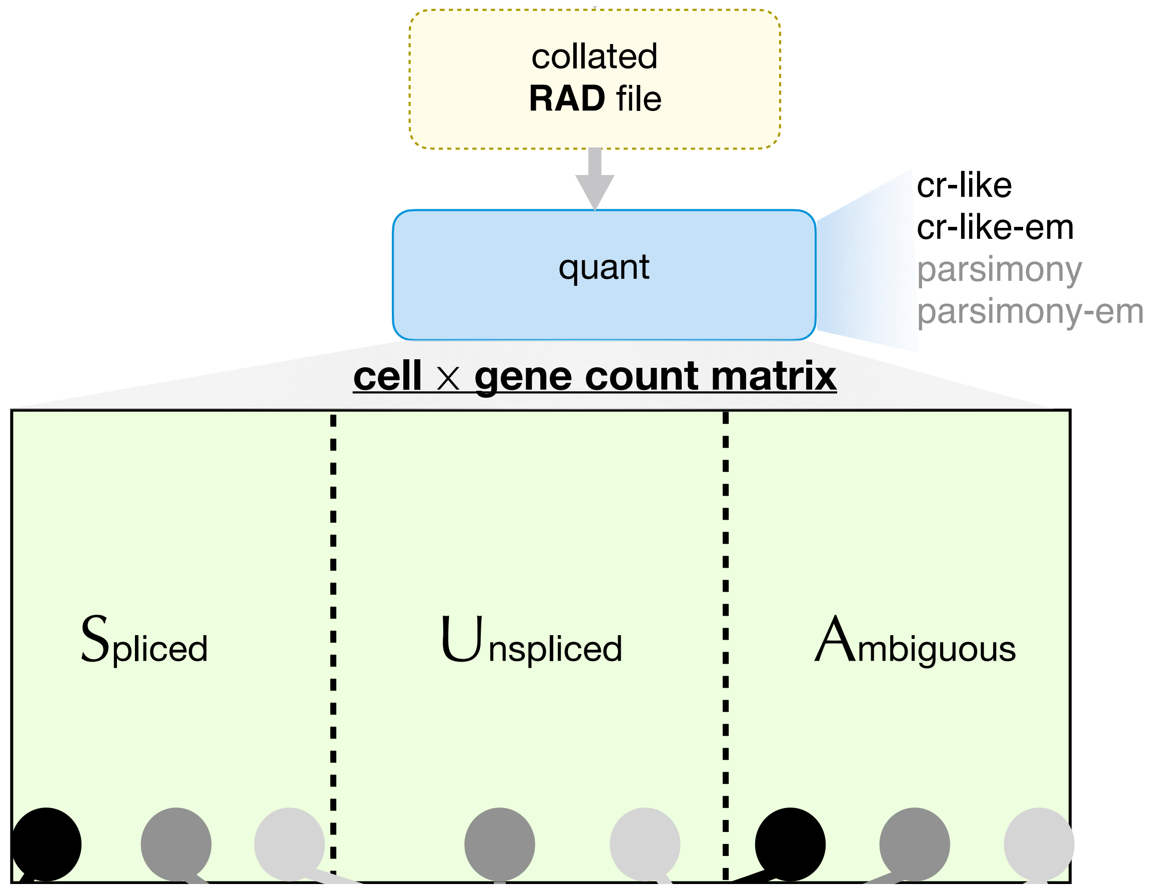


Challenges:

- UMIs can be duplicated.
- UMIs can have errors.
- Reads can map to multiple genes.
- **How to assign splicing status to UMIs**

Resolve UMIs to assign counts to genes

Characterize *the number of distinct pre-PCR* molecules arising from *each splicing status* of *each gene* within *each cell*.



- How to assign splicing status to UMIs

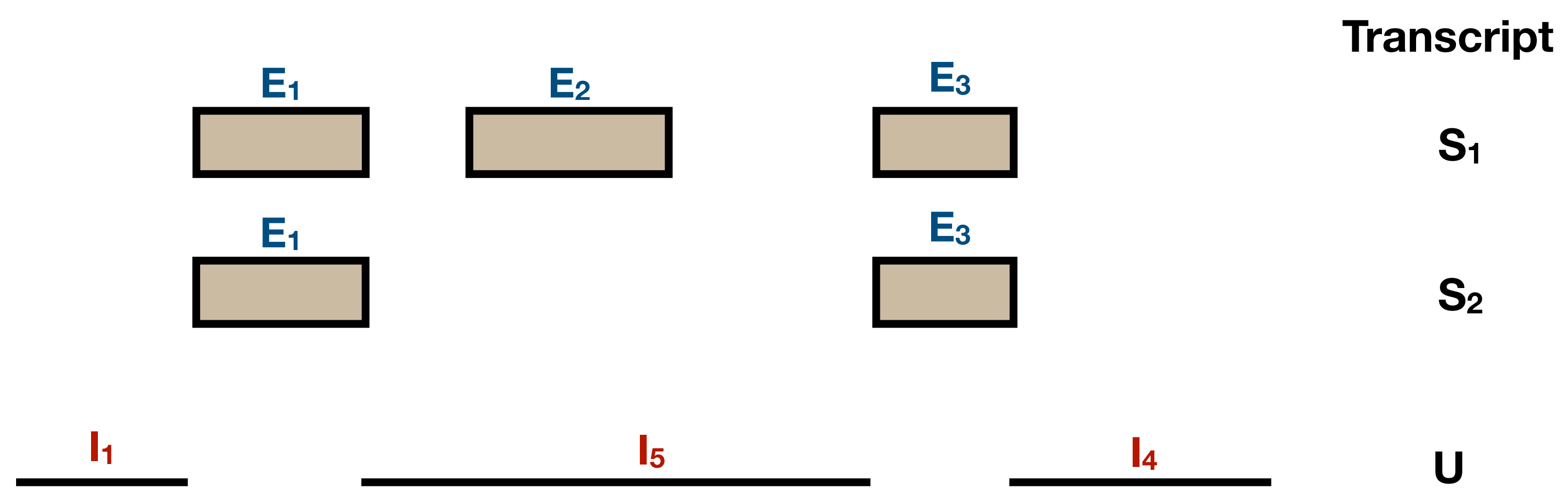
Challenges:

- UMIs can be duplicated.
- UMIs can have errors.
- Reads can map to multiple genes.

UMIs’ splicing status

The splicing status of a UMI is decided by the splicing status of its reference transcripts.

When using the *splici* reference



- Spliced UMIs:
  - UMIs with exon-exon junctional reads
  - UMIs with only exonic reads and the exon does not intersect with introns.
- Ambiguous UMIs: UMIs with only exonic reads and the exon intersects with introns.
- Unspliced UMIs: UMIs with intronic reads.

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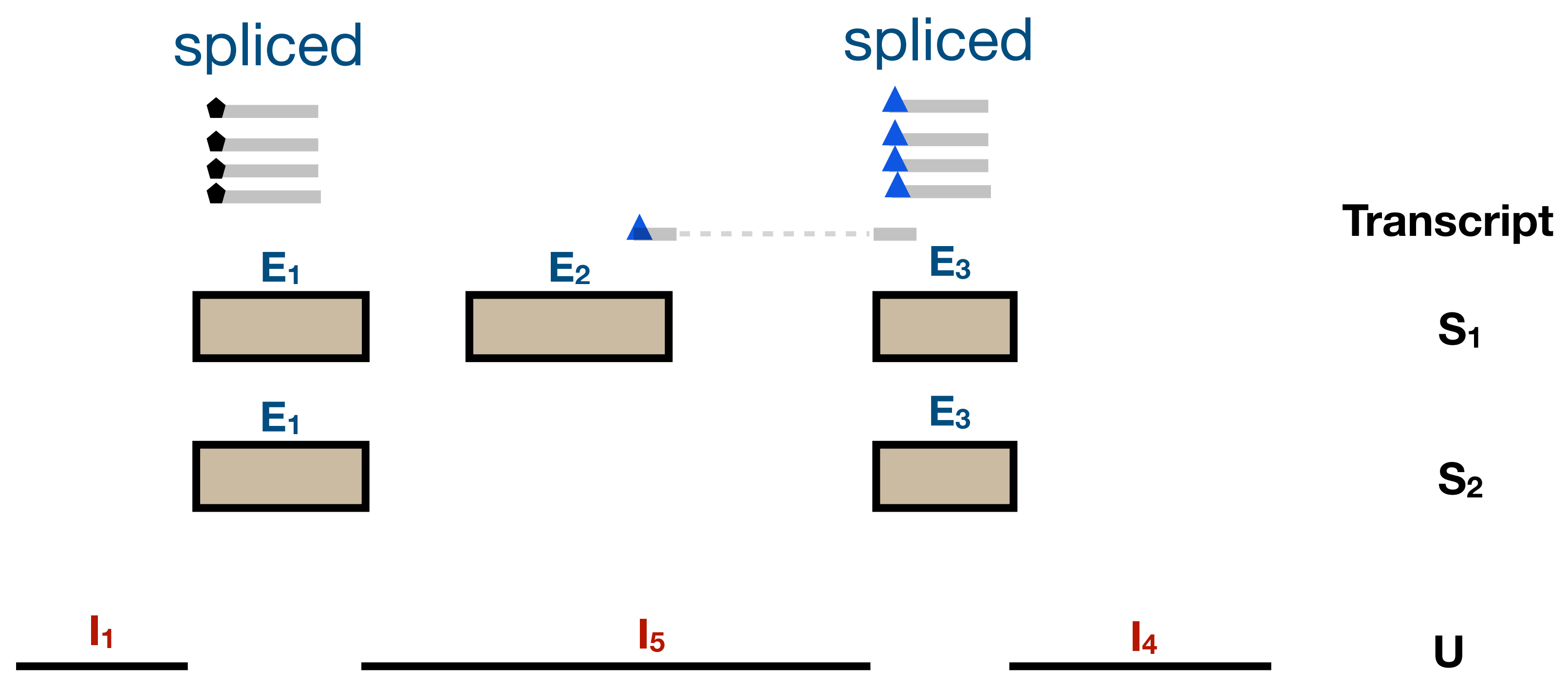

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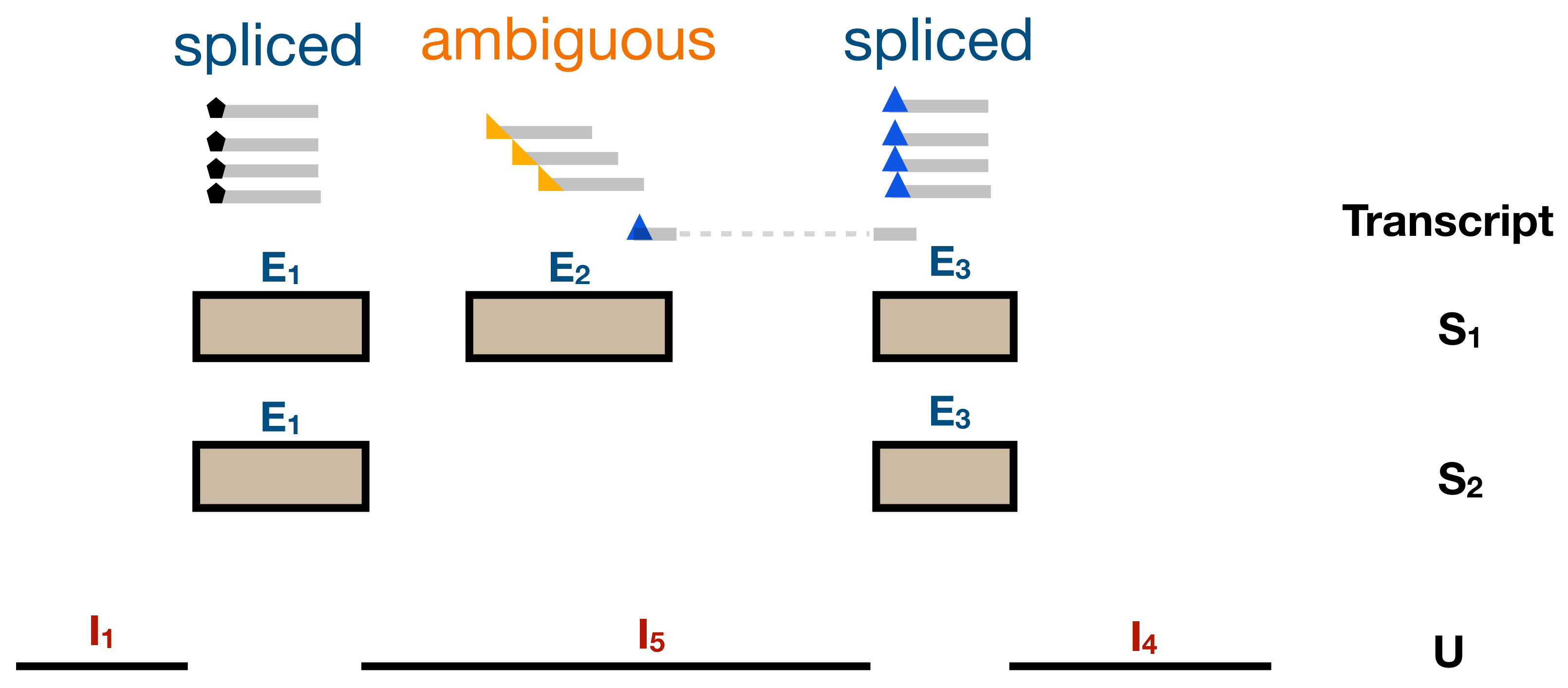

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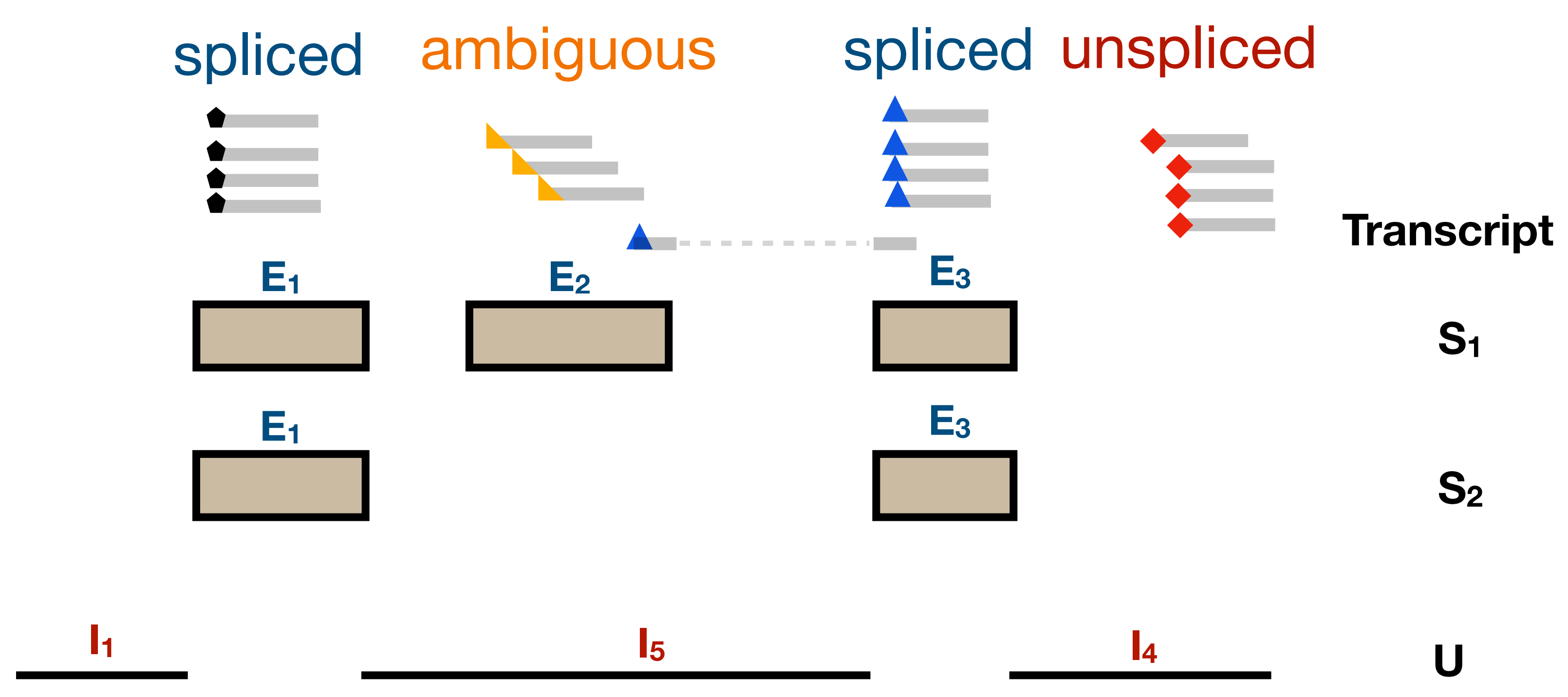

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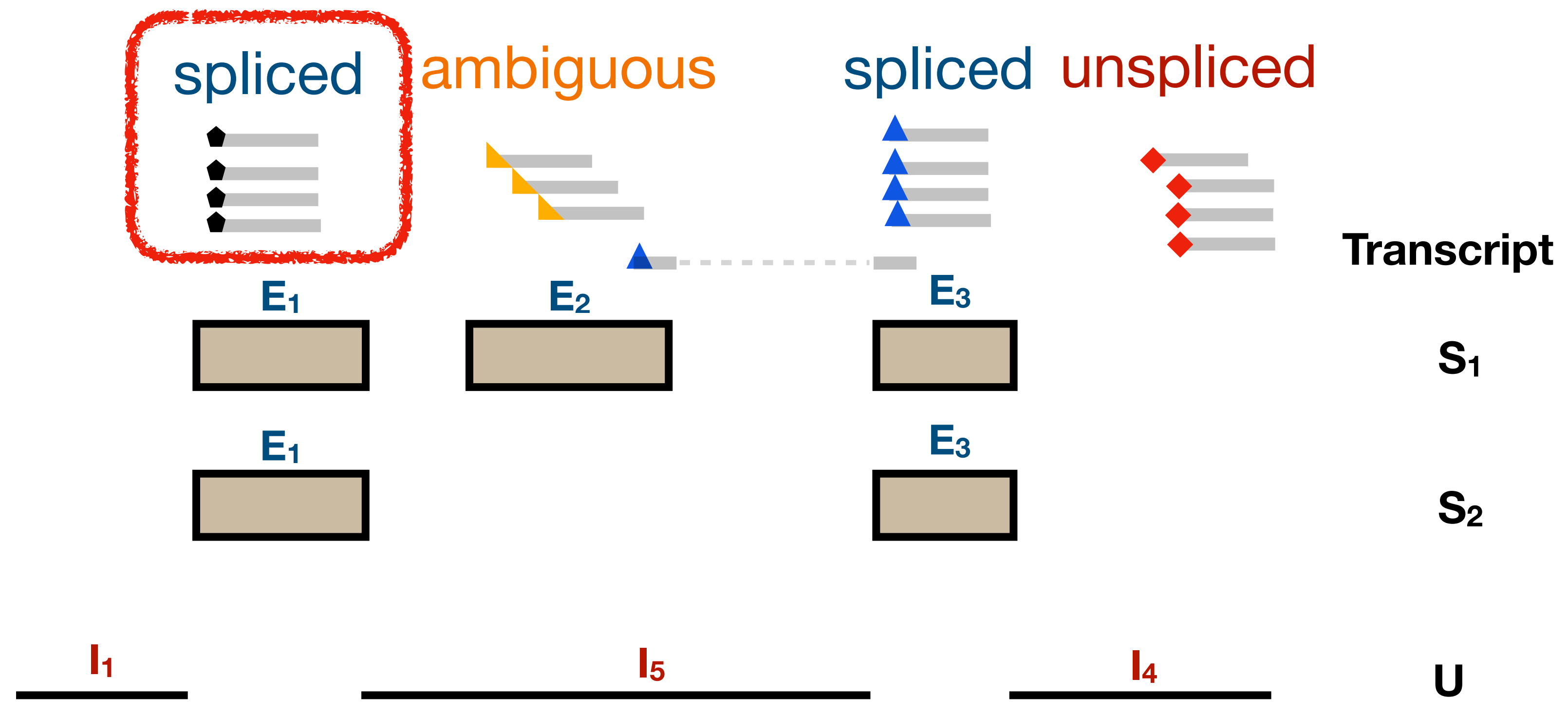

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Both spliced and unspliced transcripts contain exons!

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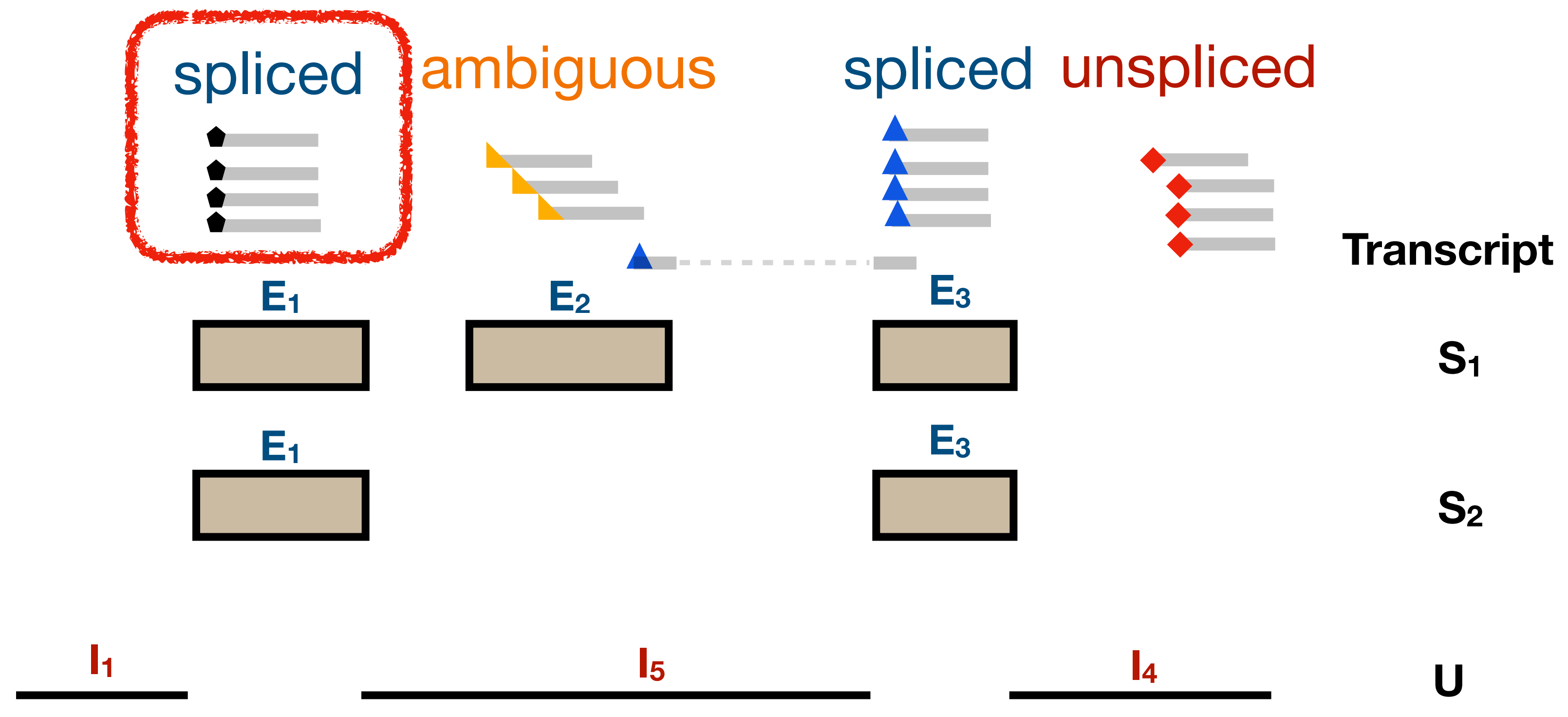
When using the *splici* reference



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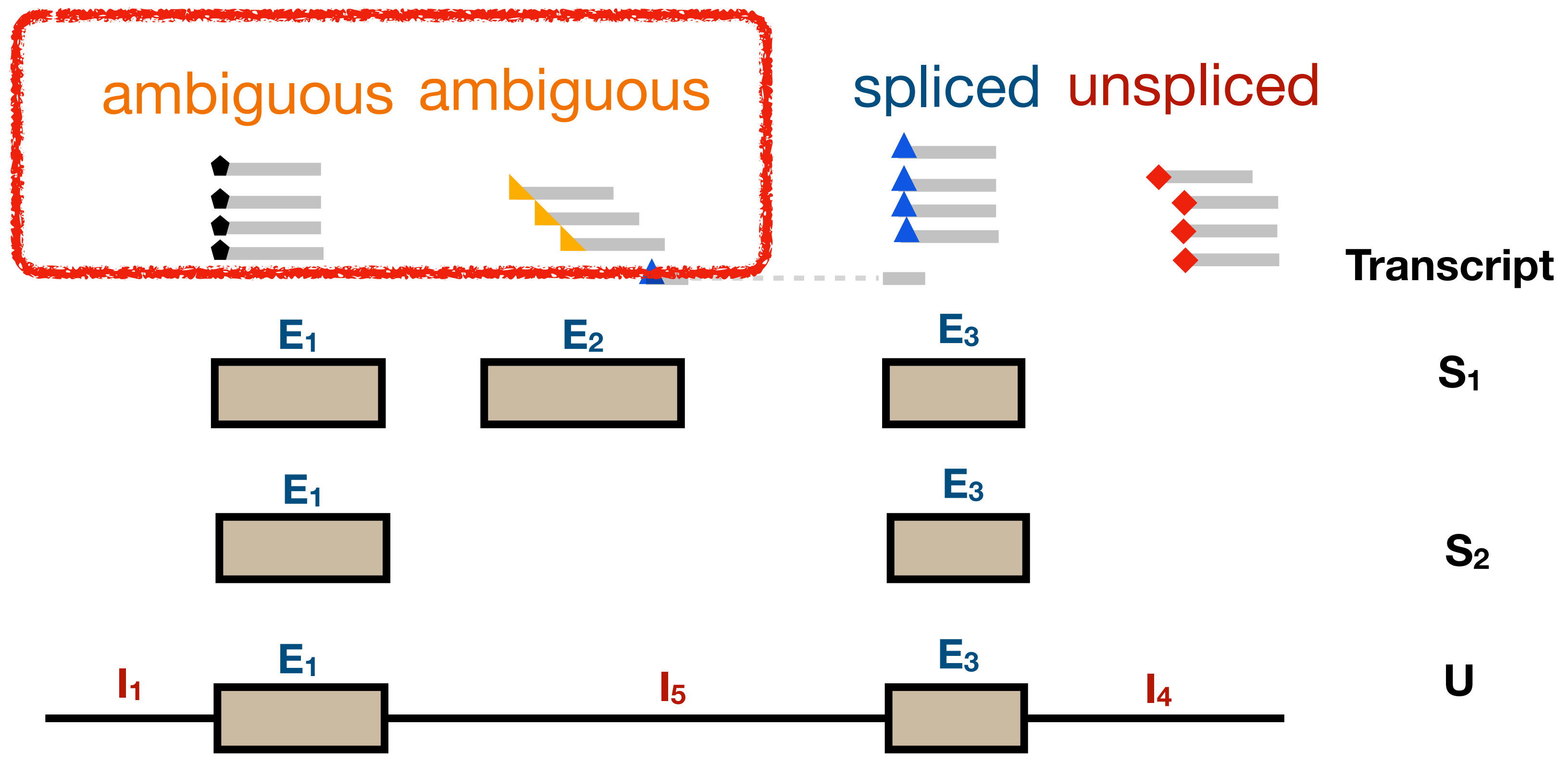
- Unspliced UMIs: UMIs with intronic reads.

Both spliced and unspliced transcripts contain exons!

When using the *spliceu* reference



- Spliced UMIs: UMIs with exon-exon junctional reads.
- Unspliced UMIs: UMIs with intronic reads.
- Ambiguous UMIs: UMIs with only exonic reads.

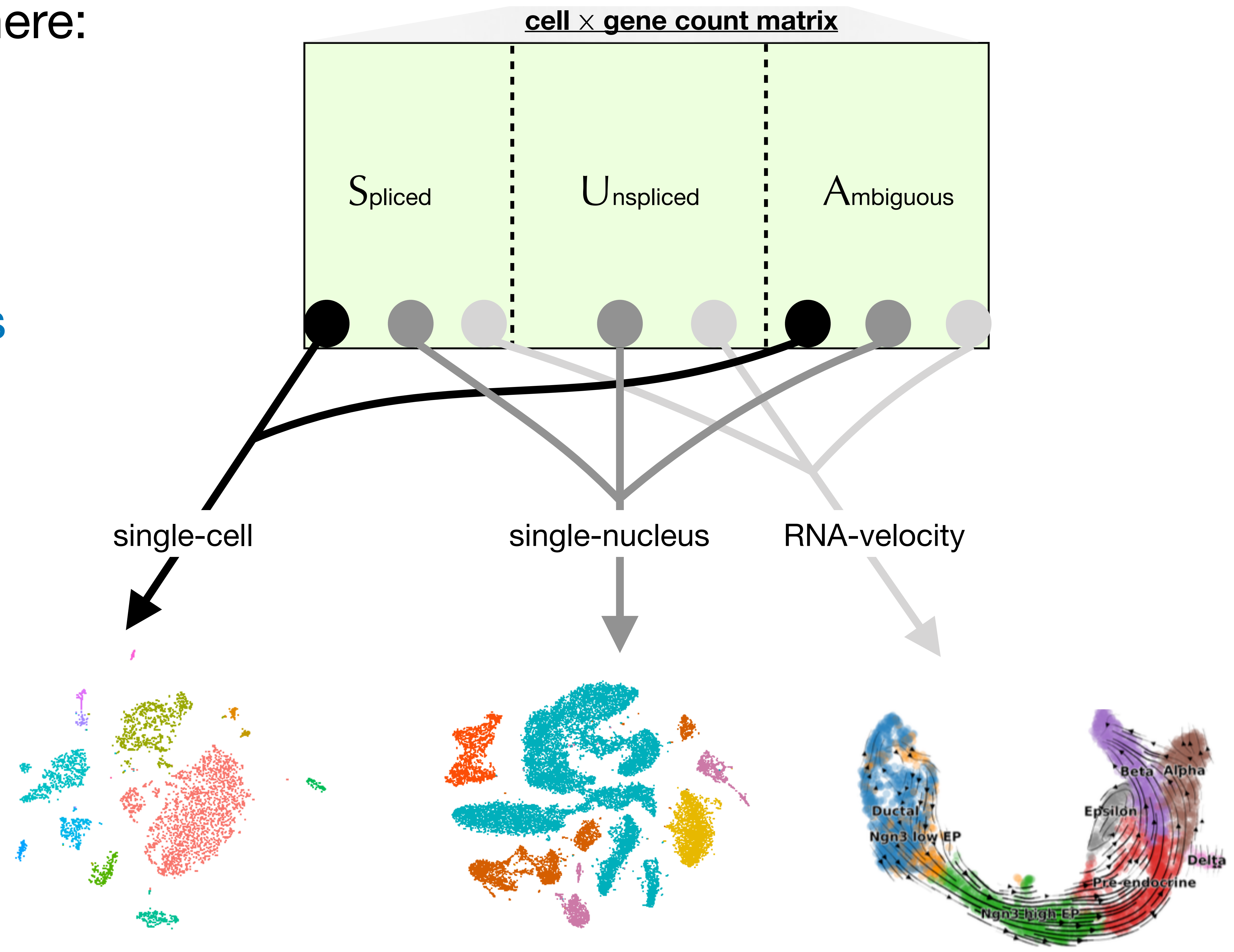
20-40% ambiguous UMIs!

What comes out of alevin-fry?

Output of this whole process is a *C* x *3G* count matrix. Where:
C = # "detected" cell barcodes and
G = # genes

It encodes the number of spliced, unspliced, or ambiguous molecules for *each gene* within *each cell*.

This unifies the pre-processing for these different types of analysis into one, common approach and gives the resolution algorithm *more information*.



Alevin-fry is accurate

Non-parametric simulated data from STARsolo[1].

Pseudoalignment results in spurious mapping (predicting many non-expressed genes per-cell); **using splici reference resolves this**!

	↑ better	↓ better	↓ better	↓ better	↓ better
method	mean Sp. corr.	MARD (drop NA)	MARD (NA=0)	mean rFP/cell	mean rFN/cell
`STARsolo`	0.997	0.031	0.002	0.001	0.005
`alevin-fry` (splici, sla)	0.992	0.019	0.001	0.004	0.011
`kallisto|bustools`	0.864	0.263	0.024	0.328	0.006

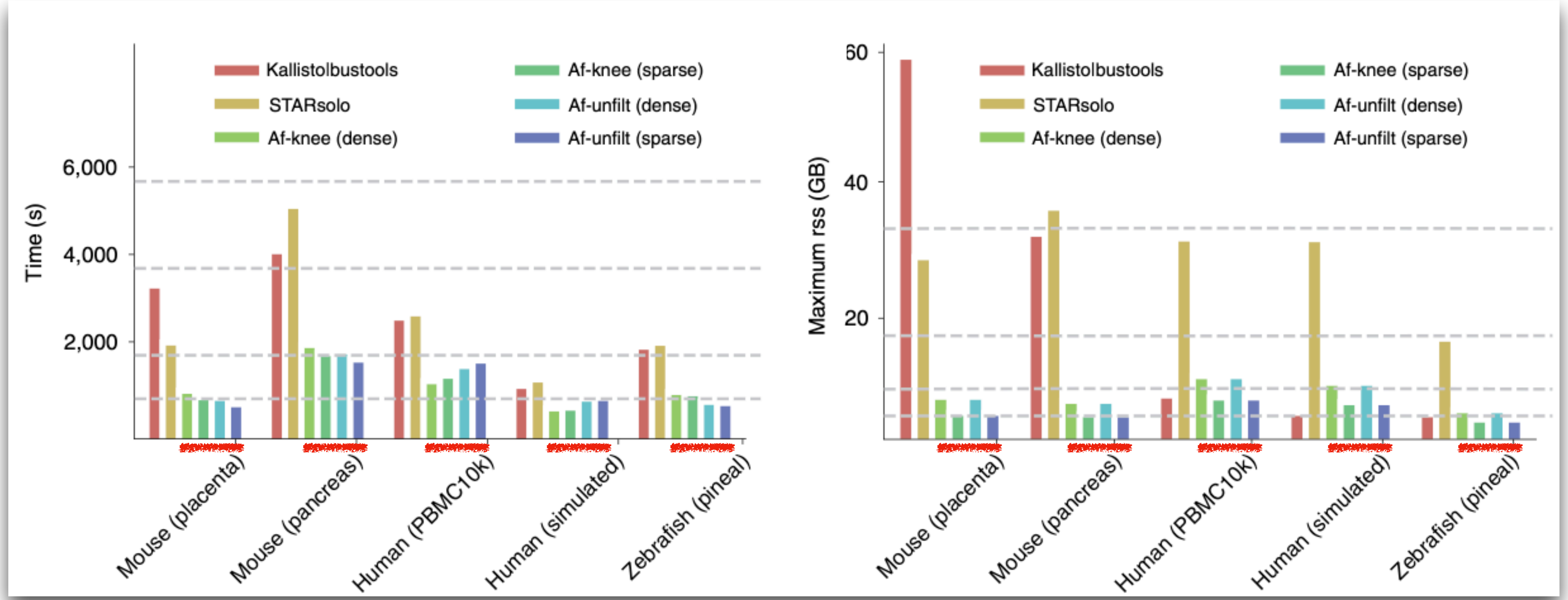
With *splici transcriptome*, alevin-fry is highly accurate.

[1] Kaminow, Benjamin, Dinar Yunusov, and Alexander Dobin. "STARsolo: accurate, fast and versatile mapping/quantification of single-cell and single-nucleus RNA-seq data." *bioRxiv* (2021).

Alevin-fry is fast and memory frugal

Alevin-fry is *fast:*
on average > 2x faster than other methods in the benchmarking

Alevin-fry is *memory frugal:*
memory footprint is low *regardless of analysis type*, since pipeline is uniform.



All analyses performed with 16-threads on a server with 512G of RAM.

JOURNAL ARTICLE

Forseti: a mechanistic and predictive model of the splicing status of scRNA-seq reads

Dongze He, Yuan Gao, Spencer Skylar Chan, Natalia Quintana-Parrilla, Rob Patro

Author Notes

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Published: 28 June 2024

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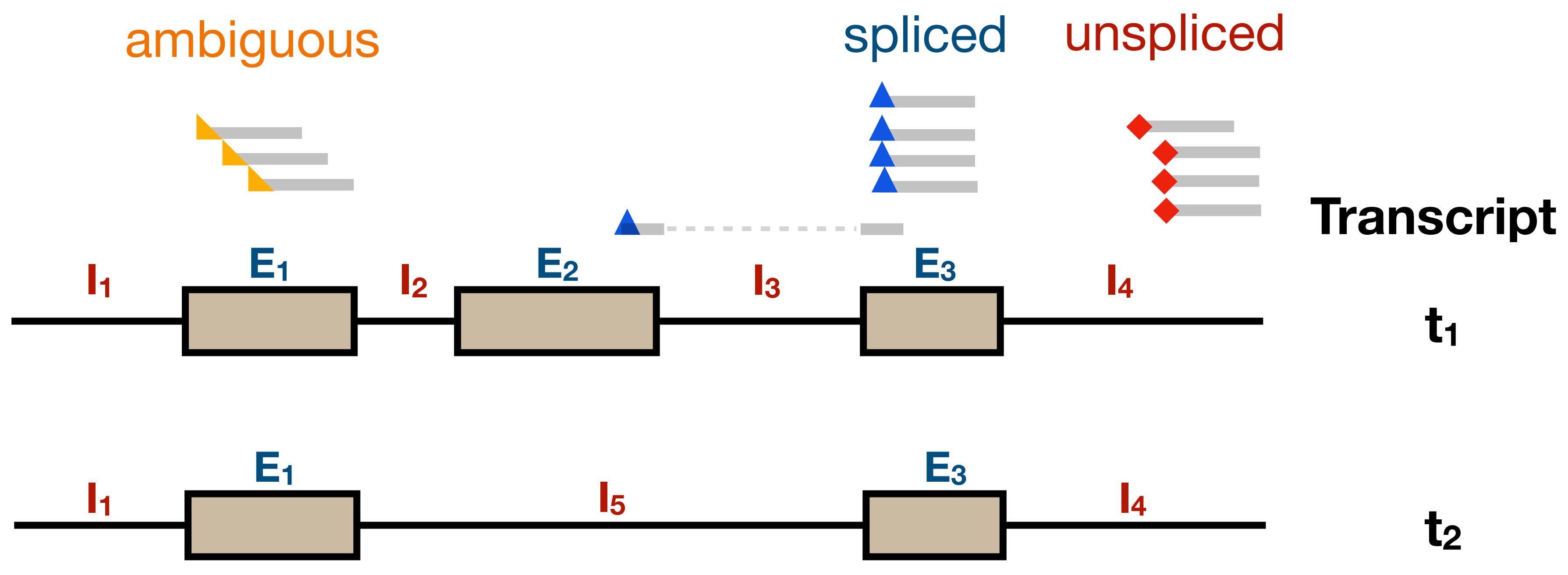
ScRNA-seq Splice Status Prediction

- A mechanistic model for predicting the splicing status of reads

Useful information for predicting splicing status

Attribute \ Name	pbmc	pbmc_multiome	bmmc	bmmc_multiome	mouse_brain_multiome	human_brain_multiome	mouse_brain
% Ambiguous Txome	37.46%	33.7%	36.34%	26.83%	31.61%	19.01%	33.35%

When using the *spliceu* reference



- Spliced UMIs: UMIs with exon-exon junctional reads.
- Unspliced UMIs: UMIs with only intronic reads.
- Ambiguous UMIs: UMIs with only exonic reads.

20-40% ambiguous UMIs!

[1] https://kb.10xgenomics.com/hc/en-us/articles/360000939852-What-is-the-difference-between-Single-Cell-3-and-5-Gene-Expression-libraries-

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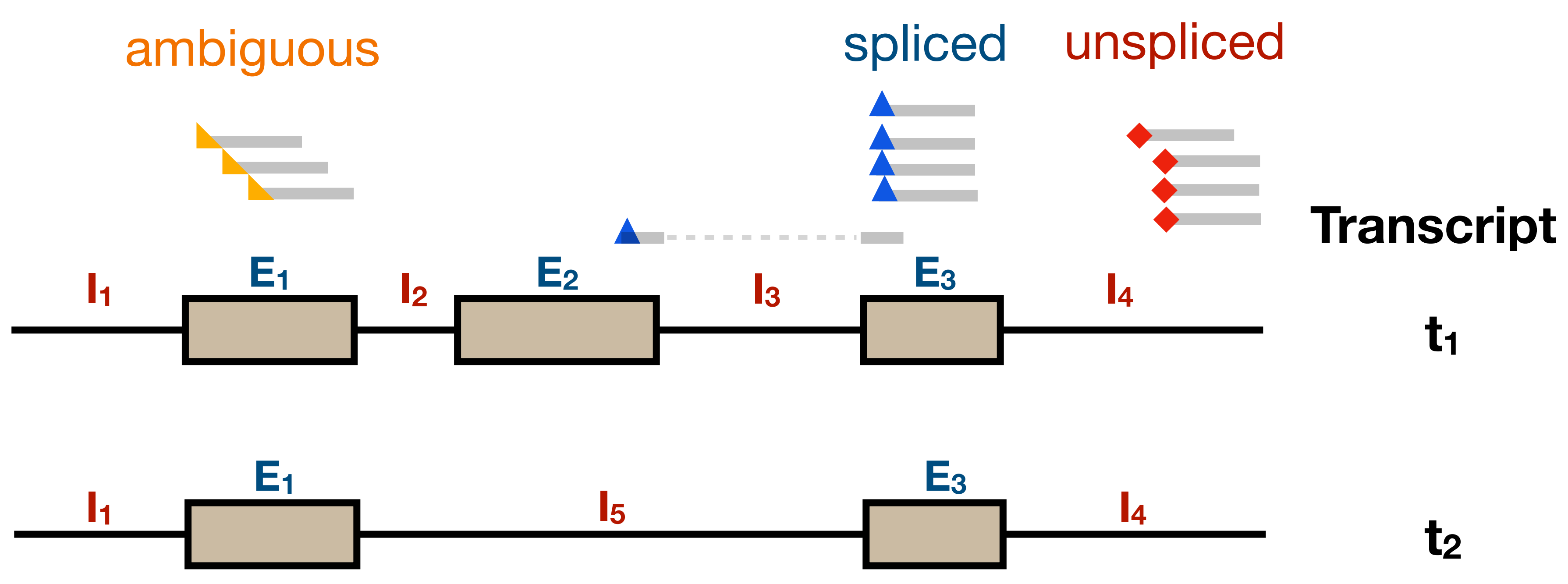

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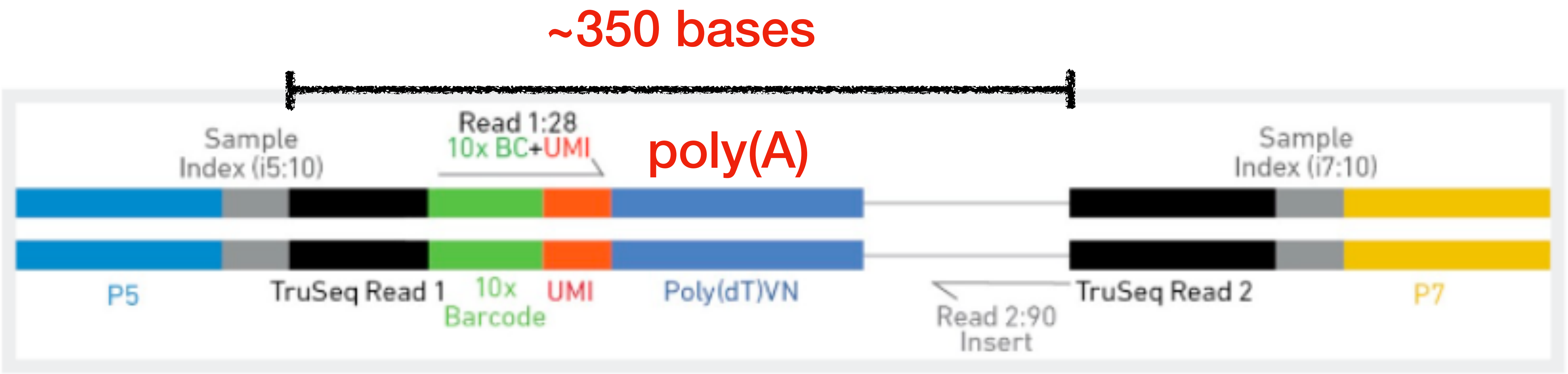

"After PCR amplification, cDNA molecules are randomly fragmented under conditions that favor **300-400 bp length fragments**."[1]

- 10x Genomics

Hidden messages behind 10X Chromium

1. cDNA fragments are 300-400 bp long.
2. The priming sites should contain a polyA stretch.

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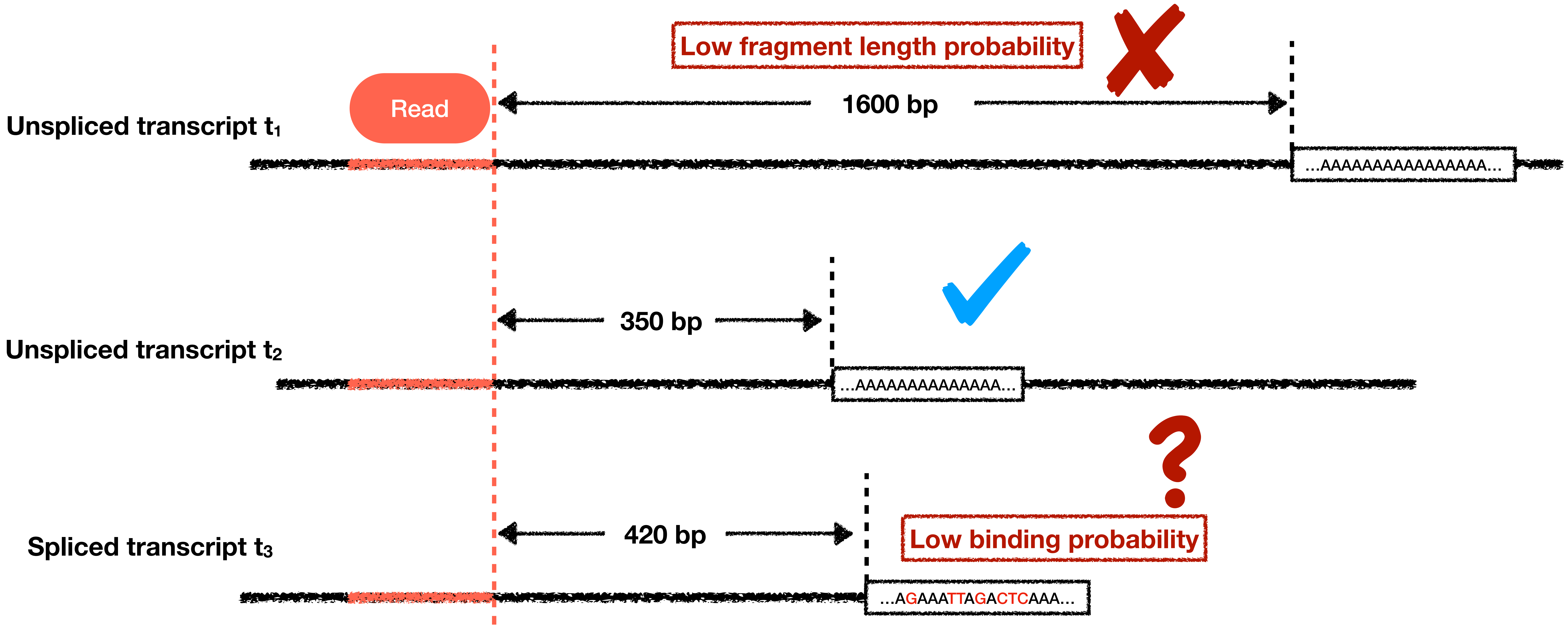
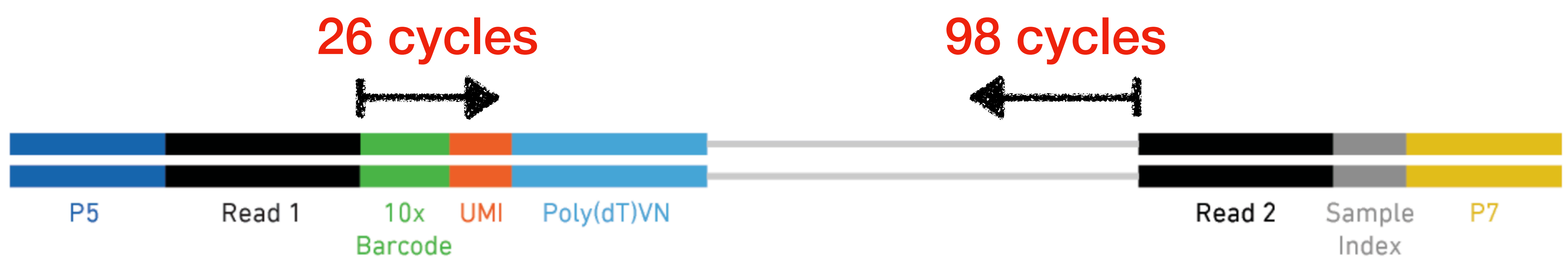


How can we get the probabilities?

Fragment length distribution

Incorporating mechanistic information in scRNA-seq

Sequencing Read	Recommended Number of Cycles
Read 1	26 cycles
i7 index	8 cycles
i5 index	0 cycles
Read 2	98 cycles



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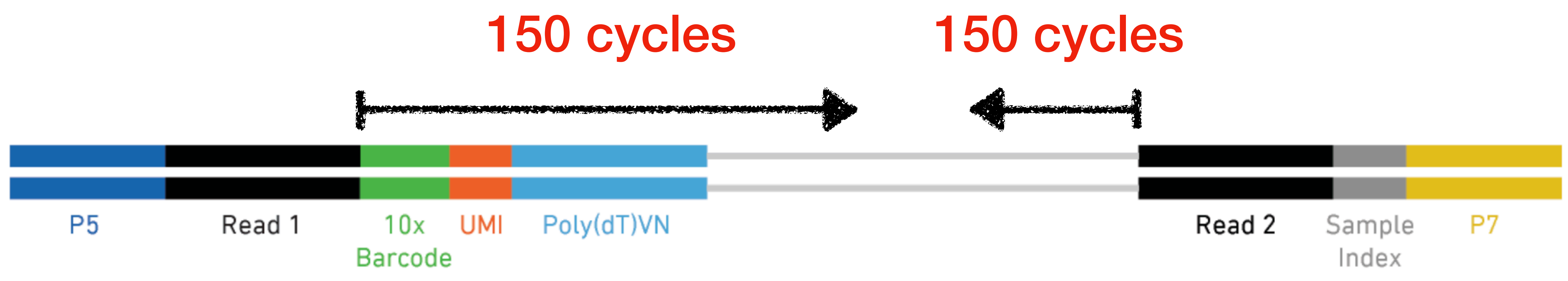

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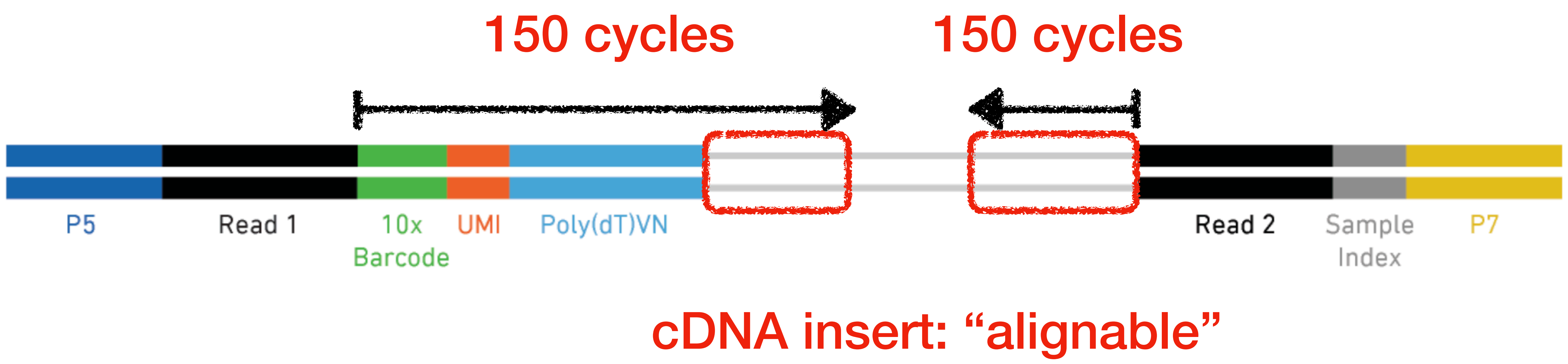

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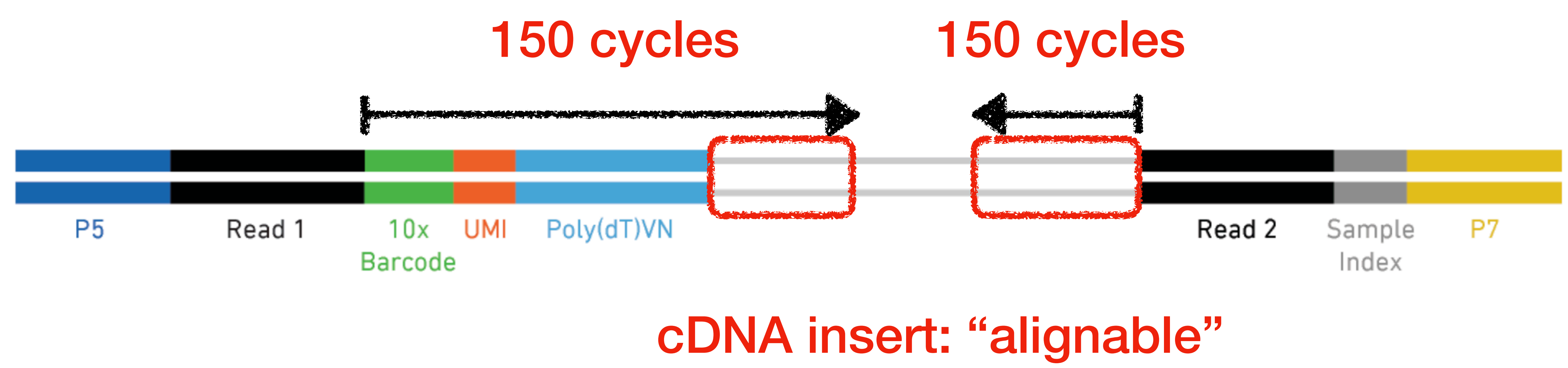

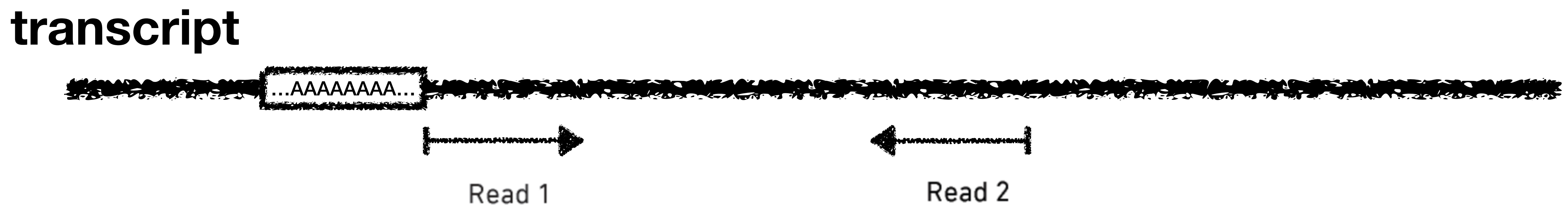

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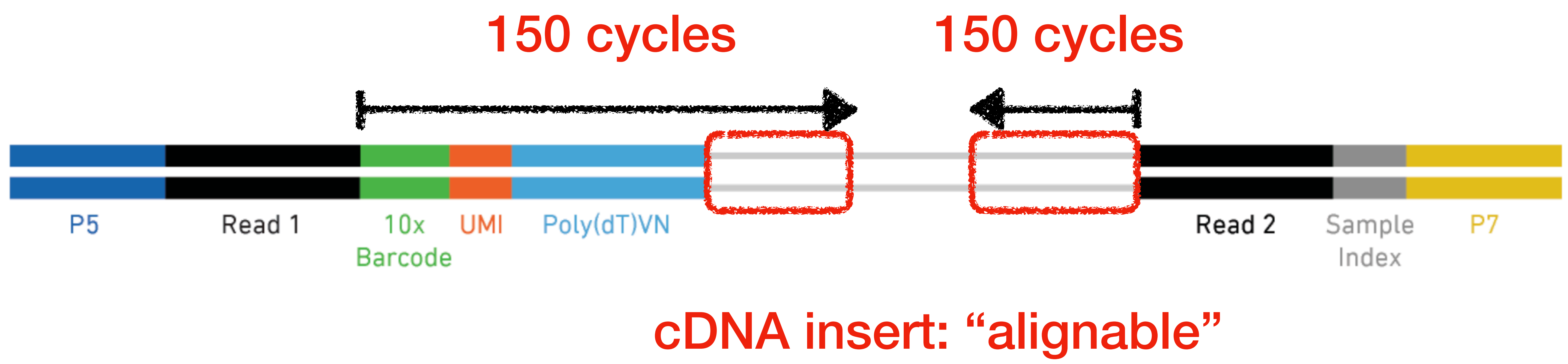

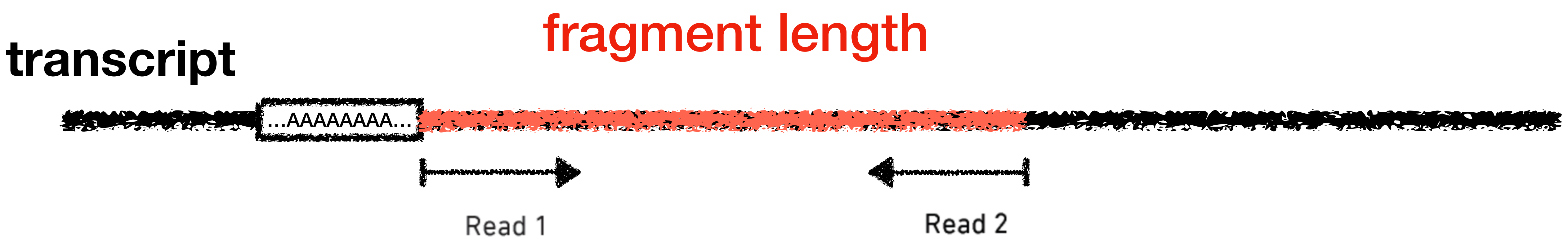

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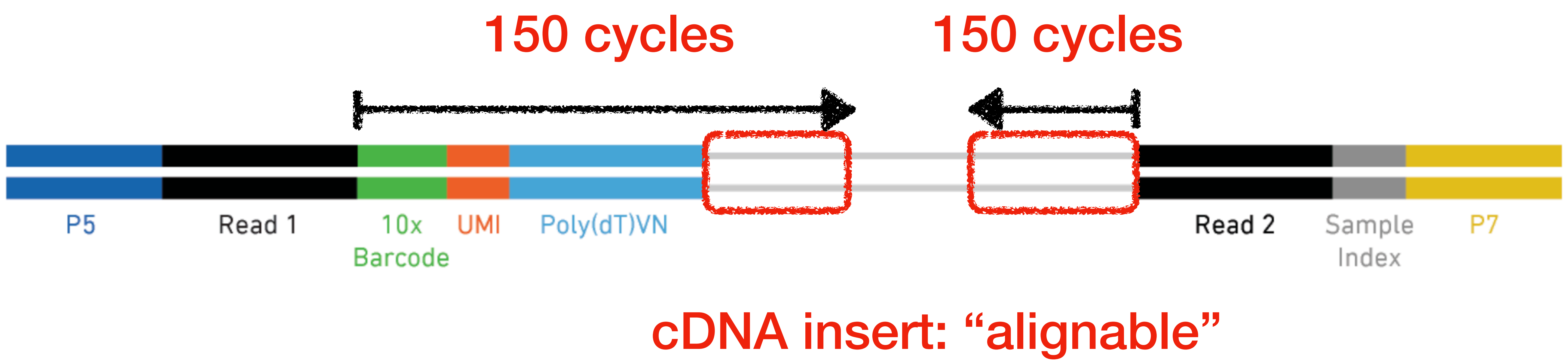

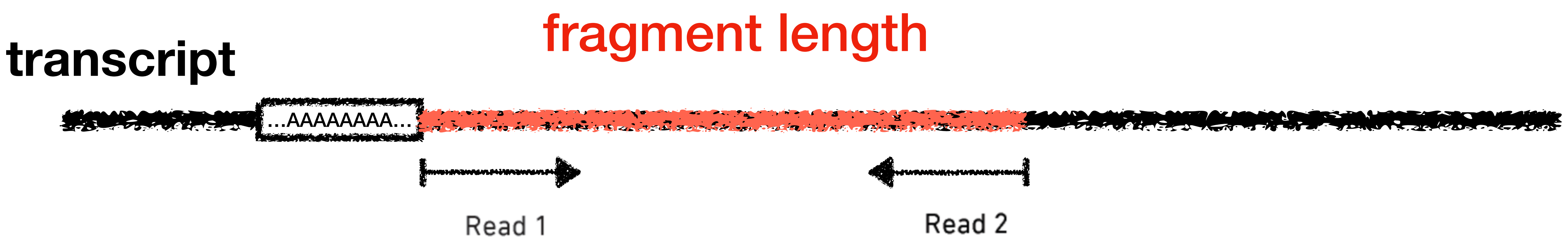

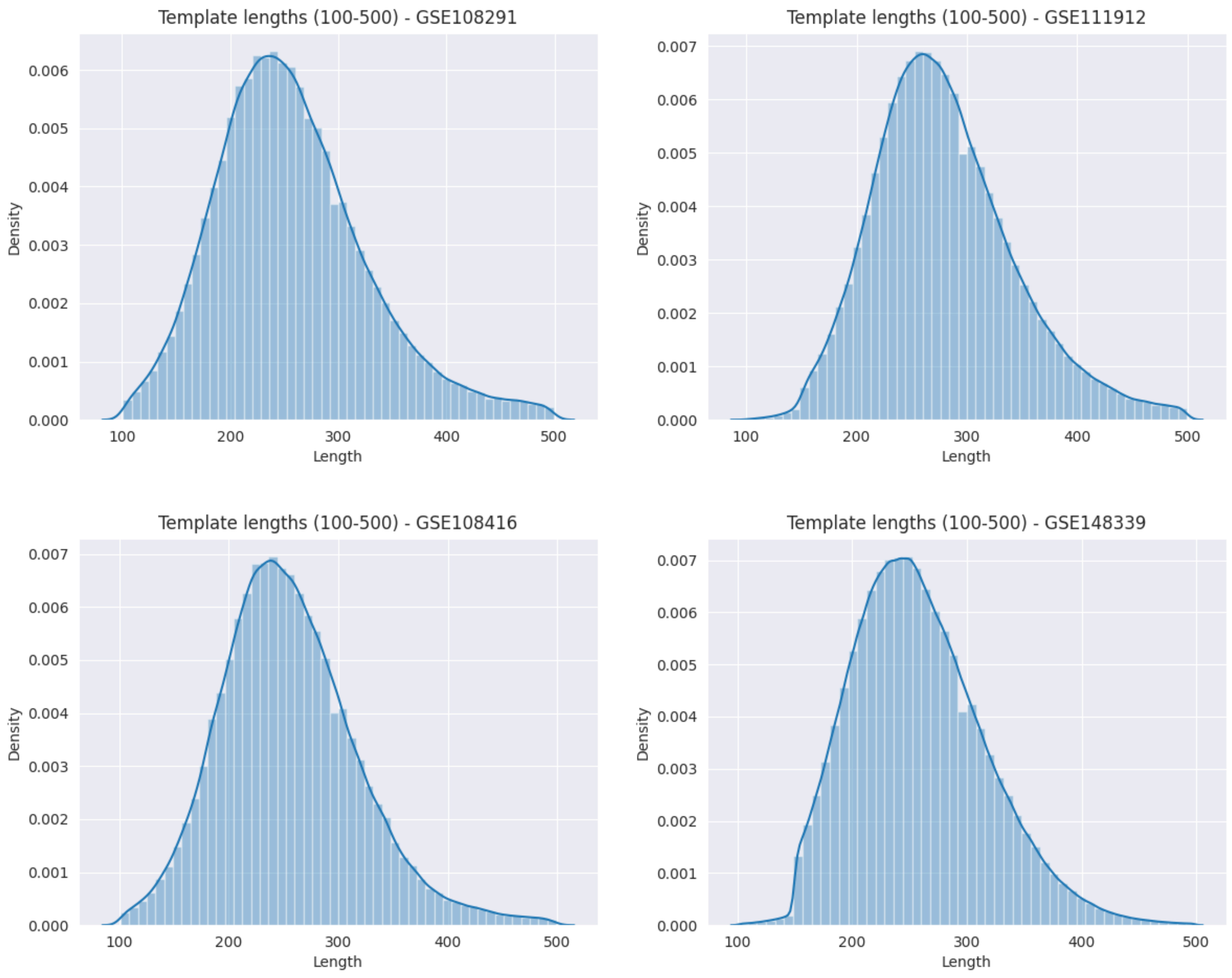

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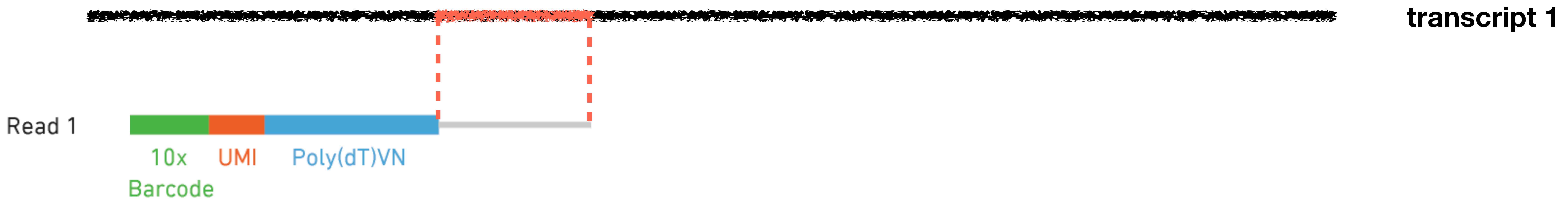
Oligo(dT) primer identification



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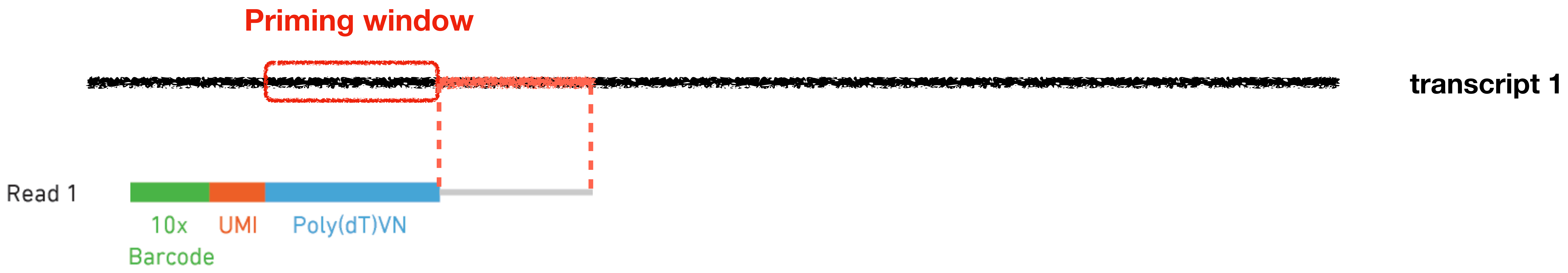

Observed priming windows tells us the probability that a given window belongs to the positive class (can be primed).

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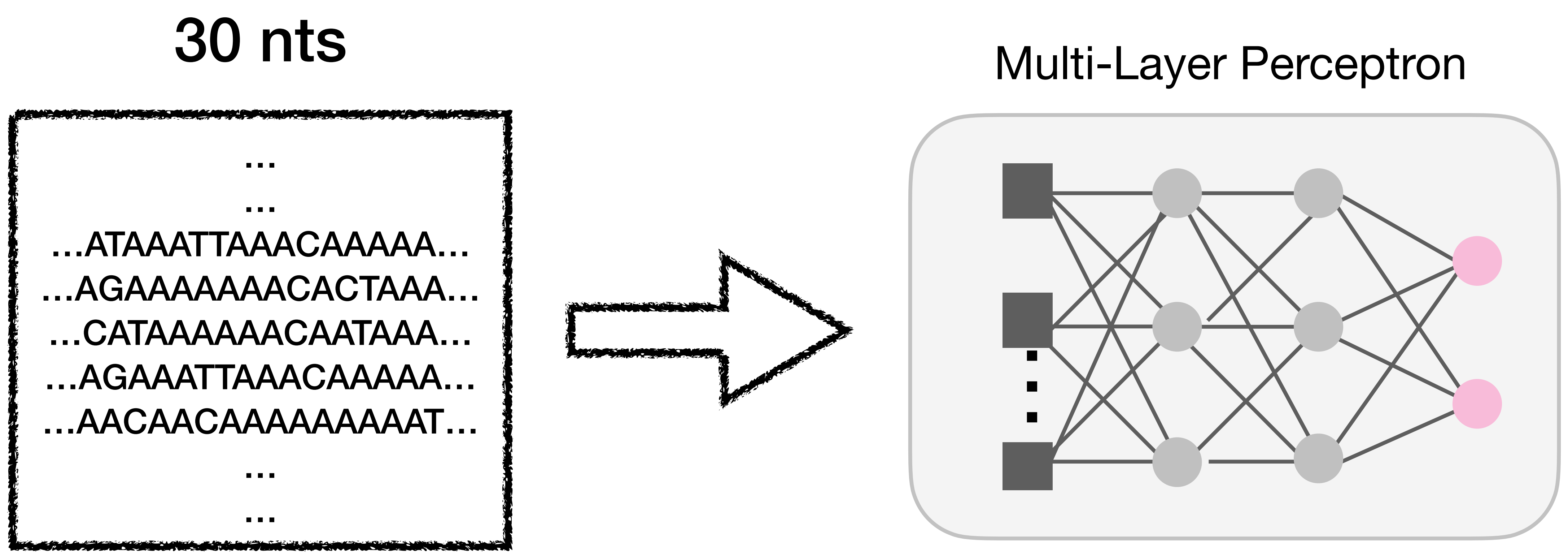

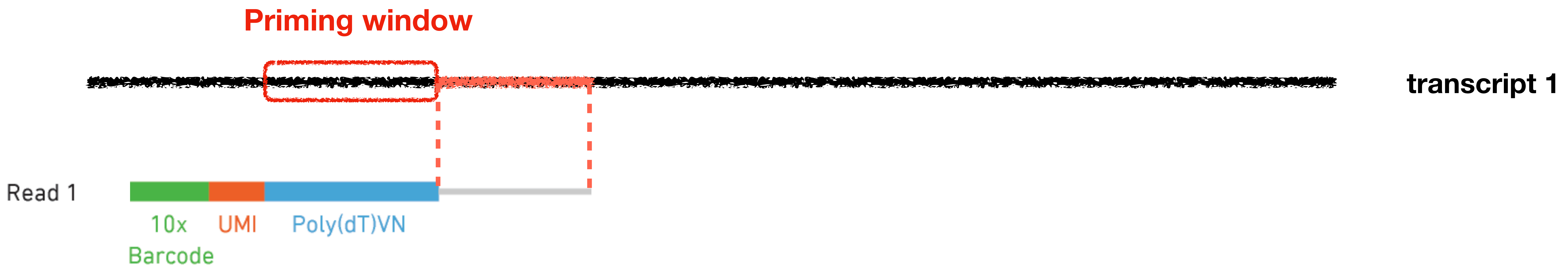

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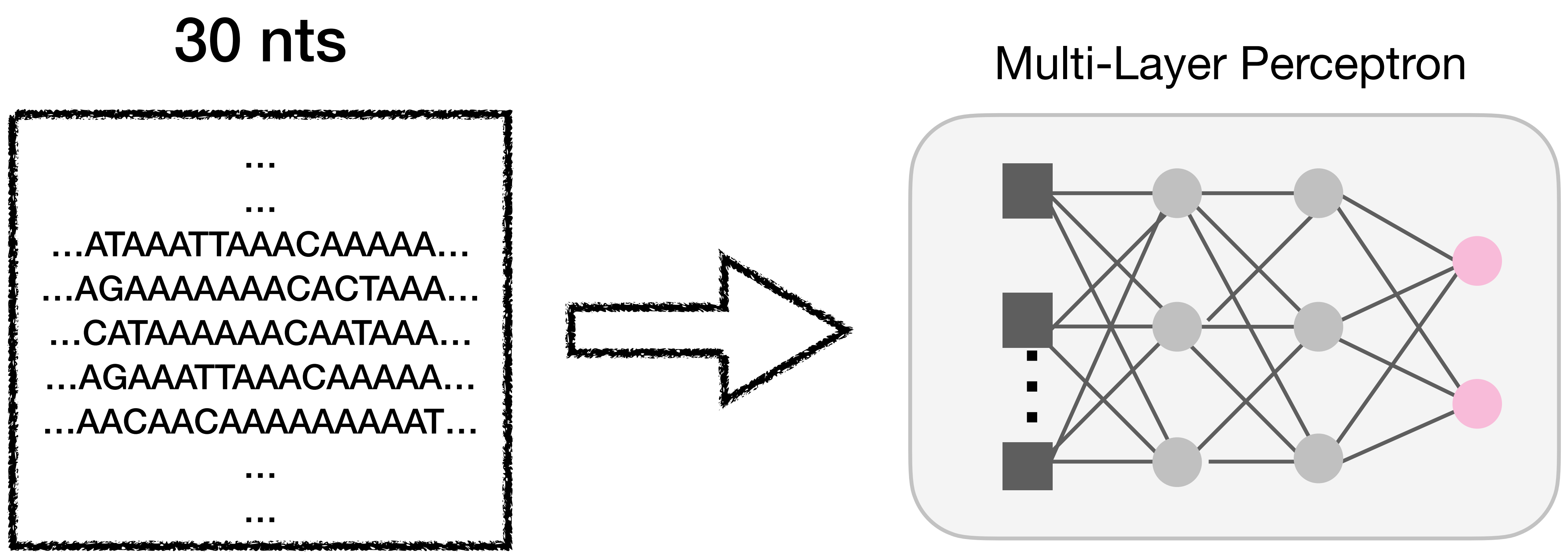
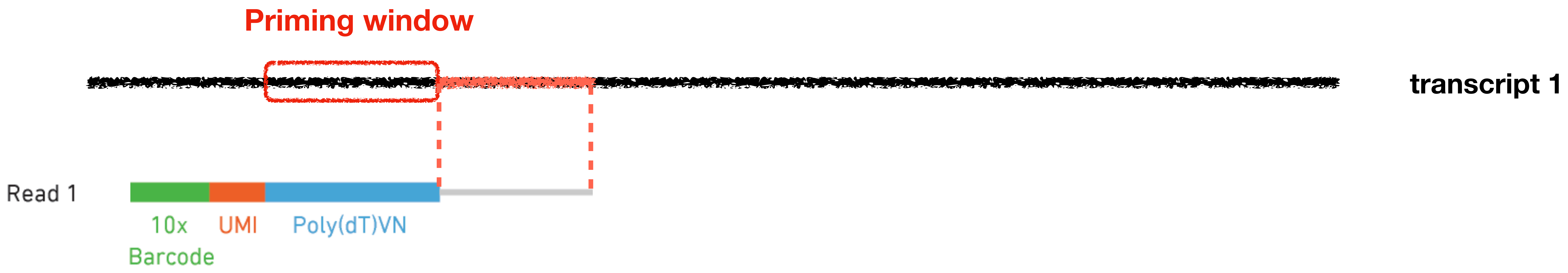


Accuracy

Training set: 0.877

Test set: 0.865

Evaluation set: 0.860



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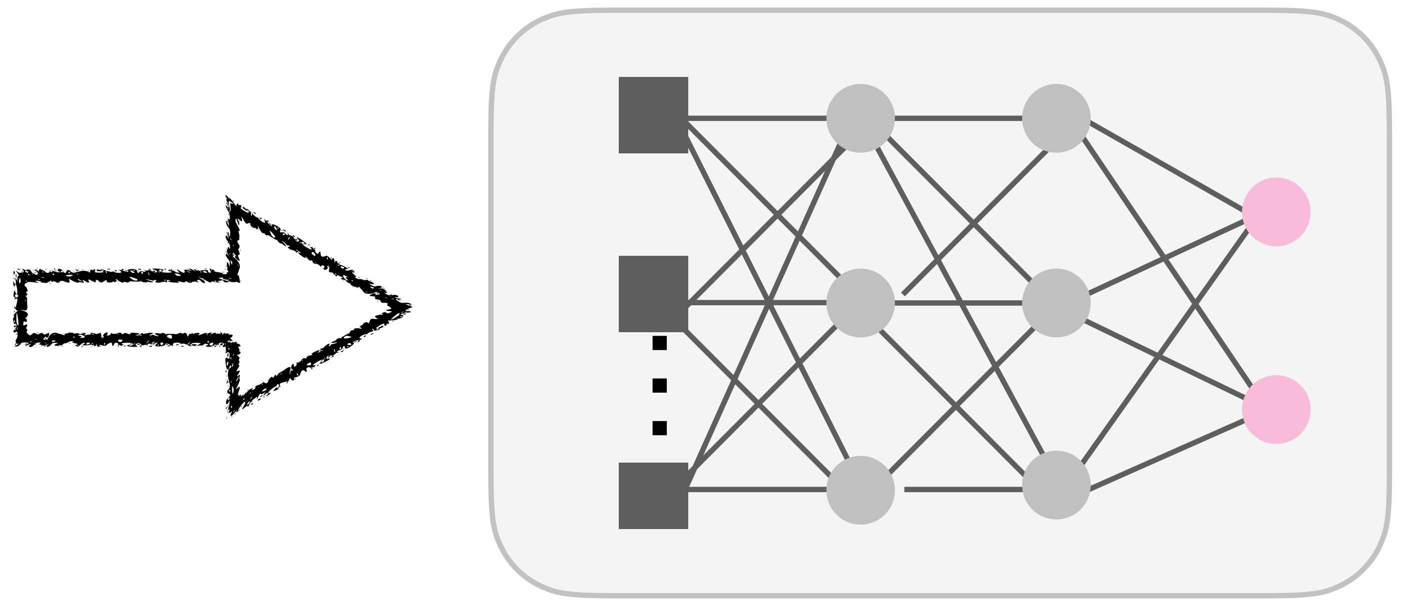
Oligo(dT) primer identification

30 nts

...
...
...ATAAATTAAACAAAAA...
...AGAAAAAAACACTAAA...
...CATAAAAAACAATAAA...
...AGAAATTAAACAAAAA...
...AACAACAAAAAAAAAT...
...
...

Multi-Layer Perceptron



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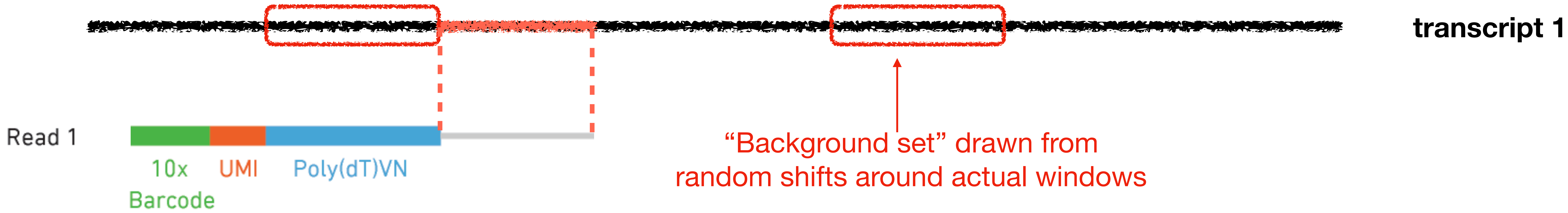
Priming window **Random window**



“Background set” drawn from
random shifts around actual windows

Observed priming windows tells us the probability that a given window belongs to the positive class (can be primed).

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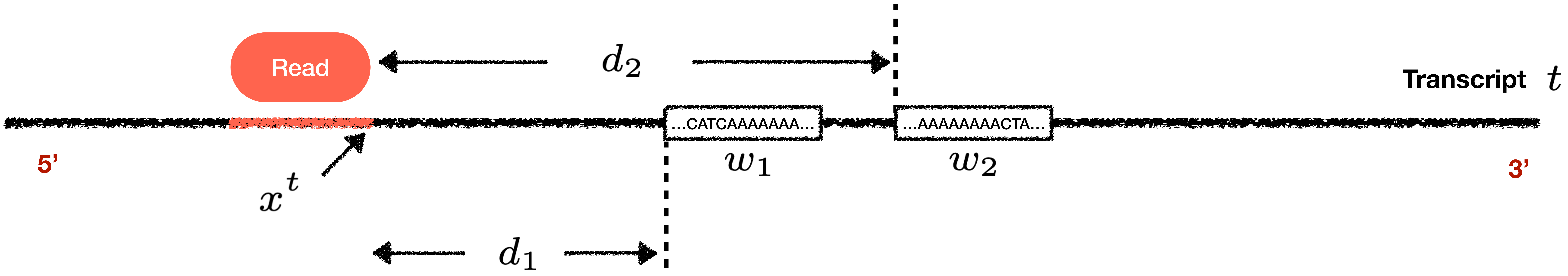
A mechanistic and predictive model



t - A reference transcript of an alignment

x^t - The alignment site

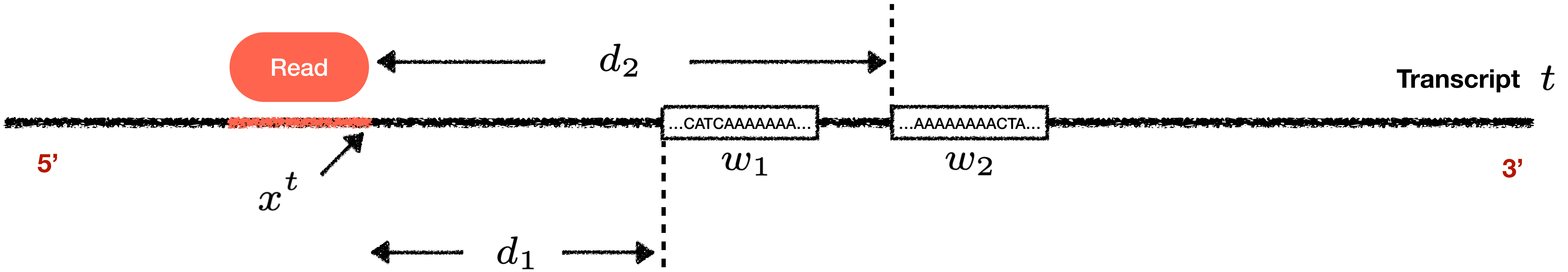
w_i - A potential priming window (30 nts)

d_i - The distance from w_i to x

fragment length probability   binding probability

$$p_f(d_i) \times p_b(w_i)$$

A mechanistic and predictive model



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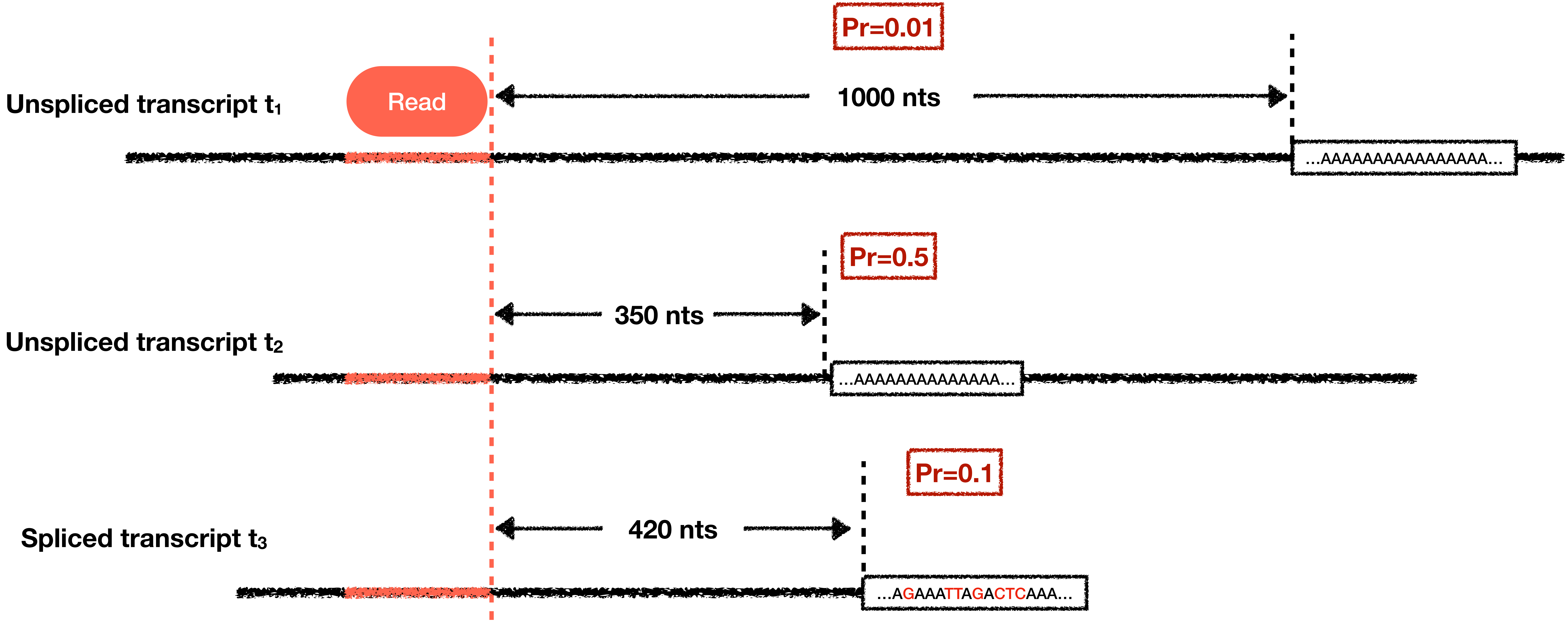
fragment length probability

binding probability

$$\Pr(t \mid x^t) = \max_{(w_i, d_i) \in \mathrm{pbs}(t, x^t)} p_f(d_i) \times p_b(w_i)$$

potential priming sites

The probability of being a spliced read



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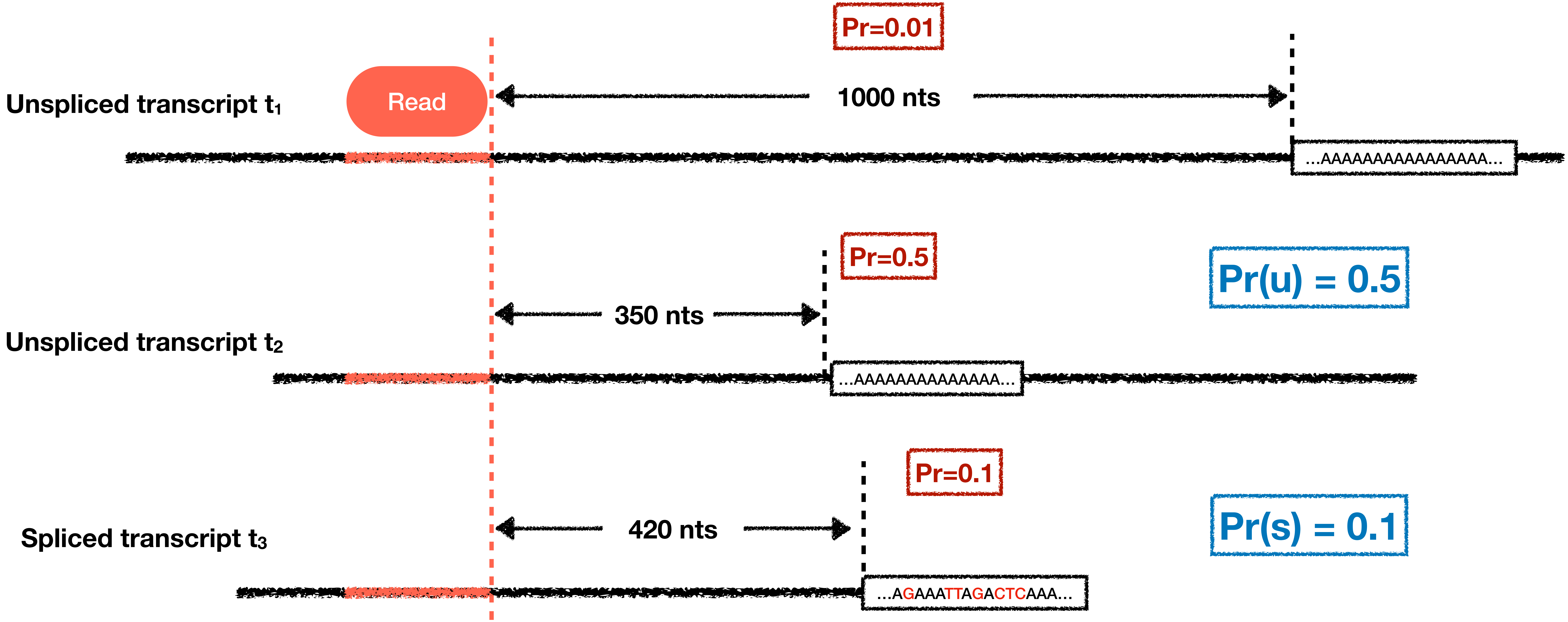
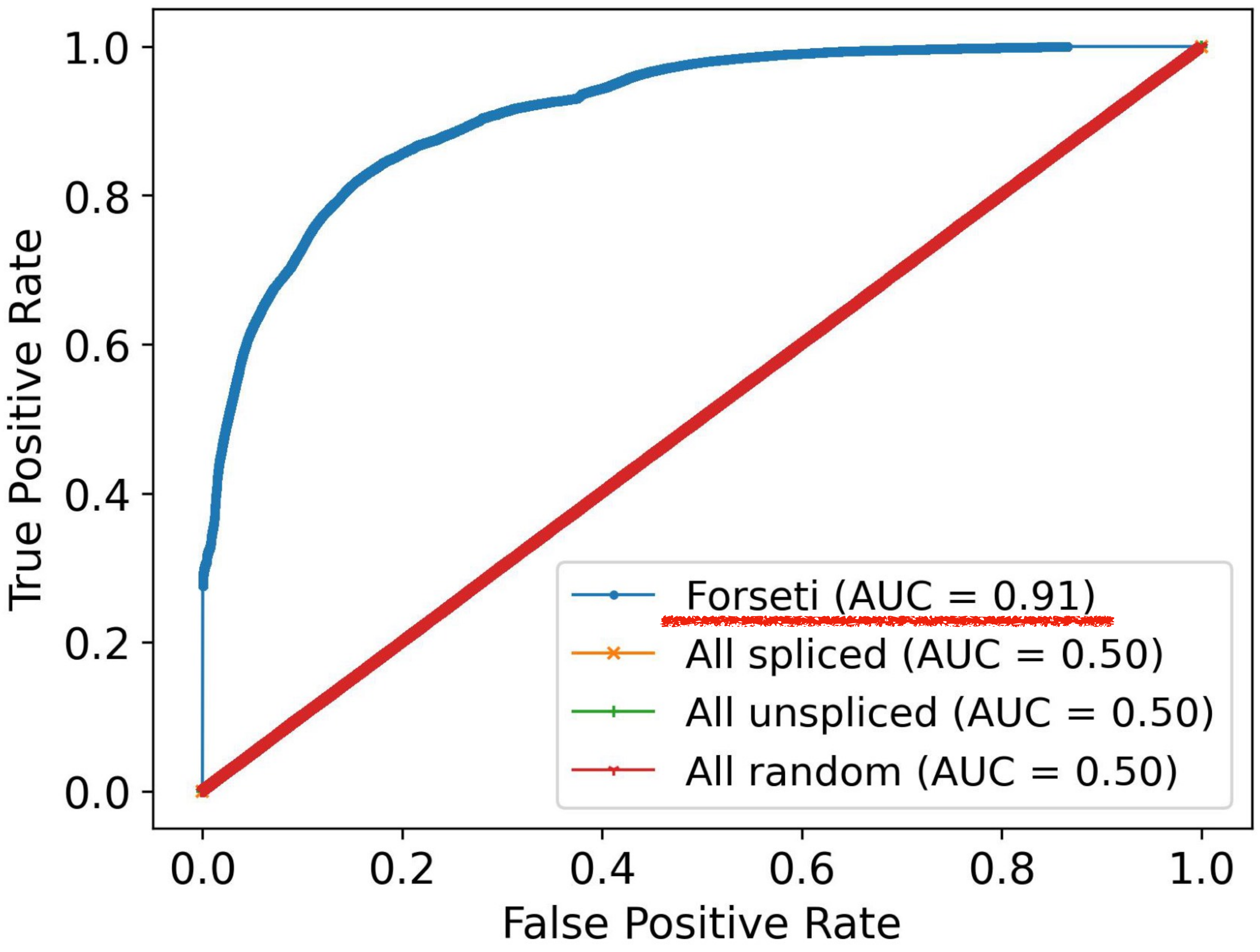
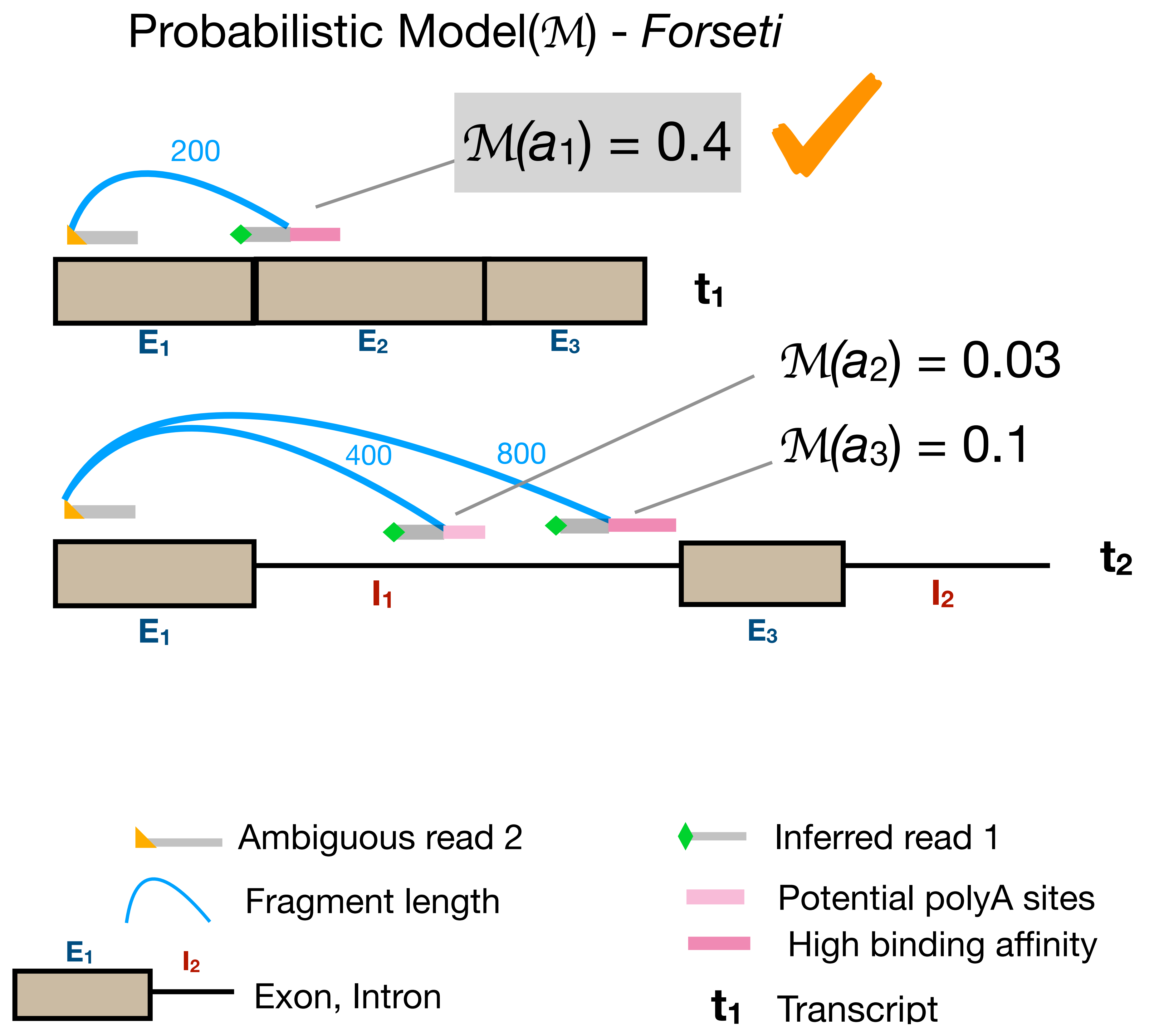


$$P(s|r,g) = \frac{0.1}{0.1+0.5} = 0.167$$

Forseti accurately predicts the splicing status of scRNA-seq reads





Note: These metrics are over reads that are not “fundamentally ambiguous”. For fundamentally ambiguous reads, no read-level recovery is possible.

Forseti can also also help resolve gene-ambiguous reads

Instead of just scoring splicing status *within* transcripts of a gene, score every potential (spliced and unspliced) priming site, including between genes.

Assign the read to the gene having the highest probability priming site. The Forseti model provides *additional information* not otherwise considered in multi-gene resolution.

When Forseti can make a prediction in the multi-gene context (when mappings for all genes aren't strictly tied), the prediction is right in ~70% of the cases. This isn't bad, but there's room for improvement.